

Algebraic Geometry

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1 Affine Varieties

Remark 1.0.1 — Throughout this course, k will be an algebraically closed field.

Example 1.0.2

Some examples of algebraically closed fields are $\mathbb{C}, \overline{\mathbb{Q}}, \mathbb{C}_p, \overline{\mathbb{F}_p} = \bigcup_n \mathbb{F}_{p^n}$.

§1.1 Affine Varieties

Definition 1.1.1. Let k be an algebraically closed field. We denote

$$\mathbb{A}_k^n := \{(c_1, \dots, c_n) : c_i \in k \text{ for } i = 1, \dots, n\}$$

the n -dimensional affine space over k .

Remark 1.1.2 — We may also just write $\mathbb{A}^n$ to mean k^n , when we don't care about the base field. There are at least two reasons for this nomenclature.

- (i) When we write k^n , it makes us think of a vector space, equipped with operations of addition and scalar multiplication. But $\mathbb{A}^n$ means just a set of points, described by coordinates $(x_1, \dots, x_n)$ with $x_i \in k$, without the vector space structure.
- (ii) Because it usually doesn't matter much what our base field k is (as long as it is algebraically closed), it is convenient to have notation which does not prominently mention k .

On occasions when it is important to specify which field k we are using, we write $\mathbb{A}_k^n$ for affine n -space.

Definition 1.1.3. If $S \subseteq k[x_1, \dots, x_n]$, we consider the **vanishing locus** of S is the set

$$V(S) := \{x \in \mathbb{A}^n : f(x) = 0 \text{ for all } f \in S\} \subseteq \mathbb{A}^n.$$

Here are some elementary facts.

- If $S_1 \subseteq S_2$, then $V(S_2) \subseteq V(S_1)$.
- $V(S) = V((S))$. In particular, we can always suppose that S is an ideal.

Notation 1.1.4. If $S = (f_1, \dots, f_n)$, then we denote $V(f_1, \dots, f_n) := V(\{f_1, \dots, f_n\})$.

Remark 1.1.5 — By the Hilbert basis theorem, every ideal of $k[x_1, \dots, x_n]$ is finitely generated, hence every affine variety can be written in the form $V(f_1, \dots, f_n)$.

Definition 1.1.6. An **affine algebraic variety** is a subset of $\mathbb{A}^n$ of the form $V(f_1, \dots, f_r)$ (for $f_i \in k[x_1, \dots, x_n]$; that is, a vanishing locus of some subset of $k[x_1, \dots, x_n]$).

Here are some examples.

Example 1.1.7

The whole space $\mathbb{A}^n$, defined by the polynomial $f = 0$ (or by the empty set of polynomials).

Example 1.1.8

The set $\{2\}$, defined by the polynomial $X - 2$. More generally, any single point in $\mathbb{A}^n$ is a variety. If $a \in \mathbb{A}^n$, then $V(x_1 - a_1, \dots, x_n - a_n)$ is just a . By the example below, finite sets of points are varieties.

Example 1.1.9

A finite union of affine algebraic varieties. In particular, $V(I) \cup V(J) = V(IJ)$, where $IJ = \{ij : i \in I, j \in J\}$.

Example 1.1.10

Intersections (possibly infinite) of varieties are varieties: $\cap_{\alpha} V(I_{\alpha}) = V(\sum_{\alpha} I_{\alpha})$.

Example 1.1.11

An algebraic curve in $\mathbb{A}^2$, that is, a set of the form

$$\{(x_1, x_2) \in \mathbb{A}^2 : f(x_1, x_2) = 0\}$$

for some polynomial $f \in k[X_1, X_2]$, is a variety.

Example 1.1.12

Embeddings of $\mathbb{A}^m$ in $\mathbb{A}^n$ where $m < n$ are varieties:

$$\{(x_1, \dots, x_m, 0, \dots, 0) \in \mathbb{A}^n\} = \{(x_1, \dots, x_n) \in \mathbb{A}^n : x_{m+1} = \dots = x_n = 0\}.$$

More generally, the image of a linear map $\mathbb{A}^m \rightarrow \mathbb{A}^n$ is of the form

$$\{(x_1, \dots, x_n) \in \mathbb{A}^n : \text{some linear conditions}\}.$$

Example 1.1.13 (Non-Example)

Any infinite proper subset of $\mathbb{A}^1$ is not a variety. This is because a one-variable polynomial with infinitely many roots must be the zero polynomial.

Example 1.1.14 (Non-Example)

It is not the case that images of maps with polynomial coordinates are subvarieties. For instance, consider $\phi: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ defined by $(x, y) \mapsto (x, xy)$. The image Z is $(\mathbb{A}^2 \setminus \{(0, y)\}) \cup \{0\}$. This is not a subvariety. Consider a polynomial f which vanishes on Z . For every fixed $y \neq 0$, $f(x, y)$ has infinitely many roots. In particular, f is the zero polynomial. However, Z is not $\mathbb{A}^2$.

§1.1.i The Zariski Topology

We have seen that (possibly infinite) intersections of varieties are varieties, finite unions of varieties are varieties, and that $\mathbb{A}^n, \emptyset$ are varieties.

Definition 1.1.15. The above implies that varieties form the closed sets of a topology on $\mathbb{A}^n$, called the **Zariski topology**. Further, any variety $V \subseteq \mathbb{A}^n$ inherits the Zariski topology, via the subspace topology.

To elaborate on this last sentence, $X \subseteq V$ is closed if and only if $X \subseteq \mathbb{A}^n$ is closed.

Remark 1.1.16 — We shall note that open subsets of V need not be open in $\mathbb{A}^n$. For instance, consider $V(X) \subseteq \mathbb{A}^n$, so $V(X)$ is a line. Look at an open subset $V(X) \setminus \{\text{point}\}$.

Example 1.1.17

The Zariski topology on $\mathbb{A}^1$ is the same as the cofinite topology; that is, the open sets are the complements of finite sets.

Remark 1.1.18 — We note that the Zariski topology has much less open and closed sets than the standard Euclidean topology.

Lemma 1.1.19

Let $k = \mathbb{C}$, with the usual Euclidean topology. If V is Zariski closed in $\mathbb{A}_{\mathbb{C}}^n$, then it is closed in the Euclidean topology.

Proof. If f is a polynomial, it defines a continuous map from $\mathbb{C}^n \rightarrow \mathbb{C}$ (evaluation map), and $V(f) = f^{-1}\{0\}$ is closed, since $\{0\}$ is closed. Thus, done. $\square$

The lemma above tells us that the Euclidean topology is finer than the Zariski topology.

Lemma 1.1.20

Every pair U_1, U_2 of Zariski open subsets of $\mathbb{A}^1$ have non-empty intersection.

Proof. We can write $U_1 = \mathbb{A}^1 \setminus V(f)$, $U_2 = \mathbb{A}^1 \setminus V(g)$. Thus, $U_1 \cap U_2 = \mathbb{A}^1 \setminus (V(f) \cup V(g))$, which is nonempty since f and g have finitely many roots. $\square$

Definition 1.1.21. If $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ are subvarieties, then $V \times W$ is a subvariety of $\mathbb{A}^{n+m}$. If $V = V(f_1, \dots, f_r)$ with $f_i \in k[x_1, \dots, x_n]$ and $W = V(g_1, \dots, g_l)$ with $g_j \in k[y_1, \dots, y_m]$, then we define the **product** $V \times W$ as

$$V(f_1(\underline{x}), \dots, f_r(\underline{x}), g_1(\underline{y}), \dots, g_l(\underline{y})) \subseteq \mathbb{A}^{n+m}.$$

Remark 1.1.22 — We note that the Zariski topology is not the same as the product topology, as shown in the following example.

Example 1.1.23

Consider $\mathbb{A}^1 \times \mathbb{A}^1$ and $\mathbb{A}^2$, which are isomorphic as sets, but not topological spaces. This is because $\mathbb{A}^2$ has sets like $V(f)$ as closed sets, which aren’t present on the left.

Exercise 1.1.24. Determine all Zariski closed subsets of the union of two lines $V(XY) \subset \mathbb{A}^2$. Deduce that $V(XY)$ is connected in the Zariski topology.

Solution. We have that $V(XY) = V(X) \cup V(Y)$, which is the union of two affine lines intersecting at the origin. Consider $W \subseteq V$, a closed subset. Then, $W = (W \cap V(X)) \cup (W \cap V(Y))$, is the union of two closed subsets of $V(X)$ and $V(Y)$. Therefore, the Zariski closed subsets of $V(XY)$ are V , $\emptyset$, and the union of $V(X)$ with a finite set of points of $V(Y)$, and the union of $V(Y)$ with a finite set of points of $V(X)$. $V(XY)$ is connected in the Zariski topology, because we cannot write it as a union of disjoint nonempty closed subsets. In particular, if we could write it as a disjoint union, then it would be a union of a subset of $V(X)$ and a subset of $V(Y)$, so contains the origin—a contradiction. $\square$

§1.1.ii Subvarieties and Ideals

Definition 1.1.25. If A is any subset of $\mathbb{A}^n$, we can define an **ideal** associated to A ,

$$I(A) = \{f \in k[X_1, \dots, X_n] : f(a) = 0, \forall a \in A\}.$$

Remark 1.1.26 — This is indeed an ideal of $k[X_1, \dots, X_n]$.

Note that we now have two functions.

$$V : \{ \text{ideals of } k[X_1, \dots, X_n] \} \leftrightarrow \{ \text{subvarieties of } \mathbb{A}^n \} : I$$

Remark 1.1.27 — These are not inverses.

Example 1.1.28

Consider $I = (X^2) \subseteq k[X]$. Then, $V(I) = \{0\}$ and $I(V(I)) = (X) \neq I$.

Lemma 1.1.29

If $V \subseteq \mathbb{A}^n$ is a subvariety, then $V(I(V)) = V$.

Proof. We always have $V \subseteq V(I(V))$, for any V . The reverse inclusion is where we use that V is a variety. Suppose that $y \notin V$. We want to show that $y \notin V(I(V))$. Let $V = V(J)$, for some ideal J . Since $y \notin V(J)$, there exists $f \in J$ such that $f(y) \neq 0$. $J \subseteq I(V)$, so $V(J) \supseteq V(I(V))$. Therefore, $f(y) \neq 0$, which implies that $y \notin V(I(V))$. $\square$

Exercise 1.1.30. Let $A \subseteq \mathbb{A}^n$ be any subset. Prove that $V(I(A))$ is the Zariski closure of A .

Solution: Firstly, it is clear that $A \subseteq V(I(A))$. This is because if $a \in A$, then $f(a) = 0$, for all $f \in I(A)$, so $a \in V(I(A))$. Further, this is closed because it is a variety. Now, we want to show that $V(I(A))$ is the smallest Zariski closed set which contains A . Let B be any other Zariski closed set which contains A ; that is, $A \subset B$. Then we have that

$$A \subseteq B \implies I(B) \subseteq I(A) \implies V(I(A)) \subseteq V(I(B)) = B,$$

where the final equality holds because B is a variety. $\square$

§1.1.iii The Nullstellensatz

Definition 1.1.31. Given an ideal I of a commutative ring R , the **radical** of I , is the ideal $\sqrt{I}$, defined by

$$\sqrt{I} := \{r \in R : \exists n, r^n \in I\}.$$

We say that I is **radical** if $I = \sqrt{I}$.

Recall the following statement.

Theorem 1.1.32 (Hilbert's Nullstellensatz)

Let I be an ideal of $k[X_1, \dots, X_n]$. Then, $I(V(I)) = \sqrt{I}$.

Remark 1.1.33 — Calculating $\sqrt{I}$ is difficult; that is, we can't compute this just from looking at the generators.

Example 1.1.34

For instance, let us try to compute the radical of the ideal $I = (X, X - Y^2) \subseteq k[X, Y]$. We note that $I = (X, Y^2)$, so $\sqrt{I} = (X, Y) \neq I$.

§1.2 Connectedness and Irreducibility

Now consider the following affine algebraic subvarieties of $\mathbb{A}^2$. Do they have one or two *pieces*? Note that here, we have deliberately not specified what is meant by pieces. There are multiple sensible interpretations, so there is not always a unique *correct* answer.

1. The union of two disjoint lines $V(X(X - 1))$.

2. The union of two intersecting lines $V(XY)$.

3. The hyperbola $V(XY - 1)$.

To begin with, the variety $V(X(X - 1))$ unambiguously has two pieces; that is, the two lines $X = 0$ and $X = 1$. Recall the following definition now.

Definition 1.2.1. A topological space X is **connected** if it is nonempty and it can't be written as $X = V_1 \cup V_2$, of two disjoint nonempty closed subset.

This notion makes sense for the Zariski topology. The set

$$V(X(X - 1))$$

is therefore not connected because

$$V(X(X - 1)) = V(X) \cup V(X - 1).$$

Next, $V(XY)$. This has more than one answer; the two axes form two pieces. However, they intersect at the origin, joining them into one piece. The set

$$V(XY)$$

is connected but reducible. To elaborate on this, we must introduce some more terminology.

Definition 1.2.2. A topological space S is **reducible** if it is empty, or there exist closed sets $S_1, S_2 \subseteq S$ such that $S = S_1 \cup S_2$, and neither S_1 nor S_2 is equal to S .

Definition 1.2.3. Conversely, a topological space S is **irreducible** if it is nonempty and it is not possible to write it as the union $S_1 \cup S_2$ for two closed sets, unless at least one of S_1 and S_2 is equal to S itself.

Remark 1.2.4 — Note that a change from the definition of connectedness is that S_1 and S_2 are not required to be disjoint.

Remark 1.2.5 — Irreducibility is not a very useful notion for the topological spaces in analysis. For example, considering the real line with the Euclidean topology, we can write it as a union of proper closed subsets:

$$\mathbb{R} = \{x \in \mathbb{R} : x \leq 0\} \cup \{x \in \mathbb{R} : x \geq 0\}.$$

These subsets are not disjoint because they intersect at 0.

It can often be convenient to rewrite the definition of irreducible spaces in terms of open sets instead of closed sets:

Lemma 1.2.6

The following conditions on a topological space X are equivalent to irreducibility:

- (i) X is nonempty, and every pair of nonempty open subsets $U_1, U_2 \subseteq X$ has non-empty intersection $U_1 \cap U_2$.
- (ii) X is non-empty, and every non-empty open subset of X is dense in X .

Proof. (i). Suppose X is irreducible and take open nonempty $U_1, U_2 \subseteq X$. Suppose that $U_1 \cap U_2 = \emptyset$. Then, $X \setminus U_i$ are closed, so $(X \setminus U_1) \cup (X \setminus U_2) = X$. By irreducibility, WLOG, suppose that $X \setminus U_1 = X$, so $U_1 = \emptyset$, a contradiction. Thus, $U_1 \cap U_2 \neq \emptyset$. For the converse now. Let V_1, V_2 be closed such that $X = V_1 \cup V_2$. Suppose that V_1, V_2 are proper and nonempty. Then, $X \setminus V_i$ are open, so $(X \setminus V_1) \cap (X \setminus V_2) \neq \emptyset$. Therefore, there is some point p in $(X \setminus V_1) \cap (X \setminus V_2)$, so there is some point p in $X \setminus (V_1 \cup V_2)$. However, $V_1 \cup V_2 = X$, so this latter space is empty, so contradiction.

(ii). First suppose X is irreducible. Let $U \subseteq X$ be open and nonempty. Now, consider $\overline{U} \cup U^c$. By irreducibility, one of these must be X , but by nonemptiness of U , it follows that $\overline{U} = X$. Thus, U is dense. For the converse, suppose that every nonempty open subset of X is dense in X . Suppose that $X = V_1 \cup V_2$, where the V_i are nonempty and closed, and $V_1 \neq \emptyset$. Let $U_1 = V_1^c$, so that $U_1 \subseteq V_2$. Therefore, $\overline{U_1} \subseteq V_2$ as V_2 is closed. However, since U_1 is dense in X , it follows that $X = V_2$, so X is irreducible. $\square$

Corollary 1.2.7

Let X be an irreducible topological space and $U \subseteq X$ a nonempty open subset. Then U is irreducible (in the subspace topology).

Proof. Suppose that we can write $U = V \cup W$, a union of two closed subsets of U . By the subspace topology, $V = V' \cap U$ and $W = W' \cap U$, where V', W' are closed in X . Focus now at the union $V' \cup W'$. We know that $(V' \cup W') \cap U = U$, implying that $V' \cup W' \supseteq U$, which implies that $V' \cup W' = X$ (since U is dense in X), so U is irreducible. $\square$

Remark 1.2.8 — Corollary 1.2.7 says that being irreducible is a very long way from Hausdorff; the Hausdorff condition says that a space has lots of pairs of disjoint non-empty open subsets, while an irreducible space has none. For example, we saw that $\mathbb{R}$ (with the Euclidean topology) is reducible in many ways.

Now finally consider $V(XY - 1)$. A drawing of it in $\mathbb{R}^2$ looks like it has two pieces. However, we are missing a lot by only looking at real solutions. Over $\mathbb{C}$, it unambiguously has one *piece*. One way to visualize this is to note that $V(XY - 1)$ looks like $\mathbb{A}^1 \setminus \{0\}$ by projecting onto the x -coordinate, which is a bijection between these sets. This is not a formal statement yet, as we have not yet defined a notion of isomorphism of affine algebraic varieties, and even if we had, $\mathbb{A}^1 \setminus \{0\}$ is not an affine algebraic variety. Later, we will develop technology to make this into a rigorous statement.

But for now we use it as a heuristic. The set $\mathbb{R} \setminus \{0\}$ unambiguously has two pieces, but $\mathbb{C} \setminus \{0\}$ unambiguously has one piece. Therefore, the hyperbola (over an algebraically closed field) should have only one piece. We prove later that

$$V(XY - 1)$$

is irreducible (and also connected).

Here is an extra fact about connected sets in the Zariski topology which will not be proven:

Theorem 1.2.9

Over $\mathbb{C}$, a variety is connected in the Zariski topology if and only if it is connected in the Euclidean topology.

Exercise 1.2.10. Let $V \subset \mathbb{A}^m$ and $W \subset \mathbb{A}^n$ be irreducible affine algebraic varieties. Prove that the product $V \times W$ is irreducible.

Solution: Suppose that $V \times W = Z_1 \cup Z_2$, with $Z_i \subset V \times W$ closed. For any $x \in V$, we have $\{x\} \times W$ is isomorphic to W as a topological space, hence irreducible. But

$$\{x\} \times W = (\{x\} \times W \cap Z_1) \cup (\{x\} \times W \cap Z_2)$$

is a decomposition into two closed subsets, hence by irreducibility of W , for each $x \in V$, there is $i \in \{1, 2\}$ such that $\{x\} \times W \subset Z_i$. Then

$$V = \{x \in V : \{x\} \times W \subset Z_1\} \cup \{x \in V : \{x\} \times W \subset Z_2\},$$

and the two subsets are closed. To see that these sets are closed, we use the projection map $\pi_V : V \times W \rightarrow V$. We have that

$$V \setminus \{x \in V : \{x\} \times W \subset Z_i\} = \pi_V((V \times W) \setminus Z_i).$$

Noting that $(V \times W) \setminus Z_i$ is open (because Z_i is closed) and that π_V is an open map, we have that $V \setminus \{x \in V : \{x\} \times W \subset Z_i\}$ is open, so we have proven true. To finish, the irreducibility of V tells us that

$$V = \{x \in V : \{x\} \times W \subset Z_i\}$$

for some $i \in \{1, 2\}$; thus, $V \times W = Z_i$. □

§1.2.i Prime Ideals and Irreducible Varieties

Definition 1.2.11. An ideal I of a commutative ring R is a **prime ideal** if I is proper and for every $f, g \in R$, if $fg \in I$, then $f \in I$ or $g \in I$.

Lemma 1.2.12

An affine variety $V \subset \mathbb{A}^n$ is irreducible if and only if $I(V)$ is a prime ideal in $k[X_1, \dots, X_n]$.

Proof. First suppose that V is irreducible. Suppose we have $f, g \in k[X_1, \dots, X_n]$ such that $fg \in I(V)$. Let

$$V_1 = V(f) \cap V, \quad V_2 = V(g) \cap V.$$

For every $x \in V$, $f(x)g(x) = 0$ and hence either $f(x) = 0$ or $g(x) = 0$. Thus for every $x \in V$, either $x \in V_1$ or $x \in V_2$. In other words, $V = V_1 \cup V_2$. Furthermore, V_1 and V_2 are closed subsets of V . Hence as V is irreducible, either $V_1 = V$ or $V_2 = V$. If $V_1 = V$ then $f \in I(V)$ and if $V_2 = V$ then $g \in I(V)$.

Now suppose that V is reducible. Then we can write it as a union $V_1 \cup V_2$ of proper closed subsets. Since V_1 is a proper closed subset of V , there exists some $f \in k[X_1, \dots, X_n]$

vanishing on V_1 but not on all of V . Similarly there exists g vanishing on V_2 but not on all of V . Thus neither f nor g is in $I(V)$, but the product fg vanishes on $V_1 \cup V_2$ and hence we have $fg \in I(V)$. Thus $I(V)$ is not prime.

V is empty if and only if $I(V) = k[X_1, \dots, X_n]$, which is explicitly defined to not be a prime ideal. So it was okay to ignore this case above. $\square$

Exercise 1.2.13. Prove that a hypersurface $\{x \in \mathbb{A}^n : f(x) = 0\}$ is irreducible if and only if f is a power of an irreducible polynomial. *Hint: You will need to use the Nullstellensatz and the fact that $k[X_1, \dots, X_n]$ is a unique factorization domain.*

Solution: Writing $V = V(f)$, we note that V is irreducible if and only if $I(V) = \sqrt{(f)}$ is prime. If we write the decomposition of f into irreducibles as $\prod_{i=1}^r f_i^{a_i}$, then we have that

$$\sqrt{(f)} = \left(\prod_{i=1}^r f_i \right) = \bigcap_{i=1}^r (f_i).$$

Geometrically, this means that $V(f) = \bigcup_{i=1}^r V(f_i)$, so this is irreducible if and only if $r = 1$. $\square$

Exercise 1.2.14. Show that the following varieties are irreducible:

1. The parabola $V(Y - X^2) \subset \mathbb{A}^2$.
2. The circle $V(X^2 + Y^2 - 1) \subset \mathbb{A}^2$. (assuming that the characteristic of k is not 2)

Solution: We just need to show that the following polynomials which define the variety are powers of irreducible polynomials.

1. $Y - X^2$ is clearly irreducible.

2. $X^2 + Y^2 - 1$ is also clearly irreducible. If it weren't, then it must be of the form $(X + A(Y))(X + B(Y))$, where $A(Y), B(Y) \in k[Y]$. Then, we just equate parts to say that $A(Y) = -B(Y)$, so we get that $-A^2(Y) = Y^2 - 1$. That is, $A^2(Y) = 1 - Y^2$. However, this is not possible because again if it were, then $A(Y) = (aY + b)$, but then $a^2 = -1$, $2ab = 0$, and $b^2 = 1$. This is inconsistent, because then one of a or b should be 0, but then $a^2, b^2 \neq 0$. $\square$

Remark 1.2.15 — Note that the 0 ideal of a ring R is prime precisely when R is an integral domain. This is true because 0 is prime if and only if $fg = 0$ implies that $f = 0$ or $g = 0$.

Definition 1.2.16. Let X be a topological space. An **irreducible component** of X is a maximal irreducible subset of X (with respect to inclusion).

Using the language between ideals and subvarieties, one finds that an irreducible component of V corresponds to a minimal prime ideal containing the radical ideal $I(V)$.

Remark 1.2.17 — We shall note that irreducible components of a variety correspond to minimal prime ideals of the polynomial ring.

Proposition 1.2.18

Any affine variety has a finite number of irreducible components, and is the union of its irreducible components.

Proof. This essentially follows (and is equivalent) from the primary decomposition theorem for radical ideals. $\square$

Remark 1.2.19 — As an easy observation, we shall note that unlike connected components, irreducible components need not be disjoint. For example, the irreducible components of

$$\{(x, y) : xy = 0\}$$

are the lines $x = 0$ and $y = 0$, which intersect in $\{(0, 0)\}$.

More generally, the irreducible components of a hypersurface $V(f)$ correspond to the irreducible factors of f : if

$$f = f_1^{a_1} \cdots f_m^{a_m}$$

(where the f_i are distinct irreducible polynomials), then the irreducible components of $V(f)$ are

$$V(f_1), \dots, V(f_m).$$

Exercise 1.2.20. Let V be the affine algebraic variety in $\mathbb{A}^2$ defined by the polynomials

$$f = X^2 + Y^2 - 1, \quad g = X - 1.$$

Describe the set V and find $I(V)$. Is $I(V) = (f, g)$?

Solution: V determines the points which vanish simultaneously at f and g . This is when $X = 1$ and $Y = 0$, so $V = \{(1, 0)\}$. $I(V) = (X - 1, Y)$. Thus, $I(V) \neq (f, g) = (X - 1, Y^2)$. $\square$

Exercise 1.2.21. Describe the irreducible components of the affine algebraic varieties

$$V := V(XY, Z), \quad W := V(XY, XZ) \subseteq \mathbb{A}^3.$$

Solution: First, V . The irreducible components here are given by $V(X, Z)$, $V(Y, Z)$.

Next, W . The irreducible components here are $V(Y, Z)$, $V(X)$. $\square$

Exercise 1.2.22. Assume that the characteristic of k is not 2. Find the irreducible components of the subset $V \subset \mathbb{A}^3$ defined by the equations

$$X^2 + Y^2 + Z^2 = 0, \quad X^2 - Y^2 - Z^2 + 2 = 0.$$

Solution: We note that $X^2 + 1 = 0$ and $Y^2 + Z^2 - 1 = 0$. We note that the curve $Y^2 + Z^2 - 1 = 0$ is smooth; hence irreducible. By algebraic closure, $\sqrt{-1} = i \in k$. Therefore,

$$V = V(X + i, Y^2 + Z^2 - 1) \cup V(X - i, Y^2 + Z^2 - 1)$$

$\square$

Exercise 1.2.23. Decompose into irreducible components the algebraic variety $V \subset \mathbb{A}^3$ defined by the polynomials

$$f = Y^2 - XZ, \quad g = Z^2 - Y^3.$$

Start by factorizing $Yf + g = h_1h_2$. Then describe $V \cap V(h_1)$ and $V \cap V(h_2)$.

Solution: By the hint, since $V \subset V(h_1) \cup V(h_2)$, we have that $V = (V \cap V(h_1)) \cup (V \cap V(h_2))$. We have that

$$Yf + g = Z^2 - XYZ = Z(Z - XY) =: h_1h_2.$$

Then, $V \cap V(h_1) = V(Y, Z)$. Next, $V \cap V(h_2)$. Imposing the condition $Y = X^2$ to $Y^2 - XZ$ gives us $Y(Y - X^2) = 0$. If $Y = 0$, then again, we get $V(Y, Z)$. If we impose $Y = X^2$, then the second condition doesn't tell us anything, so we get $V(X^2 - Y, Z)$. Therefore, $V \cap V(h_2) = V(Y, Z) \cup V(X^2 - Y, Z - XY)$. Thus,

$$V = V(Y, Z) \cup V(X^2 - Y, Z - XY).$$

□

Remark 1.2.24 — This technique works even if the irreducible factors that appear on the RHS of the 'combination' of generating polynomials is different.

Exercise 1.2.25. Show that the algebraic variety $V \subset \mathbb{A}^3$ defined by the equations:

$$f = Y^2 - XZ = 0, \quad g = X^3 - YZ = 0$$

has two irreducible components, one of which is the set:

$$C = \{(t^3, t^4, t^5) : t \in k\}.$$

Determine the other irreducible component.

Solution: We have $Xg - Yf = X^4 - Y^3$. Then, $V \subset V(X^4 - Y^3)$, so $V = V \cap V(X^4 - Y^3)$. We then consider when $X = Y = 0$ and when this isn't the case. When this is not the case, cubing the relation for g and imposing $X^4 = Y^3$, we get that $Z^3 = X^5$, which gives C . Next, what if $X = Y = 0$. This still satisfies the equations, giving $V(X, Y)$. Thus,

$$V = C \cup V(X, Y).$$

□

Exercise 1.2.26. Determine the irreducible components of $V(I) \subseteq \mathbb{A}^3$, where

$$I = (X^2 + Y^2 + Z^2, XY + YZ + ZX).$$

Solution: First, we note that $X + Y + Z \in I(V(I)) = \sqrt{I}$, as is clear. Now, in $I(V(I))$, we have that $z = -x - y$ if $(x, y, z) \in V(I)$. Therefore, the first condition tells us that

$$X^2 + Y^2 + X^2 + 2XY + Y^2 \in I(V(I)).$$

Thus, $X^2 + XY + Y^2 \in I(V(I))$, but this factorizes $(Y - aX)(Y - bX)$ for suitable $a, b \in k$, since k is algebraically closed. It is clear that every point of V satisfies $y = ax$ or $y = bx$ because this is what we just showed, and it's also clear that these satisfy the given equations. Therefore, we have that V is the union of the following sets:

$$V(Y - aX, Z + (1 + a)X) \cup V(Y - bX, Z + (1 + b)X)$$

□

§1.3 Regular Functions and Regular Maps

§1.3.i Regular Functions

So far we have only considered algebraic subvarieties as sets, sitting individually. Now we look at functions between them. Just as one uses continuous functions for topological spaces, holomorphic functions for complex manifolds, homomorphisms for groups, etc., so algebraic geometry has its own type of functions—regular functions. Of course, these are given by polynomials.

Definition 1.3.1. Let $V \subseteq \mathbb{A}^n$ be an affine variety. A **regular function** on V is a function $f: V \rightarrow k$ such that there exists a polynomial $F \in k[X_1, \dots, X_n]$ with $f(x) = F(x)$ for all $x \in V$.

Remark 1.3.2 — Note that the polynomial F is not uniquely determined by the function f . That is, $F, G \in k[X_1, \dots, X_n]$ determine the same regular function on V if and only if $F - G$ vanishes on V , that is, if $F - G \in I(V)$.

Definition 1.3.3. The regular functions on V form a k -algebra: they can be added and multiplied by each other and multiplied by scalars in k . This is called the **coordinate ring** of V and denoted $k[V]$.

Proposition 1.3.4

There is an isomorphism

$$k[V] \cong k[X_1, \dots, X_n]/I(V).$$

Proof. There is a homomorphism of rings

$$k[X_1, \dots, X_n] \rightarrow k[V]$$

which sends a polynomial F to the restriction $F|_V$. This is clearly surjective. The kernel of this ring homomorphism is $I(V)$, so we are done. $\square$

Example 1.3.5

The coordinate ring of $\mathbb{A}^n$ is $k[X_1, \dots, X_n]$.

Example 1.3.6

The coordinate ring of a point is just k . Note that a regular function on a point is just a single value.

Example 1.3.7

The coordinate ring of $\{x \in \mathbb{A}^1 : x(x-1) = 0\}$ is $k \times k$. A regular function on two points is determined by two scalars, namely its value on each of the two points. For any pair of values $(a, b) \in k \times k$, one can easily write down a polynomial $f \in k[X]$ such that $f(1) = a$ and $f(0) = b$. Alternatively, one can check algebraically that the map

$$(a, b) \mapsto (a-1)X + b \bmod (X(X-1))$$

is a k -algebra isomorphism

$$k \times k \rightarrow k[X]/(X(X-1)).$$

Remark 1.3.8 — As a generalization of the above, if V is a disconnected variety, then we can write $V = V_1 \cup V_2$ a disjoint union of proper closed subsets of V . Then, $k[V] \cong k[V_1] \times k[V_2]$. This is not true if the union is not disjoint and $k[V]$ is a subset of $k[V_1] \times k[V_2]$.

Example 1.3.9

The coordinate ring of $\{(x, y) \in \mathbb{A}^2 : xy = 0\}$ is $\{(f, g) \in k[X] \times k[Y] : f(0) = g(0)\}$. To prove this, note that

$$k[X, Y]/(XY) \cong \left\{ a_0 + \sum_{r=1}^m b_r X^r + \sum_{s=1}^n c_s Y^s : a_0, b_1, \dots, b_m, c_1, \dots, c_n \in k \right\}.$$

We can compare these two descriptions by observing that

$$k[X] = \left\{ a_0 + \sum_{r=1}^m b_r X^r \right\}, \quad k[Y] = \left\{ a_0 + \sum_{s=1}^n c_s Y^s \right\},$$

and the condition that $f(0) = g(0)$ is equivalent to insisting that these two polynomials have the same constant coefficient a_0 .

Example 1.3.10

The coordinate ring of $\{(x, y) \in \mathbb{A}^2 : xy - 1 = 0\}$ is $k[X, X^{-1}]$, as we can easily see.

Lemma 1.3.11

An affine variety V is irreducible precisely when the coordinate ring $k[V]$ is an integral domain.

Proof. Recall that $k[V] \cong k[X_1, \dots, X_n]/I(V)$, so this quotient is an integral domain if and only if $I(V)$ is prime. This is true if and only if V is irreducible. $\square$

§1.3.ii Regular Maps

Definition 1.3.12. Let $V \subseteq \mathbb{A}^n$ be an algebraic variety. A **regular map** $V \rightarrow \mathbb{A}^m$ is a m -tuple of regular functions on V ; that is, $k[V]^m = k[V] \times \dots \times k[V]$.

An equivalent formulation of the above definition is as follows. A regular map is a function $f: V \rightarrow \mathbb{A}^m$ such that there exist polynomials $F_1, \dots, F_m \in k[X_1, \dots, X_n]$ such that $f(x) = (F_1(x), \dots, F_m(x))$. This easily generalizes:

Definition 1.3.13. Let $V \subseteq \mathbb{A}^m$ and $W \subseteq \mathbb{A}^n$ be affine varieties. A **regular map** is a function $\varphi: V \rightarrow W$ such that there exist polynomials $F_1, \dots, F_n \in k[X_1, \dots, X_m]$ such that

$$\varphi(x) = (F_1(x), \dots, F_n(x)), \quad \forall x \in V.$$

Regular maps are often called **morphisms**.

Remark 1.3.14 — In order to check that a given list of polynomials $F_1, \dots, F_n$ defines a regular map $V \rightarrow W$, it is necessary to check that $(F_1(x), \dots, F_n(x)) \in W$ for every $x \in V$. Equivalently, we need to check that the regular functions $F_1|_V, \dots, F_n|_V \in k[V]$ satisfy the equations $g(F_1|_V, \dots, F_n|_V) = 0$ in the coordinate ring $k[V]$, for each polynomial $g \in I(W)$.

Example 1.3.15

The projection map $\mathbb{A}^n \rightarrow \mathbb{A}^m$ (for $m < n$) is a regular map:

$$(x_1, \dots, x_n) \mapsto (x_1, \dots, x_m).$$

Example 1.3.16

The inclusion map $\mathbb{A}^m \rightarrow \mathbb{A}^n$ (for $m < n$) is a regular map:

$$(x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0, \dots, 0).$$

Example 1.3.17

A regular function on V is the same as a regular map $V \rightarrow k$.

Example 1.3.18

Consider $SL_n(k)$. This is an affine subvariety of $\mathbb{A}^{n^2}$ because the determinant is a polynomial in the entries of a matrix. The claim is that the inverse map $A \mapsto A^{-1}$ (for $A \in SL_n(k)$) is a regular map from $SL_n(k) \rightarrow SL_n(k)$. Cramer’s rule tells us how to write each entry of a^{-1} as a polynomial in the entries of a divided by $\det a$ (which is equal to 1).

§1.3.iii Regular Maps and the Zariski Topology

Lemma 1.3.19

A regular map $\varphi: V \rightarrow W$ is a continuous function with respect to the Zariski topology.

Proof. Given a regular map $\varphi: V \rightarrow W$, we need to check that preimages of closed sets are closed. Let $A \subseteq W$ be Zariski closed. Then, A is defined by polynomials $f_1, \dots, f_r$. Then, $\varphi^{-1}(A)$ is the zero set of the polynomials $f_i \circ \varphi$, for all $i = 1, \dots, r$. That is, $\varphi^{-1}(A) = V(f_1 \circ \varphi, \dots, f_r \circ \varphi) \cap V$. This tells us that $\varphi^{-1}(A)$ is Zariski closed. $\square$

In complex analysis, being holomorphic is a much stricter condition than being continuous in the Euclidean topology, and similarly being regular is much stricter than being continuous in the Zariski topology. The following fact is very useful.

Lemma 1.3.20

Let $\varphi, \psi: V \rightarrow W$ be regular maps. If there exists a Zariski dense subset $A \subseteq V$ such that

$$\varphi|_A = \psi|_A,$$

then $\varphi = \psi$ on all of V .

Remark 1.3.21 — Note that, if X and Y are Hausdorff topological spaces, then any continuous maps $X \rightarrow Y$ which agree on a dense set must agree everywhere. However, the lemma does not follow immediately from the fact that regular maps are continuous, because the Zariski topology is not Hausdorff. Further, the lemma is definitely false if we try to generalize it to all continuous maps with respect to the Zariski topology. Thus in order to prove the lemma, we have to use something special about regular maps as opposed to general continuous maps.

Proof of Lemma 1.3.20. Write $\varphi = (F_1, \dots, F_m)$, $\psi = (G_1, \dots, G_m)$, where the F_i, G_j are polynomials. Then $F_i - G_i$ is also a polynomial for each i , and so

$$V_{eq} = \{x \in V : \varphi(x) = \psi(x)\} = \{x \in V : (F_i - G_i)(x) = 0 \text{ for all } i\}$$

is a Zariski closed subset of V . But we know that V_{eq} contains A , which is Zariski dense in V . Hence $V_{eq} = V$. $\square$

§1.4 Dictionary between Algebra and Geometry

To fully understand the relationship between affine algebraic varieties and k -algebras, we now turn to answer the question: Which k -algebras can occur as $k[V]$, where V is an affine algebraic variety?

We write down some algebraic properties which obviously hold for $A = k[V]$:

- A is finitely generated, because $k[V] = k[X_1, \dots, X_n]/I(V)$.
- A is reduced, meaning that if $f \in A$ and $f^k = 0$ for some $k > 0$, then $f = 0$. This is because A is a ring of functions in the usual set-theoretic sense: if $f^k = 0$ then $f(x)^k = 0$ for all $x \in V$, so $f(x) = 0$ for all $x \in V$.

Using the Nullstellensatz, we can prove that these properties are enough to characterise the k -algebras which are coordinate rings of affine algebraic varieties.

Proposition 1.4.1

Let A be a finitely generated reduced k -algebra. Then there exists an affine algebraic variety V such that $k[V] \cong A$.

Proof. Pick a finite set $f_1, \dots, f_n \in A$ which generates A as a k -algebra. We can define a homomorphism $\alpha: k[X_1, \dots, X_n] \rightarrow A$ by $X_i \mapsto f_i$. Let $I = \ker \alpha$ and let $V = V(I) \subseteq \mathbb{A}^n$.

The homomorphism α is surjective because $f_1, \dots, f_n$ generate A , and so

$$A \cong k[X_1, \dots, X_n]/I.$$

All we need to do is then show that I is radical, since by the Nullstellensatz, it would then follow that $I = I(V(I))$. Since $k[X_1, \dots, X_n]/I$ is a reduced k -algebra, if $f^m \in I$, then $f \in I$, so I is a radical ideal. As stated, by the Nullstellensatz we have $I = I(V)$. Thus

$$k[V] \cong k[X_1, \dots, X_n]/I(V) \cong k[X_1, \dots, X_n]/I \cong A.$$

□

§1.4.i Dictionary Between Algebraic Subsets and Ideals

We now ask if we can do something similar with Zariski closed subsets of V , and work them out from the algebra of $k[V]$.

Suppose that $V \subset \mathbb{A}^n$. In $\mathbb{A}^n$, the Nullstellensatz tells us that the functions I and V are bijections

$$\{\text{Zariski closed subsets of } \mathbb{A}^n\} \longleftrightarrow \{\text{radical ideals in } k[X_1, \dots, X_n]\}.$$

Since I and V reverse the direction of inclusions, we deduce that they restrict to bijections

$$\{\text{Zariski closed subsets of } V\} \longleftrightarrow \{\text{radical ideals in } k[X_1, \dots, X_n] \text{ containing } I(V)\}.$$

We know that we have the isomorphism $k[V] \cong k[X_1, \dots, X_n]/I(V)$. Further, we have the inclusion reversing property of ideals and their quotients:

$$\{\text{ideals in } k[X_1, \dots, X_n] \text{ containing } I(V)\} \longleftrightarrow \{\text{ideals in } k[X_1, \dots, X_n]/I(V)\}$$

Under this correspondence, radical ideals on one side correspond to radical ideals on the other side, and similarly for prime ideals. This allows us to conclude that the natural maps are bijections

$$\{\text{Zariski closed subsets of } V\} \longleftrightarrow \{\text{radical ideals in } k[V]\},$$

and

$$\{\text{irreducible Zariski closed subsets of } V\} \longleftrightarrow \{\text{prime ideals in } k[V]\}.$$

Furthermore, we have

$$\{\text{points of } V\} \longleftrightarrow \{\text{maximal ideals in } k[V]\}.$$

§1.4.ii Isomorphisms

Definition 1.4.2. A regular map $\varphi: V \rightarrow W$ is an **isomorphism** if there exists a regular map $\psi: W \rightarrow V$ such that $\psi \circ \varphi = \text{id}_V$ and $\varphi \circ \psi = \text{id}_W$.

Remark 1.4.3 — Note that a regular map that is an isomorphism is *not* a regular map which is a bijection.

Example 1.4.4

If V is the parabola

$$V(y - x^2) = \{(x, y) \in \mathbb{A}^2 : y - x^2 = 0\},$$

then the regular map $\varphi: V \rightarrow \mathbb{A}^1$ given by

$$\varphi(x, y) = x$$

is an isomorphism because it has an inverse $\psi: \mathbb{A}^1 \rightarrow V$ given by

$$\psi(x) = (x, x^2).$$

Example 1.4.5

On the other hand, if H is the hyperbola

$$\{(x, y) \in \mathbb{A}^2 : xy = 1\},$$

then the projection $(x, y) \mapsto x$ is not an isomorphism $H \rightarrow \mathbb{A}^1$ because it is not surjective (it misses 0), so it cannot possibly have an inverse. This is not enough to prove that H is not isomorphic to $\mathbb{A}^1$, because maybe there is some other regular map $H \rightarrow \mathbb{A}^1$ which is an isomorphism. We will soon prove that H is not isomorphic to $\mathbb{A}^1$.

Example 1.4.6

Consider the affine algebraic subvariety

$$W = \{(x, y) \in \mathbb{A}^2 : y^2 - x^3 = 0\}.$$

The regular map $\varphi: \mathbb{A}^1 \rightarrow W$ given by

$$\varphi(t) = (t^2, t^3)$$

is a bijection but it is not an isomorphism. Note that we should expect W not to be isomorphic to $\mathbb{A}^1$ because it has a singularity at the origin.

To prove that $\varphi: \mathbb{A}^1 \rightarrow W$ is not an isomorphism, consider a regular map

$$\psi: W \rightarrow \mathbb{A}^1.$$

It must be given by a polynomial $g(X, Y) \in k[X, Y]$, and so

$$\psi \circ \varphi(t) = g(t^2, t^3)$$

is a polynomial in t which can have constant term and terms of degree 2 or greater, but no term of degree 1. Hence we cannot find ψ such that $\psi \circ \varphi(t) = t$.

Exercise 1.4.7. Prove that a regular function $\phi: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ is an isomorphism if and only if it is given by a polynomial of degree 1. (Do not use differentiation because it does not behave nicely when the characteristic of the base field is not zero).

Solution: Let $\phi = f(X)$, where $f \in k[X]$. First suppose that ϕ is an isomorphism. Then there is a regular function $\psi: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ defined by $g \in k[X]$ such that $(g \circ f)(X) = X$. Writing

$$f(X) = a + bX + O(X^2), \quad g(X) = c + dX + O(X^2),$$

we have that

$$(f \circ g)(X) = a + b(c + dX) + O(X^2).$$

Doing the same composition the other way around, it is then clear that the $O(X^2)$ terms must be zero, so f is a polynomial of degree 1.

For the converse, suppose that f is a polynomial of degree one. Explicitly, write $f(X) = a + bX$. Then, the map $g(X) = -\frac{a}{b} + \frac{1}{b}X$ is an inverse for f :

$$(f \circ g)(X) = a + b\left(-\frac{a}{b} + \frac{1}{b}X\right) = X$$

and

$$(g \circ f)(X) = -\frac{a}{b} + \frac{1}{b}(a + bX) = X$$

Thus, ϕ is an isomorphism since we have found a left and right inverse for f . $\square$

§1.5 Regular maps and the Coordinate Ring

Definition 1.5.1. Suppose we have a regular map $\varphi: V \rightarrow W$ between affine algebraic varieties. For each regular function g on W , we define the **pullback** of $g \in k[W]$ as the

regular function φ^*g on V defined by

$$(\varphi^*g)(x) = g(\varphi(x)),$$

where $x \in V$.

Exercise 1.5.2. Show that φ^* induces a k -algebra homomorphism

$$\varphi^*: k[W] \rightarrow k[V].$$

Solution: Note that a k -algebra homomorphism is a k -linear map which is also a ring homomorphism. Let $f, g \in k[W]$. Then,

$$\phi^*(f \cdot g) = (f \cdot g)(\varphi) = f(\varphi) \cdot g(\varphi) = \phi^*(f) \cdot \phi^*(g).$$

k -linearity and the additive property is clear. $\square$

Remark 1.5.3 — Note that φ^* goes in the opposite direction to φ .

Exercise 1.5.4. Let $\phi: V \rightarrow W$ be a regular map between affine algebraic varieties. Prove that $\phi^*: k[W] \rightarrow k[V]$ is injective if and only if the image of ϕ is dense in W .

Solution: Consider $f \in k[W]$. If $\phi^*(f) = 0$, it means that $f \circ \phi = 0$, and then that $f|_{\text{Im}(\phi)} = 0$. If $\text{Im}(\phi)$ is dense in W , because f is regular, it means that $f = 0$, hence ϕ^* is injective.

Suppose that the image of ϕ is not dense in W . Then there is some nonempty open set $U \subset V$ which does not meet $\text{Im}(\phi)$. The closed complement $V \setminus U$ is defined by some polynomials: consider such a polynomial f , which vanishes on $V \setminus U$, and hence on $\text{Im}(\phi)$, but which is not identically zero on U . Then $f \neq 0$ is regular, and $\phi^*(f) = 0$, hence ϕ^* is not injective. $\square$

Proposition 1.5.5

Suppose we have two regular maps $\varphi: V \rightarrow W$ and $\psi: W \rightarrow Z$. Then, we can compose them to get $\psi \circ \varphi: V \rightarrow Z$ which satisfies

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$$

from $k[Z] \rightarrow k[V]$.

Proof. Take $g \in k[Z]$, $x \in V$. Then,

$$((\psi \circ \varphi)^*g)(x) = g(\psi \circ \varphi)(x) = g(\psi(\varphi(x))) = \varphi^*(g\psi)(x) = (\varphi^*(\psi^*(g)))(x).$$

$\square$

Proposition 1.5.6

The map $\varphi \mapsto \varphi^*$ is a bijection

$$\{\text{regular maps } V \rightarrow W\} \longleftrightarrow \{k\text{-algebra homomorphisms } k[W] \rightarrow k[V]\}.$$

Proof. Consider $V = V(I) \subset \mathbb{A}^n$ and $W = V(J) \subset \mathbb{A}^m$, such that

$$k[V] = k[X_1, \dots, X_n]/I \quad \text{and} \quad k[W] = k[X_1, \dots, X_m]/J.$$

We will build an inverse to $\varphi \mapsto \varphi^*$.

Consider a ring homomorphism

$$\Phi: k[W] = k[X_1, \dots, X_m]/J \rightarrow k[X_1, \dots, X_n]/I = k[V].$$

Choose polynomials $F_1, \dots, F_m$ such that

$$\Phi(X_i) = F_i + I \in k[X_1, \dots, X_n]/I.$$

For any $g \in k[X_1, \dots, X_m]$, we have that $\Phi(g) = g(F_1, \dots, F_m)$. In particular, if $g \in J$, then $\Phi(g) = g(F_1, \dots, F_m) \in I$. This is because Φ is a k -algebra map. Hence, for $x \in V$, the map

$$x \mapsto (F_1(x), \dots, F_m(x))$$

defines a regular map $\tilde{\Phi}: V \rightarrow W$. We now need to check that these maps are inverses; that is, $(\tilde{\Phi})^* = \Phi$ and that $\widetilde{(\varphi^*)} = \varphi$.

First $(\tilde{\Phi})^* = \Phi$. We compute

$$(\tilde{\Phi})^*(g)(x) = g\tilde{\Phi}(x) = g(F_1(x), \dots, F_m(x)) = \Phi(g)(x).$$

Thus, $(\tilde{\Phi})^* = \Phi$. Conversely, take $\varphi: V \rightarrow W$ to be a regular map. Suppose it is defined by polynomials $F_1, \dots, F_m$. We compute

$$(\varphi^*)(g) = g(F_1, \dots, F_m)(x).$$

Thus,

$$\varphi^*(x_i) = F_i.$$

In constructing $\widetilde{(\varphi^*)}$, we use the F_i . □

Remark 1.5.7 — In the language of categories, this proposition says that we have an equivalence of categories

$$\{\text{affine algebraic varieties}\} \longrightarrow \{\text{reduced } k\text{-algebras of finite type}\}^{\text{op}}.$$

given by the contravariant functor

$$V \longmapsto k[V].$$

Corollary 1.5.8

Affine algebraic varieties V and W are isomorphic if and only if their coordinate rings $k[V]$ and $k[W]$ are isomorphic as k -algebras.

The moral is: if we only care about affine algebraic varieties up to isomorphism, then the coordinate rings contain exactly the same information as the algebraic varieties themselves.

Example 1.5.9

Now we can prove that the hyperbola

$$H = \{(x, y) \in \mathbb{A}^2 : xy = 1\}$$

is not isomorphic to $\mathbb{A}^1$. We know that

$$k[H] = k[X, X^{-1}],$$

and this is not isomorphic to

$$k[\mathbb{A}^1] = k[X],$$

because in $k[X]$ the only invertible elements are the scalars, while $k[X, X^{-1}]$ contains non-scalar invertible elements, such as X .

§1.6 Defining Regular Maps Locally**§1.6.i Distinguished Open Subsets**

Note that if V is an affine variety, we understand regular functions on closed subsets of v . However, we now ask whether there is a good notion of regular functions on open subsets. We begin by giving complements on open subsets and open coverings of affine algebraic varieties. There is a certain class of open subsets of an affine algebraic variety V , known as *distinguished open subsets* of V , that are particularly easy to work with.

Definition 1.6.1. Let V be an affine algebraic variety, and let $f \in k[V]$. The **distinguished open subset** $D_V(f)$ is defined to be the open subset $V \setminus V(f)$ of V ; that is,

$$D_V(f) = \{x \in V : f(x) \neq 0\}.$$

Remark 1.6.2 — Note that since $V(fg) = V(f) \cup V(g)$, we have

$$D_V(f) \cap D_V(g) = D_V(fg).$$

Similarly, $V(f, g) = V(f) \cap V(g)$, so we have

$$D_V(f) \cup D_V(g) = V \setminus V(f, g) = D_V(f, g).$$

Every point of V contains a basis of neighborhoods consisting of distinguished open sets. In other words:

Lemma 1.6.3

Let U be an open subset of V , and let p be a point of U . Then there exists an $f \in k[V]$ such that $p \in D_V(f) \subseteq U$.

Proof. Since U is open in V , it is a set of the form $V \setminus V(I)$ for some ideal $I \subseteq k[V]$. Since $p \notin V(I)$ (since $p \in U$), there exists an $f \in I$ such that $p \notin V(f)$. Then $p \in D_V(f)$. On the other hand, since $f \in I$, we have $V(I) \subseteq V(f)$, so $D_V(f)$ (which equals $V \setminus V(f)$) is contained in U . $\square$

A rephrasing of the above is the following.

Corollary 1.6.4

Any open subset U in V is a union of finitely many distinguished open sets.

Proof. Using the above lemma, one can cover U by distinguished open sets, and from Noetherianity a finite union of them covers U . To write everything out, let $U = V \setminus V(I)$, and write $I = (f_1, \dots, f_r)$, for some r (by Noetherian). Then, look at $D_V(f_1) \cup \dots \cup D_V(f_r)$, which is equal to $V \setminus V(f_1, \dots, f_r) = U$. $\square$

Corollary 1.6.5

Distinguished open sets form a basis for the topology on V .

The Nullstellensatz gives us a nice criterion for when a collection of distinguished open sets covers V :

Proposition 1.6.6

Let $f \in k[V]$, and let $g_1, \dots, g_r \in k[V]$ such that $D_V(g_i) \subseteq D_V(f)$. Then the $D_V(g_i)$ cover $D_V(f)$ if and only if f is in the radical of $(g_1, \dots, g_r)$. In particular, since $D_V(1) = V$, the $D_V(g_i)$ cover V if and only if they generate the unit ideal of $k[V]$.

Proof. The union of the $D_V(g_i)$ is the union of $V \setminus V(g_i)$, which is $V \setminus V(g_1, \dots, g_r)$. This contains $D_V(f)$ exactly when $V(g_1, \dots, g_r) \subseteq V(f)$. Taking ideals on both sides, we find that

$$I(V(f)) \subseteq I(V(g_1, \dots, g_r)).$$

By the Nullstellensatz we then have

$$\sqrt{(f)} \subseteq \sqrt{(g_1, \dots, g_r)}.$$

Since f is an element of $\sqrt{(f)}$ the first claim follows. The second claim follows by taking $f = 1$, so that $D_V(f) = V$. Then we have $1 \in (g_1, \dots, g_r)$ in $k[V]$. $\square$

§1.6.ii Regular Functions on Open Subsets

We have studied regular functions and maps defined everywhere on an affine variety. In geometry, it is, in general, also useful to consider functions and maps which are defined only locally, that is, on an open subset. The basic idea is that we can invert a nonvanishing polynomial on the open locus where it does not vanish. That is, we must consider that, given $f, g \in k[V]$,

$$\varphi: V \setminus V(g) \rightarrow \mathbb{A}^1, \quad x \mapsto \frac{f(x)}{g(x)} \quad (1.1)$$

must be regular on the open subset $V \setminus V(g)$ where g does not vanish, hence one can work locally with rational fractions instead of polynomials. Moreover, we want to be able to check regularity locally near each point, i.e. we must not ask for a global presentation of φ as a global fraction, but only for a local presentation. That is, regularity of a function is a property that, like continuity, differentiability, holomorphy, etc., can be checked in small neighborhoods of every point in the domain. This leads to the following definition:

Definition 1.6.7. Let U be an open subset of an affine algebraic variety V . We say that $f: U \rightarrow k$ is a **regular function on U** if there exists an open cover $\{U_i\}$ of U , together with regular functions g_i, h_i such that h_i is nonvanishing on U_i and such that

$$f(x) = \frac{g_i(x)}{h_i(x)} \quad \text{for all } x \in U_i,$$

and for all i .

Notation 1.6.8. The k -algebra (or ring) of regular functions on $U \subseteq V$ will be denoted $\mathcal{O}_V(U)$.

Note that it is not immediately obvious that this agrees with our definition of a regular function on V in the case $U = V$ (i.e. that $\mathcal{O}_V(V) = k[V]$); we will verify this shortly.

Remark 1.6.9 — We observe that if f is a regular function on U , then the set $f^{-1}(0)$ of zeros of f on U is closed in U . Indeed, we can fix an open cover U_i of U on which f is given by $\frac{g_i}{h_i}$, with g_i and h_i polynomials such that h_i is nonvanishing on U_i . Then

$$f^{-1}(0) \cap U_i = V(g_i) \cap U_i,$$

and the latter is clearly closed in U_i .

A disadvantage of the above definition is that it is typically quite hard to describe the ring of regular open functions on an arbitrary open subset U of V . However, when U is a distinguished open subset, things are easier:

Proposition 1.6.10

Let V be an affine algebraic variety, $h \in k[V]$, and let $U = D_V(h)$. Then every regular function on U can be globally written in the form $\frac{g}{h^n}$, where $g \in k[V]$ and $n > 0$. In particular the natural map

$$k[V][1/h] \rightarrow \mathcal{O}_V(D_V(h))$$

is an isomorphism.

Proof. By definition, a regular function $f: U \rightarrow k$ admits an open cover $\{U_i\}$ of U , and polynomial functions g_i, h_i on U_i with h_i nonvanishing on U_i , such that

$$f(x) = \frac{g_i(x)}{h_i(x)} \quad \text{for all } x \in U_i.$$

Our first step is to replace this cover (and the g_i and h_i) with a more convenient form. Note we can cover U_i with distinguished opens $D_V(h_{ij}) \subseteq U_i$. Then, since $D_V(h_{ij})$ is contained in U_i , and h_i is nonvanishing on U_i , we have $V(h_i) \subseteq V(h_{ij})$. Therefore, $h_{ij} \in \sqrt{(h_i)} \subseteq \sqrt{(h_i)}$. Thus, we have $h_{ij}^r = g'_{ij} h_i$ for some r . Then

$$f(x) = \frac{g_i(x)}{h_i(x)} = \frac{g_i(x) g'_{ij}(x)}{h_{ij}^r(x)}$$

on U_i . Since $D_V(h_{ij}) = D_V(h_{ij}^r)$, we can set $h'_{ij} = h_{ij}^r$, and we have attained a cover of U_i by $D_V(h'_{ij})$ and we have that $f(x) = \frac{\text{polynomial}}{h'_{ij}}$ on U_i .

Doing this for all of the U_i 's, we end up with an open cover of U by distinguished opens of the form $D_V(h_i)$ (not the same h_i as in the last paragraph), and expressions for f of the form $f(x) = \frac{g_i(x)}{h_i(x)}$ on $D_V(h_i) = U_i$ for some polynomials $g_i(x)$.

On $D_V(h_i) \cap D_V(h_j)$ we have $\frac{g_i(x)}{h_i(x)} = \frac{g_j(x)}{h_j(x)}$, which implies that $h_j g_i - g_j h_i$ vanishes identically on $D_V(h_i) \cap D_V(h_j) = D_V(h_i h_j)$. On $V \setminus (D_V(h_i) \cap D_V(h_j)) = V(h_i h_j)$, we have $h_i(x) h_j(x) = 0$. On V , the product $(g_i(x) h_j(x) - g_j(x) h_i(x)) h_i(x) h_j(x)$ vanishes. Since the $D_V(h_i)$ cover $D_V(h)$, and $D_V(h_i^2) = D_V(h_i)$ for all i , the sets $D_V(h_i^2)$ cover $U = D_V(h)$. Thus, $h \in \sqrt{(h_1^2, \dots, h_s^2)}$. Thus, there exists a positive integer r such that

$$h^r = a_1 h_1^2 + \dots + a_s h_s^2,$$

for $a_i \in k[V]$. The claim is that

$$f(x) = \frac{1}{h^r(x)} \sum_{i=1}^s a_i(x) g_i(x) h_i(x),$$

for all $x \in U = D_V(h)$. If this claim is true, the theorem is proven.

Indeed, if $x \in D_V(h_i)$, then $f(x) = \frac{g_i(x)}{h_i(x)}$ on $D_V(h_i)$. Thus

$$\begin{aligned} f(x) h_i(x)^2 h(x)^r &= \frac{g_i(x)}{h_i(x)} h_i(x)^2 \sum_{j=1}^s a_j(x) h_j(x)^2 \\ &= \sum_{j=1}^s g_i(x) h_i(x) a_j(x) h_j(x)^2. \end{aligned}$$

Using that

$$g_i(x) h_i(x) h_j(x)^2 = g_j(x) h_i(x)^2 h_j(x),$$

for all j , we find that the right-hand side is equal to

$$h_i(x)^2 \sum_{j=1}^s a_j(x) g_j(x) h_j(x).$$

Dividing both sides by $h_i(x)^2 h(x)^r$, the claim follows. $\square$

Corollary 1.6.11

Taking $h = 1$, we have that $\mathcal{O}_V(V) = k[V]$.

Remark 1.6.12 (Not part of the course) — Let V be an affine variety. We have then for each $U \subseteq V$ a ring $\mathcal{O}_V(U)$, but there is a structure relating these different rings. If $U' \subseteq U$ is open, we can restrict regular functions from U to U' . That is, we have a ring map

$$\text{res}_{U,U'} : \mathcal{O}_V(U) \rightarrow \mathcal{O}_V(U').$$

If we have a third open set U'' such that $U'' U' \subseteq U$, then we can have

$$\text{res}_{U,U''} = \text{res}_{U',U''} \circ \text{res}_{U,U'}.$$

This tells us that $\mathcal{O}_V$ defines a functor

$$\text{Open}(V)^{\text{op}} \rightarrow \mathbf{Ring}.$$

Moreover, we can glue regular functions. That is, given an open cover $\{U_i\}$ of U and regular functions $f_i: U_i \rightarrow k$ regular such that

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} \quad \text{for each } i, j,$$

there is a unique regular function $f: U \rightarrow k$ such that $f|_{U_i} = f_i$. Uniqueness follows from the covering, and existence comes from local existence of the f_i . This means that the collection of $\mathcal{O}_V(U)$ and restriction map form a **sheaf**. We note that sheaf theory lies at the core of the foundation of scheme theory, which is the modern approach to algebraic geometry.

Definition 1.6.13. If $\mathcal{C}$ is a category a **$\mathcal{C}$ -valued sheaf** on V is a functor $\text{Open}(V)^{\text{op}} \rightarrow \mathcal{C}$ satisfying gluing. Just the category $\mathcal{C}$ is called a **presheaf**.

Note that we will not require any knowledge of sheaf theory in these lectures.

§1.6.iii Regular Maps on Open Subsets

As for regular maps on affine varieties, we can define regular maps on open subsets as maps having regular coordinates:

Definition 1.6.14. Let $V \subseteq \mathbb{A}_k^n$ and $W \subseteq \mathbb{A}_k^m$ be affine algebraic subvarieties, and let $U \subseteq V$, $U' \subseteq W$ be open subsets. A map $f: U \rightarrow U'$ is a **regular map** if there exist regular functions $f_1, \dots, f_m$ on U such that

$$f(x) = (f_1(x), \dots, f_m(x)) \quad \text{for all } x \in U.$$

Remark 1.6.15 — Note that, since being a regular function is a local property, being a regular map is also local: a map $f: U \rightarrow U'$ is regular if and only if for every point $p \in U$, there is a neighborhood on which f is regular.

Lemma 1.6.16

If $g: U' \rightarrow k$ is a regular function and $f: U \rightarrow U'$ is a regular map, then the composition $g \circ f: U \rightarrow k$ is also regular.

Proof. One can check this locally near $p \in U'$. We choose a neighborhood of p , U'_p , on which g is given by a rational function $\frac{t(x)}{s(x)}$, with s a polynomial nonvanishing on this neighborhood. Pick $q \in U$ such that $p = f(q)$, and pick a neighborhood U_q of q such that f is given by $\left(\frac{f_1}{h_1}, \dots, \frac{f_m}{h_m}\right)$, where the f_i and h_i are polynomials and each h_i is nonvanishing on U_q . Then,

$$g \circ f = \frac{t\left(\frac{f_1}{h_1}, \dots, \frac{f_m}{h_m}\right)}{s\left(\frac{f_1}{h_1}, \dots, \frac{f_m}{h_m}\right)}.$$

This may not be defined when $h_i = 0$, but this is fine on U_q , as the h_i do not vanish. On $U_q \setminus (V(f_1) \cup \dots \cup V(f_m))$, $g \circ f$ is a regular function. $\square$

Corollary 1.6.17

If $f: U \rightarrow U'$ is a regular map, we get a pullback map

$$\begin{aligned} f^*: \mathcal{O}_W(U') &\rightarrow \mathcal{O}_V(U) \\ g &\mapsto g \circ f. \end{aligned}$$

Corollary 1.6.18

Any regular map $f: U \rightarrow U'$ is continuous in the Zariski topology.

Proof. It suffices to check that preimages of closed sets are closed. Note that any closed subset of U' are intersections of sets of the form $V(g)$ for a regular function $g \in k[W]$ ($U' \subseteq W$). The preimage of such a set is the zero set of $g \circ f = f^*g$. That is,

$$(g \circ f)^{-1}(0),$$

which is closed in U . Previously, we have seen zero sets of regular functions are closed. $\square$

In particular, since a regular function on U is the same as a regular map $U \rightarrow \mathbb{A}_k^1$, we see that regular functions are also continuous with respect to the Zariski topology on U and $\mathbb{A}_k^1$. Note that there are certainly many continuous functions on U that are not regular.

Proposition 1.6.19

A map $f: U \rightarrow U'$ is regular if and only if for all regular functions g on U' , the composition $g \circ f$ is regular on U . In particular, any composition of regular maps is regular.

Proof. Note that we have proven the forward direction already. Recall that we have $U' \subseteq W \subseteq \mathbb{A}_k^m$ for some affine algebraic subvariety W . In particular, we have the coordinate functions $x_1, \dots, x_m$ on W , which are certainly regular functions on U' (by restriction). If $f: U \rightarrow U'$ is such that $x_i \circ f$ is regular, for all i , then the claim is that f is regular. Writing $f(x) = (f_1(x), \dots, f_m(x))$, then $x_i \circ f = f_i$. If each f_i is regular, then f is regular (by definition), proving the proposition. $\square$

Definition 1.6.20. Consider $U \subseteq V$ and $U' \subseteq W$, two open subsets of affine varieties.

- A regular map $f: U \rightarrow U'$ is said to be an **isomorphism** if there exists a regular map $g: U' \rightarrow U$ that is a two-sided inverse to f .
- A regular map $f: U \rightarrow U'$ is said to be an **open** (respectively **closed**) **embedding** if its image $f(U)$ is open (respectively closed) in U' and the induced map $U \rightarrow f(U)$ is an isomorphism.
- A regular map $f: U \rightarrow U'$ is said to be **dominant** if $f(U)$ is dense in U' .

Exercise 1.6.21. Given a dominant map $f: U \rightarrow U'$, where $U \subseteq V$ and $U' \subseteq W$ are two open subsets of affine varieties, the pullback map

$$f^*: \mathcal{O}_W(U') \rightarrow \mathcal{O}_V(U)$$

is injective.

Solution: Let $g \in \mathcal{O}_W(U')$ be nonzero such that $f^*(g) = 0$. That is, $g(f) = 0$. Then, $f(U) \subseteq V(g)$, so $\overline{f(U)} \subseteq V(g)$. However, $f(U)$ is dense in U' , so $V(g) = U'$. Thus, $g = 0$. $\square$

§1.7 Rational Functions and Rational Maps

§1.7.i Rational Maps and Domains

Now let V be an open subset of an affine variety, and suppose that V is irreducible. In particular, every nonempty open subset U of V is dense in V , and so the intersection of any two nonempty open subsets is nonempty and dense.

Definition 1.7.1. Let V be an irreducible open subset of an affine variety. A **rational function** $f: V \dashrightarrow k$ is an equivalence class of pairs (U, f) , where U is a nonempty open subset of V and f is a regular function on U under the following equivalence: $(U, f) \sim (U', f')$ if the restrictions of f and f' to $U \cap U'$ agree.

Remark 1.7.2 — Indeed, this is an equivalence relation. Symmetry and reflexivity are clear. The only subtle point is transitivity. If $(U, f) \sim (U', f') \sim (U'', f'')$, then

$$f|_{U \cap U' \cap U''} = f|_{U \cap U'}|_{U''} = f'|_{U \cap U'}|_{U''} = f'|_{U' \cap U''}|_U = f''|_{U' \cap U''}|_U = f''|_{U \cap U' \cap U''},$$

but $U \cap U' \cap U''$ is dense in $U \cap U''$ by the irreducibility of V . This means that $f|_{U \cap U''}$ and $f''|_{U \cap U''}$ are two regular functions (resp. maps) agreeing on an open dense subset, hence they are equal. This shows that $(U, f) \sim (U'', f'')$.

Definition 1.7.3. Let V be irreducible open subsets of an affine variety and W be an open subset of an affine variety. A **rational map** $f: V \dashrightarrow W$ is an equivalence class of pairs (U, f) , where U is a nonempty open subset of V and f is a regular map on U under the following equivalence: $(U, f) \sim (U', f')$ if the restrictions of f and f' to $U \cap U'$ agree.

Remark 1.7.4 — We use the broken arrow symbol $\dashrightarrow$ instead of the usual arrow because a rational map is not a function on V in the usual set-theoretic sense.

Remark 1.7.5 — From the definition, a rational function is a rational map $V \dashrightarrow \mathbb{A}^1$.

Example 1.7.6

Given a regular map $f: V \rightarrow W$, it is trivially a rational map. Since V is an irreducible open subset of an affine variety, then take the representative (V, f) .

We observe that if $(U, f) \sim (U', f')$, then we can, in particular, glue f and f' to obtain a regular map $f'': U \cup U' \rightarrow W$, and then a representative $(U \cup U', f'')$ of the same rational function. In particular, we can find a maximal representative, defined on the union of all the open subsets on which representatives are defined. This motivates the following definition.

Definition 1.7.7. The **domain** $\text{dom}(\phi)$ of a rational map ϕ is the maximal open subset on which ϕ is defined; that is, the set of points $x \in V$ (called **regular points**) such that there exists an open neighborhood $x \in U \subseteq V$ and a regular map $f: U \rightarrow W$ representing ϕ .

Remark 1.7.8 — We note that both definitions here generalize to rational maps.

Lemma 1.7.9

The domain of ϕ is the union of all open sets U on which ϕ is defined.

Proof. Write D to be the union of opens in which ϕ is defined. By definition, $D \subseteq \text{dom}(\phi)$. However, $\text{dom}(\phi)$ appears in the union defining D , so $\text{dom}(\phi) \subseteq D$. $\square$

We can add and multiply rational functions by restricting to where they are both defined:

$$(U, f) + (U', f') := (U \cap U', f|_{U \cap U'} + f'|_{U \cap U'}),$$

and similarly for multiplication. Note that if f is a function on V , then $(D_V(f), \frac{1}{f})$ is a rational function. If f is defined on U , then we can multiply this with (U, f)

$$(U, f) \cdot \left(U \setminus V(f), \frac{1}{f} \right) = (U \setminus V(f), 1) \sim (V, 1),$$

so that the rational functions have the structure of a field. Therefore, any nonzero rational function has a multiplicative inverse. The k -algebra of rational fractions on V is a field extension of k , called the **function field** of V , and is denoted $k(V)$.

Proposition 1.7.10

Let V be an irreducible affine algebraic variety. Then $\text{Frac}(k[V]) \cong k(V)$.

Proof. We certainly have a map from the field of fractions of $k[V]$ to $k(V)$, taking $\frac{f}{g}$ to the equivalence class of $(D_V(g), f/g)$. This is a map of fields, so it suffices to show that this is a surjection. Every regular function (after passing to a sufficiently small open) has a description as $(D_V(g), \frac{f}{g})$. Thus this map is a surjection of fields, hence an isomorphism. $\square$

Corollary 1.7.11

A rational function on V is an equivalence class of fractions $\frac{f}{g}$ with $f, g \in k[V]$, where $\frac{f}{g} \sim \frac{f'}{g'}$ if $fg' = f'g$. The set of regular points $\text{dom}(\phi)$ is the set of points $x \in V$ such that there exists a representative $\frac{f}{g}$ of ϕ with $g(x) \neq 0$.

Example 1.7.12

Consider the algebraic subvariety $V \subset \mathbb{A}^4$ defined by the equation $XY = ZT$. Let

$$\phi = \frac{X}{Z} \in k(V).$$

The defining equation implies that we also have

$$\phi = \frac{T}{Y}.$$

Looking at the fraction $\frac{X}{Z}$ shows us that ϕ is regular at (X, Y, Z, T) wherever $Z \neq 0$, and looking at the fraction $\frac{T}{Y}$ shows us that ϕ is regular at (X, Y, Z, T) wherever $Y \neq 0$. In fact, $\text{dom} \phi = V \setminus \{Z = 0 \text{ and } Y = 0\}$. To show this, we would need to show that ϕ has no rational representative when $Z = 0$ and $Y = 0$.

An important example of a rational map is projection from a point onto a hyperplane, which is as follows.

Example 1.7.13

Let H be a hyperplane in $\mathbb{A}^n$ (that is, the vanishing set defined by a single linear equation). Let p be a point in $\mathbb{A}^n \setminus H$. Let us write H_p for the hyperplane parallel to H . For each point $x \in \mathbb{A}^n \setminus H_p$, let L_x denote the unique line which passes through p and x . Since $x \notin H_p$, L_x intersects H in exactly one point. Call this point $\varphi(x)$.

Now, let us choose coordinates. We can choose H to be the hyperplane $H = \{(X_1, \dots, X_n) : X_n = 1\}$ and p to be the origin. Then, the hyperplane H_p is $H_p = \{(X_1, \dots, X_n) : X_n = 0\}$. Then, we have that

$$\varphi(x_1, \dots, x_n) = \left(\frac{x_1}{x_n}, \dots, \frac{x_{n-1}}{x_n}, 1 \right).$$

It suffices to check that $p, x, \varphi(x)$ is collinear for completeness of the argument.

This tells us that φ is a rational map $\mathbb{A}^n \dashrightarrow H$; it is called the **projection from p onto H** . We have

$$\text{dom}(\phi) = \mathbb{A}^n \setminus H_p,$$

which we have not yet proved, because we have not proved that there is no other list of fractions which define the same rational map but have non-zero denominators at points in H_p .

For any affine algebraic subvariety $V \subseteq \mathbb{A}^n$ such that $V \not\subseteq H_p$, we can restrict ϕ to V get a rational map $V \dashrightarrow H$. Note that p might be in V , or it might not.

Example 1.7.14

Let V be the circle $V = \{(x, y) \in \mathbb{A}^2 : x^2 + y^2 = 1\}$. Consider the projection from the point $p = (1, 0)$ onto the line $x = 0$. This is a rational map $\pi: V \dashrightarrow \mathbb{A}^1$ with the formula

$$\pi(x, y) = \frac{y}{1-x}.$$

We can see geometrically that this projection induces a bijection between the $V \setminus \{p\}$ and the line, which we call H . If we compute the formula for the inverse map, we get

$$\psi(t) = \left(\frac{t^2 - 1}{t^2 + 1}, \frac{2t}{t^2 + 1} \right),$$

a well-known parameterization of the circle. Thus we see that the inverse is a rational map $\psi: \mathbb{A}^1 \dashrightarrow V$. Note that ψ is not regular at $t = \pm i$; note that we do not see this on the picture, which only shows the real points.

Exercise 1.7.15. Suppose k is of characteristic not equal 2. Let C denote the circle $V(X^2 + Y^2 - 1) \subset \mathbb{A}^2$. Let ϕ be the rational function $\frac{1-Y}{X}$ on C . Find a point $a \in C$ where ϕ is not regular (you need to prove that there is no function $\frac{f}{g}$ which is equal to ϕ in $k(C)$ and satisfies $g(a) \neq 0$). Prove that ϕ is regular at all points of C other than a .

Solution: V is defined by $X^2 + Y^2 = 1$, which provides $X^2 = 1 - Y^2$. Thus, in $k(C)$, we have

$$\phi = \frac{1-Y}{X} = \frac{X}{1+Y}.$$

This representation tells us that ϕ is regular except at $(0, -1)$. Suppose that we can write $\phi = \frac{f}{g}$ at $(0, -1)$. Then, suppose.

$$(1-Y)g(X, Y) = Xf(X, Y), \quad g(0, -1) \neq 0.$$

Then, $2g(0, -1) = 0 \cdot f(0, -1) = 0$, which is a contradiction. $\square$

Next, we will define formally what it means to say that the rational maps π and ψ are inverse to each other, taking into account that they are not true functions between the sets V and $\mathbb{A}^1$ because they are not regular everywhere.

§1.7.ii Composing Rational Maps

Last time we defined rational maps $\pi: V \dashrightarrow \mathbb{A}^1$ and $\psi: \mathbb{A}^1 \dashrightarrow V$, where V is the circle. These maps are inverses in that composing them (either way round) gives the identity, if we ignore the points where the maps are not regular.

In order to rigorously define composition of rational maps, we need to notice that sometimes the set of points where a composite map is undefined is ‘everywhere’, and exclude that situation. For example, consider the rational map $\xi: \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ defined by

$$\xi(x, y) = \frac{1}{x^2 + y^2 - 1}.$$

This map is not regular anywhere on the circle V , and hence it does not make sense to try to define the composite map $\xi \circ \psi: \mathbb{A}^1 \dashrightarrow \mathbb{A}^1$ (it is not defined anywhere).

This problem can occur because the image of ψ is not dense in $\mathbb{A}^2$. To rule it out, we make the following definition.

Definition 1.7.16. The **image** of a rational map $\varphi: V \dashrightarrow W$ is the set of points

$$\{\varphi(x) \in W : x \in \text{dom}(\varphi)\}.$$

A rational map is **dominant** if its image is Zariski dense in W .

Remark 1.7.17 — We note that the notion of dominance depends on the codomain.

For example, the map ψ in the example above is dominant if we consider it as a rational map $\mathbb{A}^1 \dashrightarrow V$ but it is not dominant if we consider it as a rational map $\mathbb{A}^1 \dashrightarrow \mathbb{A}^2$. (This is like surjectivity: whether a function is surjective or not depends on what codomain you declare it to have.)

Let V, W, T be irreducible affine algebraic varieties. If $\varphi: V \dashrightarrow W$ is a dominant rational map and $\psi: W \dashrightarrow T$ is a rational map (not necessarily dominant), then it makes sense to compose them because we know that $\text{dom}(\psi)$ is a Zariski open subset of W , while $\text{Im}(\varphi)$ is a Zariski dense subset of W , and so

$$\text{dom}(\psi) \cap \text{Im}(\varphi) \neq \emptyset.$$

Therefore, there are at least some points where $\psi \circ \varphi$ is defined. One can check (by writing ψ in terms of fractions of polynomials, then substituting in fractions of polynomials representing φ) that $\psi \circ \varphi$ is a rational map $V \dashrightarrow T$.

Definition 1.7.18. Rational maps $\varphi: V \dashrightarrow W$ and $\psi: W \dashrightarrow V$ are **rational inverses** if both are dominant and $\varphi \circ \psi = \text{id}_W$ and $\psi \circ \varphi = \text{id}_V$, everywhere these composite rational maps are well-defined.

Definition 1.7.19. A rational map $\varphi: V \dashrightarrow W$ is a **birational equivalence** if it is dominant and has a rational inverse. We say that irreducible algebraic varieties V and W are **birational** (or **birationally equivalent**) if there exists a birational equivalence $V \dashrightarrow W$.

Example 1.7.20

Our example from the previous lecture showed that the circle is birational to $\mathbb{A}^1$.

Example 1.7.21

Consider the cuspidal cubic

$$W = \{(x, y) : y^2 = x^3\}.$$

This is also birational to $\mathbb{A}^1$, as shown by the rational maps

$$\begin{aligned} W \dashrightarrow \mathbb{A}^1 : (x, y) &\mapsto \frac{y}{x} \\ \mathbb{A}^1 \dashrightarrow W : t &\mapsto (t^2, t^3). \end{aligned}$$

Remark 1.7.22 — If two varieties are isomorphic, then they are certainly birationally equivalent, because regular maps are rational maps.

Remark 1.7.23 — Birationally equivalent affine algebraic varieties look the same ‘almost everywhere’. For example, the cuspidal cubic is the same as the affine line everywhere except at the origin.

Example 1.7.24 (Non-Example)

$\mathbb{A}^1$ is not birationally equivalent to $\mathbb{A}^2$ or to an elliptic curve

$$\{(x, y) : y^2 = f(x)\}, \text{ where } f \text{ is a cubic polynomial with no repeated roots.}$$

We will prove this later in the course once we have more tools.

Exercise 1.7.25. Determine (with proof) the irreducible components of the affine algebraic variety:

$$V = V(Y^2 - XZ, X^2Y - Z) \subset \mathbb{A}^3.$$

Solution: We have that $Y^2 - XZ - X(X^2Y - Z) = Y(Y - X^3)$. It is clear that $V \subset V(Y(Y - X^3))$. Therefore, $V = (V \cap V(Y)) \cup (V \cap V(Y - X^3))$, since V is closed. If $Y = 0$, then we get that $Z = 0$, and X can be anything. Thus, $V(Y, Z)$ is one component. Next, if $Y = X^3$, then we get the condition $X(X^5 - Z)$. When $X = 0$, this produces a degenerate case, If $Z = X^5$, this satisfies the second condition too, so we get another component, which is a curve:

$$C := \{(t, t^3, t^5) : t \in \mathbb{R}\}.$$

Thus,

$$V = V(Y, Z) \cup C.$$

We note that these are indeed irreducible. First, $V(Y, Z)$ is irreducible because the ideal $(Y, Z) \subseteq k[X, Y, Z]$ is prime; this is because $k[X, Y, Z]/(Y, Z) \cong k[X]$, which is an integral domain. As for C , this is also irreducible. The ideal $(Y - X^3, Z - X^5)$ is the kernel of the map $k[X, Y, Z] \mapsto k[t]$ defined by

$$X \mapsto t, \quad Y \mapsto t^3, \quad Z \mapsto t^5.$$

This is clearly a surjection, since $X \mapsto t$, so we get the isomorphism $k[X, Y, Z]/(Y - X^3, Z - X^5) \cong k[t]$, which is an integral domain again. $\square$

Exercise 1.7.26. Consider the map $\phi: \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ defined by $(x, y) \mapsto \frac{x}{y}$. Is this rational? If so, what is its domain? Is it regular? What if we replace $\mathbb{A}^2$ with $\mathbb{A}^2 \setminus V(y)$.

Solution: This is a rational function with domain $\text{dom} \phi = \mathbb{A}^2 \setminus V(y)$. To see if it's rational, suppose that $xg(x, y) = yf(x, y)$, with $g(x, y) \neq 0$ for $(x, y) \in \mathbb{A}^2$. Then, consider $(t, 0)$, for some nonzero t . Then,

$$tg(t, 0) = 0 \cdot f(t, 0),$$

so $g(t, 0) = 0$, which is not possible. Thus, is not a regular map. If we replace $\mathbb{A}^2$ with $\mathbb{A}^2 \setminus V(y)$, then this is actually regular, since it is well-defined everywhere. $\square$

§1.7.iii Correspondence between Algebra and Geometry

Definition 1.7.27. If $\varphi: V \dashrightarrow W$ is a dominant rational map between irreducible affine varieties, then we can use it to define the **pull back** rational functions from W to V (just like we earlier used regular maps to pull back regular functions). We get a k -homomorphism of fields

$$\varphi^*: k(W) \longrightarrow k(V),$$

defined by

$$\varphi^*(g) = g \circ \varphi.$$

This is well-defined because φ is dominant.

Remark 1.7.28 — This is a k -homomorphism, meaning that φ^* restricts to the identity on copies of k contained in $k(W)$ and $k(V)$, namely the constant functions.

Remark 1.7.29 — If φ is a birational equivalence, then the pullback φ^* is a k -isomorphism of fields.

Proof. The reason for this is that if we have a birational equivalence, we have that $\varphi\psi = \text{id}$. Therefore,

$$\text{id} = \text{id}^* = (\varphi\psi)^* = \psi^*\varphi^*.$$

This clearly implies that φ^* is a k -isomorphism then. □

One can do the same thing that for regular functions:

Proposition 1.7.30

The map $\varphi \mapsto \varphi^*$ is a bijection

$$\{\text{dominant rational maps } V \dashrightarrow W\} \longrightarrow \{k\text{-algebra homomorphisms } k(W) \rightarrow k(V)\}.$$

Proof. The proof is similar to the proof of Proposition 1.5.6. To construct the inverse, given $h: k(W) \rightarrow k(V)$, we look at $h(x_i) = \frac{a_i}{b_i}$ (both a_i, b_i polynomials). We attain a rational map $\tilde{h}: V \rightarrow W$ defined by $x \mapsto \left(\frac{a_1(x)}{b_1(x)}, \dots, \frac{a_m(x)}{b_m(x)}\right)$. What remains is to just show that $\tilde{h}$ is dominant. If $\tilde{h}$ is not dominant, then $\overline{\text{Im}\tilde{h}} \subsetneq W$, so there exists some nonzero rational function on W , $g \in k[W]$, such that $\overline{\text{Im}\tilde{h}} \subseteq V(g)$ (this is simply because $\overline{\text{Im}\tilde{h}}$ is a closed subset of the affine variety W). Considering the pullback, this means that $h(g) = 0$, so h being an injection implies $g = 0$, which is a contradiction. Thus, $\overline{\text{Im}\tilde{h}} = W$. □

One can deduce an equivalent of Corollary 1.5.8:

Corollary 1.7.31

Two irreducible affine algebraic varieties V and W are birationally equivalent if and only if their function fields $k(V)$ and $k(W)$ are k -isomorphic.

Hypersurfaces and Birational Equivalence

We now use the algebraic description of birational equivalence to give a geometric interpretation of this algebraic theorem.

Lemma 1.7.32

Let k be an algebraically closed field and let K be a finitely generated extension field of k . Then there exist $t_1, \dots, t_d, u \in K$ such that

1. $K = k(t_1, \dots, t_d, u)$;
2. $t_1, \dots, t_d$ are algebraically independent over k (that is, there is no non-zero polynomial in d variables with coefficients in k whose value at $(t_1, \dots, t_d)$ is zero);
3. u is algebraic over $k(t_1, \dots, t_d)$ (that is, there exists a non-zero polynomial in one variable with coefficients in the field $k(t_1, \dots, t_d)$ which is zero at u).

Proof. This follows from the primitive element theorem. For a full proof, see Proposition A.7 in the Appendix of [1]. $\square$

Proposition 1.7.33

Let K be a finitely generated extension of k . Then there exists an irreducible hypersurface $H \subset \mathbb{A}^{d+1}$ for some d such that K is isomorphic to the field of functions $k(H)$.

Proof. Write $K = k(t_1, \dots, t_d, u)$ as in Lemma 1.7.32, and let $K' = k(t_1, \dots, t_d)$. Let $p(U) \in K'[U]$ be the minimal polynomial of u over K' . Each coefficient of $p(U)$ is a fraction whose numerator and denominator are polynomials in $t_1, \dots, t_d$. We can multiply up by a suitable element of $k[t_1, \dots, t_d]$ to clear the denominators, and also replace $t_1, \dots, t_d$ by indeterminates $T_1, \dots, T_d$, we get a polynomial $g \in k[T_1, \dots, T_d, U]$ such that

$$g(t_1, \dots, t_d, u) = 0$$

in the field K . Assuming we multiplied up by a lowest common denominator for the coefficients of p , g is irreducible.

Let H be the hypersurface $V(g) \subset \mathbb{A}^{d+1}$. Because g is irreducible, it generates a radical ideal in $k[T_1, \dots, T_d, U]$, and so the (Strong) Nullstellensatz implies that

$$I(H) = (g).$$

Thus the coordinate ring is given by

$$k[H] = k[T_1, \dots, T_d, U]/(g).$$

There is a k -algebra homomorphism

$$\alpha: k[T_1, \dots, T_d, U] \rightarrow K$$

defined by

$$T_1 \mapsto t_1, \dots, T_d \mapsto t_d, U \mapsto u.$$

The claim is now that $\ker \alpha = (g)$. Suppose that $\alpha(f) = 0$. Then, $p(U)$ divides $f(t_1, \dots, t_d, U)$ in $K'[U]$ because $p(U)$ is the minimum polynomial of u over K' ; thus, $g(U)$ divides $f(t_1, \dots, t_d, U)$ in $K'[U]$. Then, Gauss's lemma shows that $g \mid f$ in $K[T_1, \dots, T_d, U]$. Therefore, $f \in (g)$, so $\ker \alpha = (g)$. Thus, α induces an injection $k[H] \hookrightarrow K$. Furthermore, the image of α generates K as a field, so α induces an injection from the fraction field of $k[H]$ to K . However, the image of α generates K as a field, so α induces an isomorphism from $k(H) \cong \text{Frac} k[H]$ to K . $\square$

Corollary 1.7.34

Let $V \subset \mathbb{A}^n$ be an irreducible affine algebraic subvariety. Then there exists an irreducible hypersurface $H \subset \mathbb{A}^{d+1}$ for some d such that V is birationally equivalent to H .

Proof. We apply the previous proposition with $K = k(V)$. Then, we have that

$$k(H) \cong k(V).$$

By Corollary 1.7.31, this implies that V is birationally equivalent to H . $\square$

This tells us that, even if V is a complicated algebraic variety defined by many equations, provided we only care about properties of V which are preserved by birational equivalence, we can replace V by a simpler set defined by just one equation, that is, a hypersurface. Note that it is not true that every affine algebraic variety is isomorphic to a hypersurface.

Exercise 1.7.35. Let $V = V(XY)$ and $W = V(Y(Y - X^2))$.

1. Describe the irreducible components of W (a proof is not required).
2. Which points lie in more than one irreducible component of W ?
3. Consider the function $f: W \rightarrow k$ defined by

$$f(x, 0) = x^2, \quad f(x, y) = 0 \text{ if } y \neq 0.$$

Prove that f is a regular function on W by finding a polynomial $F \in k[X, Y]$ such that $F|_W = f$.

4. Consider the function $g: W \rightarrow V$ defined by

$$g(x, 0) = x, \quad g(x, y) = 0 \text{ if } y \neq 0.$$

Prove that g is not regular.

5. Write down polynomials in $k[X, Y]$ which define a surjective map $V \rightarrow W$. (Note: you only need to write down such a map, and you do not need to prove that its image is equal to W .)
6. Is W isomorphic to V ? Give a proof. You may use any facts or examples from the lecture notes without proof.

Solution: 1. The irreducible components of W are $V(Y)$ and $V(Y - X^2)$.

2. The point $(0, 0)$ lie in both irreducible components.

3. Consider $F(X, Y) = X^2 - Y$. Then, on $Y = 0$, we get $F(X, 0) = X^2$. For $Y = X^2$, then $F(X, Y) = 0$. Thus, $F|_W = f$.

4. If g were regular, then we should find some polynomial $G \in k[X, Y]$ such that $G|_W = g$. Then,

$$G(X, 0) = X,$$

so that $G(X, Y) = X + H(Y)$, where $H \in k[Y]$. Then, $G(X, X^2) = X + H(X^2)$. We need this to equal zero, so H cannot have a constant term. Moreover, $H(X^2)$ must kill X , so $H(X^2) = -X$, but this is not possible, since H would not be a polynomial then.

5. Consider the map $\phi: (x, y) \mapsto (x + y, x^2)$. Then, $(x, 0) \mapsto (x, x^2) \in V(Y - X^2)$. Further, $(0, y) \mapsto (y, 0) \in V(Y)$. This is clearly surjective.

6. W and V are not isomorphic. Let $V = V_1 \cup V_2$ and $W = W_1 \cup W_2$, with W_1 being the line and W_2 being the parabola. Suppose that $\phi: W \rightarrow V$ is an isomorphism. The decomposition into irreducible components being intrinsic for topology, this isomorphism will exchange the irreducible components of W and V . Because the role of V_1 and V_2 is symmetric in V , up to relabeling the variables, one can suppose that ϕ induce an isomorphism between W_1 and V_1 , and between W_2 and V_2 , and sends the point $(0, 0)$ of W to the point $(0, 0)$ of V . The restriction to V_1 gives an automorphism of the affine line fixing the point 0 from 1), it must be of the form $x \mapsto \lambda x$, for $\lambda \in k^*$. Denoting $\phi: (x, y) \mapsto (g(x, y), h(x, y))$, we have $\phi: (x, 0) \mapsto (\lambda x, 0)$, hence $g(x, 0) = x$, and, for $y \neq 0$, $\phi(x, y) \in V_2$, hence $g(x, y) = 0$, which is not possible by 4. $\square$

Exercise 1.7.36. Let $C = V(Y^2 - X^3) \subset \mathbb{A}^2$ (you may use without proof the fact that C is irreducible). Let $\phi \in k(C)$ denote the rational function Y/X . Prove that ϕ is not regular at $(0, 0)$.

Suggested method (you are free to use a different method if you wish): Suppose that we can write $\phi = f/g$ for some regular functions $f, g \in k[C]$. Observe that for every $t \in k$, we have $(t^2, t^3) \in C$, and use this to prove that

$$f(T^2, T^3) = Tg(T^2, T^3)$$

in the ring of polynomials $k[T]$. Deduce that T^2 divides $f(T^2, T^3)$ and then that $g(0, 0) = 0$.

Solution: Let

$$\phi = \frac{Y}{X} = \frac{X^3}{XY} = \frac{X^2}{Y}.$$

Suppose that ϕ were regular here. Then, $\phi = \frac{f}{g}$, so we get that

$$Xf(X, Y) = Yg(X, Y).$$

Then, note that C can be parametrized by the points (t^2, t^3) , for $t \in k$. Then, we attain that

$$f(t^2, t^3) = tg(t^2, t^3).$$

This holds for infinitely many values of t because k is infinite, so we have that

$$f(T^2, T^3) = Tg(T^2, T^3).$$

Then, $T \mid f$, but this cannot have a degree 1 term, so $T^2 \mid f(T^2, T^3)$. Therefore, $T \mid g(T^2, T^3)$, so $g(0, 0) = 0$. Therefore, $\phi = \frac{f}{g}$ is not well-defined. $\square$

2 Projective Varieties

§2.1 Projective Space

Projective space consists of affine space together with ‘points at infinity’, one for each direction. The purpose of adding extra points is that it avoids special cases where a point ‘disappears to infinity’. For example, a pair of parallel lines do not intersect in affine space but they do intersect at a point at infinity in projective space.

Definition 2.1.1. The **projective n -space**, $\mathbb{P}^n$, is the quotient of $k^{n+1} \setminus \{(0, \dots, 0)\}$ by the equivalence relation

$$(x_0, \dots, x_n) \sim (\lambda x_0, \dots, \lambda x_n)$$

where $\lambda \in k \setminus \{0\}$. We call a representative for an equivalence class the **homogeneous coordinates** of that point in $\mathbb{P}^n$.

Remark 2.1.2 — Note that there are many choices for each point, by scaling by λ . To avoid confusion between homogeneous coordinates for $\mathbb{P}^n$ and ordinary coordinates for $\mathbb{A}^n$, we usually write homogeneous coordinates as

$$[x_0 : x_1 : \dots : x_n].$$

Observe that we can embed $\mathbb{A}^n$ into $\mathbb{P}^n$ by the map

$$(x_1, \dots, x_n) \mapsto [1 : x_1 : \dots : x_n].$$

Any other homogeneous coordinates where the first coordinate is non-zero can be re-scaled to have first coordinate 1. So we are left with the points with first coordinate equal to 0: these are the ‘points at infinity’. A point $[0 : x_1 : \dots : x_n]$ can be seen as a point in $\mathbb{P}^{n-1}$, just dropping the initial zero. Thus

$$\mathbb{P}^n = \mathbb{A}^n \cup \mathbb{P}^{n-1}.$$

Similarly,

$$\mathbb{P}^1 = \mathbb{A}^1 \cup \{\text{a point}\}.$$

Now, we can (reverse) induct, to get that

$$\mathbb{P}^n = \mathbb{A}^n \cup \mathbb{A}^{n-1} \cup \dots \cup \mathbb{A}^0.$$

Thinking about projective space as affine space plus points at infinity can be useful if we want to make use of our geometric intuition about affine space or the algebraic tools we have developed for working with affine algebraic varieties. On the other hand, thinking about projective space in terms of homogeneous coordinates emphasizes that all points of projective space look the same: we can only distinguish points at infinity from points in affine space after choosing a convention for how we embed $\mathbb{A}^n$ into $\mathbb{P}^n$ (for example, we could have used $[x_1 : \dots : x_n : 1]$ instead); throughout this lecture we will use the convention above.

§2.2 Zariski Topology on $\mathbb{P}^n$ and Homogeneous Ideals

Definition 2.2.1. We define the **Zariski topology** on $\mathbb{P}^n$ to be the topology induced from the Zariski topology on $\mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$ by the quotient map

$$\pi: \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\} \rightarrow \mathbb{P}^n.$$

Unwrapping the definition, one has:

$$V \subset \mathbb{P}^n \text{ is closed} \iff \pi^{-1}(V) \subset \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\} \text{ is closed}$$

Definition 2.2.2. Let $X \subseteq \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$. We define the **affine cone** $\hat{X} \subseteq \mathbb{A}^{n+1}$ to be the set $\hat{X} = X \cup \{(0, \dots, 0)\}$ if $X \neq \emptyset$ and $\hat{X} = X = \emptyset$ if X is empty.

Lemma 2.2.3

Let $X \subseteq \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$. Then, X is closed if and only if $\hat{X}$ is closed in $\mathbb{A}^{n+1}$.

Proof. Suppose $X = \emptyset$. Then, the result is clear. If $X \neq \emptyset$, then write $U = \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$. X is closed, so $X = U \cap V$, for $V \subseteq \mathbb{A}^{n+1}$ closed. In particular, $\hat{X} = V \cup \{(0, \dots, 0)\}$ is closed, so clear. For the converse, if $\hat{X}$ is closed, then $X = U \cap \hat{X}$ is closed. $\square$

Corollary 2.2.4

$X \subseteq \mathbb{P}^n$ is closed if and only if $\widehat{\pi^{-1}(X)} \subseteq \mathbb{A}^{n+1}$ is closed.

Definition 2.2.5. Let $V \subseteq \mathbb{P}^n$ be closed. Then, $\widehat{\pi^{-1}(V)}$ is called the **affine cone** on V .

Remark 2.2.6 — Note that if $x \in \widehat{\pi^{-1}(V)}$, then $\lambda x \in \widehat{\pi^{-1}(V)}$ for all $\lambda \in k$. Equivalently, if $V \neq \emptyset$, $0 \in \widehat{\pi^{-1}(V)}$ and if $x \in \widehat{\pi^{-1}(V)}$, then if L is the line through x and 0 , then $L \subseteq \widehat{\pi^{-1}(V)}$.

Lemma 2.2.7

If $V \subseteq \mathbb{P}^n$ is closed, then $\widehat{\pi^{-1}(V)}$ is the (affine) closure of $\pi^{-1}(V) \subseteq \mathbb{A}^{n+1}$.

Proof. When $V = \emptyset$, this is trivial, so assume that V is nonempty. Now, by the closure of V , $\pi^{-1}(V)$ is already closed, so $\widehat{\pi^{-1}(V)} \subseteq \pi^{-1}(V)$. We work on the reverse containment. Pick $x \in \pi^{-1}(V)$, which exists by nonemptiness of V . If L is the line through x and 0 , put $L' = L \setminus \{0\}$. Then, $L' \subseteq \pi^{-1}(V)$. Thus, $\widehat{\pi^{-1}(V)} \cap L$ is a closed subset of L that contains L' . Therefore, $\widehat{\pi^{-1}(V)} \cap L = L$, and $L \subseteq \widehat{\pi^{-1}(V)}$. We note that $\pi^{-1}(V) \subseteq \widehat{\pi^{-1}(V)}$, so they must agree. $\square$

Remark 2.2.8 — In general, this proof shows that $\widehat{\pi^{-1}(X)} \subseteq \overline{\pi^{-1}(X)}$, and we have equality precisely when X is closed.

Definition 2.2.9. An ideal $I \subset k[X_0, \dots, X_n]$ is said to be **homogeneous** if, for any $f \in I$ and $\lambda \in k$, one has $f(\lambda x_0, \dots, \lambda x_n) \in I$.

Notation 2.2.10. The homogeneous part $f^{(d)}$ of a polynomial $f \in k[X_0, \dots, X_n]$ is the sum of all terms of degree d in f , and a polynomial is called **homogeneous** of degree d if every term appearing in it has degree d .

In particular, if $f \in k[X_0, \dots, X_n]$ is homogeneous, $(f) \subset k[X_0, \dots, X_n]$ is obviously a homogeneous ideal. More generally, we have:

Proposition 2.2.11

Let I be an ideal of $k[X_0, \dots, X_n]$. The following are equivalent:

- (i) I is a homogeneous ideal.
- (ii) For any polynomial $f \in I$, every homogeneous part $f^{(d)}$ of f is also an element of I .
- (iii) I is generated by homogeneous polynomials.

Proof. It is clear that (ii) implies (iii). Take generators $f_1, \dots, f_r$ of I then their homogeneous parts are also elements of I by (ii), so these homogeneous parts are a generating set for I .

It is also easy to see that (iii) implies (i): if I is generated by homogeneous polynomials $f_1, \dots, f_r$ of degree d_i , for each i . Take any $f \in I$, and write $f = \sum_i a_i f_i$. Then,

$$f(\lambda x) = \sum_i a_i(\lambda x) f_i(\lambda x) = \sum_i a_i(\lambda x) \lambda^{d_i} f_i(x) \in I,$$

for all λ .

We are left with (i) implies (ii). Let $f \in I$ and write

$$f = f^{(0)} + f^{(1)} + \dots + f^{(d)}.$$

Since k is infinite we can find $c_0, \dots, c_d \in k^*$ such that the matrix

$$(c_i^j)_{0 \leq i, j \leq d}$$

is invertible by considering the Vandermonde determinant. On the other hand we have:

$$f(c_i x_0, \dots, c_i x_n) = \sum_{j=0}^d c_i^j f^{(j)}(x_0, \dots, x_n).$$

Using (i), the left hand side is in I , hence, applying the inverse of the matrix

$$(c_i^j)_{0 \leq i, j \leq d}$$

we obtain that each $f^{(j)}$ is in I by writing $f^{(j)}$ as sums of $f(c_i x_0, \dots, c_i x_n)$ as i varies. Notice that we have only used the fact that k is an infinite field here. This shows that each of the $f^{(j)}$ lie in I . $\square$

Remark 2.2.12 — In particular, from the Hilbert basis theorem, a homogeneous ideal has a finite basis of homogeneous polynomials.

§2.3 Dictionary Between Homogeneous Ideals and Subvarieties of $\mathbb{P}^n$

One can now define maps:

$$\tilde{V}: \{\text{homogeneous ideals of } k[X_0, \dots, X_n]\} \longleftrightarrow \{\text{closed subsets of } \mathbb{P}^n\}: \tilde{I}$$

by:

$$\begin{aligned}\tilde{V}(I) &= \pi(V(I) \setminus \{(0, \dots, 0)\}) = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : f(x_0, \dots, x_n) = 0 \text{ for all } f \in I\} \\ \tilde{I}(V) &= I(\widehat{\pi^{-1}(V)}) = \{f \in k[X_0, \dots, X_n] : f(x_0, \dots, x_n) = 0 \text{ for all } [x_0 : \dots : x_n] \in V\}.\end{aligned}$$

Exercise 2.3.1. Show that $\tilde{V}(\tilde{I}(V)) = V$. More generally, if V is not closed, $\tilde{V}(\tilde{I}(V))$ is the Zariski closure of V .

The converse of the exercise above is given by the projective Nullstellensatz:

Theorem 2.3.2 (Projective Nullstellensatz)

Let I be a homogeneous ideal of $k[X_0, \dots, X_n]$. Then

$$\tilde{I}(\tilde{V}(I)) = \sqrt{I}$$

unless $\sqrt{I}$ is the **irrelevant ideal** $(x_0, \dots, x_n)$ (in which case $\tilde{V}(I) = \emptyset$ and $\tilde{I}(\tilde{V}(I)) = k[X_0, \dots, X_n]$). In particular, the maps $\tilde{I}$ and $\tilde{V}$ give a bijection between closed subsets of $\mathbb{P}^n$ and non-irrelevant homogeneous radical ideals of $k[X_0, \dots, X_n]$.

Proof. If $V(I) \neq \{(0, \dots, 0)\}$, i.e. $\sqrt{I} \neq (x_0, \dots, x_n)$, one has that $\tilde{I}(\tilde{V}(I)) = I(V(I))$, hence it suffices to apply the affine Nullstellensatz. $\square$

§2.3.i Projective Varieties

Definition 2.3.3. A **projective variety** of $\mathbb{P}^n$ is a closed subset $V = \tilde{V}(I) \subset \mathbb{P}^n$, i.e. a subset of the form

$$V = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : f_1(x_0, \dots, x_n) = \dots = f_r(x_0, \dots, x_n)\},$$

where $f_1, \dots, f_r$ are homogeneous polynomials. The Zariski topology on V is the topology induced from the Zariski topology on $\mathbb{P}^n$.

Example 2.3.4

An example of a projective algebraic subvariety is

$$V = \{[w : x : y] \in \mathbb{P}^2 \mid wx - y^2 = 0\}.$$

What is $V \cap \mathbb{A}^2$? (Using the embedding $\mathbb{A}^2 \hookrightarrow \mathbb{P}^2$ which we considered before.) To find this, we just substitute $w = 1$ into the equation for V :

$$V \cap \mathbb{A}^2 = \{(x, y) \in \mathbb{A}^2 \mid x - y^2 = 0\},$$

that is, a parabola. We can also work out the intersection of V with the ‘ $\mathbb{P}^1$ at infinity’: it is the points where $w = 0$. Substituting into the equation for V , we get

$$\{[x : y] \in \mathbb{P}^1 : -y^2 = 0\} = \{[1 : 0]\}.$$

Thus V consists of the parabola together with a point at infinity ‘in the direction $(1, 0)$ ’, i.e. along the x -axis (informally, the two arms of the parabola close up at infinity).

We can also consider the embedding $\mathbb{A}^2 \subset \mathbb{P}^2$ given by $(w, x) \mapsto [w : x : 1]$, with the ‘ $\mathbb{P}^1$ at infinity’ given by $\{z = 0\}$. One has

$$V \cap \mathbb{A}^2 = \{(w, x) \in \mathbb{A}^2 : wx - 1 = 0\},$$

a hyperbola. The intersection with the ‘ $\mathbb{P}^1$ at infinity’ is given by

$$\{[w : x] \in \mathbb{P}^1 : wx = 0\} = \{[1 : 0], [0 : 1]\}.$$

Thus V consists of the hyperbola together with a point at infinity ‘in the direction $(1, 0)$ ’, i.e. along the w -axis, and one ‘in the direction $(0, 1)$ ’, i.e. along the x -axis, corresponding to the two asymptotes of the hyperbola.

§2.3.ii Homogenization

We would like to reverse this process, and go from an affine algebraic subvariety to a projective algebraic subvariety. One way to do this is via the process of homogenization.

Example 2.3.5

Consider the affine hyperbola

$$H = \{(w, x) \in \mathbb{A}^2 \mid wx - 1 = 0\}.$$

We need to turn the polynomial $WX - 1$ into a homogeneous polynomial, using a new variable Y . To do this, note that the highest degree term in $WX - 1$ has degree 2. We multiply each term by an appropriate power of Y to get all terms of degree 2. Thus we get $WX - Y^2 = 0$.

The process we went through above to obtain V' from V and H' from H can be generalized as follows.

Definition 2.3.6. For any polynomial $f \in k[X_1, \dots, X_n]$, we define the **homogenization** of f to be the polynomial $\bar{f} \in k[X_0, \dots, X_n]$ obtained by the following procedure: let d

be the maximum degree of terms of f . Then multiply each term of f by X_0^{d-e} , where e is the degree of this term in f .

Example 2.3.7

For example: if $f(X_1, X_2, X_3) = X_1^3 + 4X_1X_2X_3 - X_2^2 - X_2 + 5X_3 + 8$, then its homogenization is

$$\bar{f}(X_0, X_1, X_2, X_3) = X_1^3 + 4X_1X_2X_3 - X_2^2X_0 - X_2X_0^2 + 5X_3X_0^2 + 8X_0^3.$$

Definition 2.3.8. Let I be an ideal of $k[X_1, \dots, X_n]$. The **homogenization** of I is the ideal of $k[X_0, X_1, \dots, X_n]$ is generated by the $\bar{f}$ for $f \in I$.

Example 2.3.9

Here is a more sophisticated example. Consider the *twisted cubic* curve

$$C := \{(t, t^2, t^3) \in \mathbb{A}^3\} = V(Y - X^2, Z - XY).$$

Homogenizing polynomials, we get

$$C' = \{[w : x : y : z] \in \mathbb{P}^3 : wy - x^2 = wz - xy = 0\}.$$

It is still true that we can reverse this by just setting $w = 1$, so $C' \cap \mathbb{A}^3 = C$. However, what happens at infinity? Substituting in $w = 0$, we get

$$\{[0 : x : y : z] \in \mathbb{P}^3 : -x^2 = -xy = 0\} = \{[0 : 0 : y : z] \in \mathbb{P}^3\}.$$

Thus the intersection of C' with the plane at infinity is a copy of $\mathbb{P}^1$. This is not what we should expect: if C' were the smallest possible projective algebraic subvariety containing C , the dimension of the intersection with the plane at infinity should be smaller than the dimension of the initial affine algebraic subvariety (speaking informally).

In fact, the smallest possible projective algebraic subvariety containing C' (homogenize $I = (y - x^2, z - xy)$) is

$$C'' = \{[w : x : y : z] \in \mathbb{P}^3 : wy - x^2 = wz - xy = zx - y^2 = 0\}.$$

The extra polynomial involves only x, y, z and is in the ideal generated by $Y - X^2$ and $Z - XY$. You can calculate:

$$C'' = C \cup \{[0 : 0 : 0 : 1]\}.$$

Here, we are not giving a procedure to find the smallest projective algebraic subvariety containing a given affine algebraic subvariety—we just assert that this happens to work in this case. There is an algorithm but you would not want to have to use it by hand.

Let $V \subset \mathbb{A}^n$ be an affine algebraic subvariety. Let $W \subset \mathbb{P}^n$ be the set defined by the homogenizations of all polynomials in $I(V)$. Then W is the smallest projective algebraic

subvariety containing V . When we substitute $x_0 = 1$ into the polynomials defining W , we just get $I(V)$, so

$$W \cap \{[1 : x_1 : \cdots : x_n]\} = V.$$

The example above shows that if we just use homogenizations of a generating set, instead of all of $I(V)$, we still get a projective algebraic subvariety V' such that $V' \cap \mathbb{A}^n = V$, but it might not be the smallest such set.

Exercise 2.3.10. Prove that a quasi projective variety U is irreducible if and only if its projective closure $\bar{U}$ is irreducible.

Solution: The reverse implication is clear, since a nonempty open subset of an irreducible space is irreducible. As for the forward, this relies on the fact that the closure of an irreducible space is irreducible. If we can write $\bar{U} = A \cup B$, where A, B are nonempty and closed, then we have that $U = (U \cap A) \cup (U \cap B)$. Note that each of these intersections are closed in U because they're closed in $\bar{U}$. By irreducibility of U , we find that $U \cap A = U$ or $U \cap B = U$, so $U \subseteq A$ or B . However, then $\bar{U} = A$ or B , so $\bar{U}$ is irreducible. $\square$

Exercise 2.3.11. Prove that the variety $X = \mathbb{A}^2 \setminus \{(0,0)\}$ is not isomorphic to an affine variety.

Solution: For this, we employ the ring of regular functions. A regular function on X is, locally, a quotient of polynomials. Because X misses $(0,0)$, any regular function on X is also regular on the opens $D_x = \{x \neq 0\}$ and $D_y = \{y \neq 0\}$. Note these cover X . Therefore, if $f \in k[X]$, then $f \in k[D_x] \cap k[D_y]$. Now, $k[D_x] = k[x, y]_x$ and $k[D_y] = k[x, y]_y$. Conversely, if $h \in k[D_x] \cap k[D_y]$, then h is regular on D_x and D_y , so it glues to some regular function on X , so we have $X = k[D_x] \cap k[D_y] = k[x, y]_x \cap k[x, y]_y = k[x, y]$. Now, if X is isomorphic to some affine variety W , then every nonempty proper ideal of $k[W]$ has nonempty vanishing locus. However, if we take the ideal $(x, y) \subseteq k[X] = k[x, y]$, it has empty vanishing locus inside $k[X]$ because $(0,0)$ is not in X . $\square$

Exercise 2.3.12. Consider the map $\varphi: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ defined by $(x, y) \mapsto (x^2y, y)$. Is the image a quasi-projective variety?

Solution: We note that the image of φ is $\{(0,0)\} \cup \{(a,b) : b \neq 0\}$. Write $S := \text{Im} \varphi$. Then, $S = D_{\mathbb{A}^2}(y) \cup \{(0,0)\}$. Then, S is open inside its affine closure. However, the affine closure of S is $\mathbb{A}^2$ because $D_{\mathbb{A}^2}(y)$ is an open subset of $\mathbb{A}^2$, which is irreducible, so dense, and S contains $D_{\mathbb{A}^2}(y)$. Therefore, S is dense, and $\bar{S} = \mathbb{A}^2$, but S isn't open in $\bar{S}$ because it contains the point $(0,0)$. If S was open, then there would be some distinguished open $D(f)$ such that $f(0,0) \neq 0$. Then, $f(t,0)$ is not identically zero, and doesn't vanish at $t = 0$. Thus, there is some t such that $f(t,0) \neq 0$, so $(t,0) \in D(f)$. However, such a point is not in S . $\square$

Exercise 2.3.13. Is the subset

$$W := \{(a : b : c) : a \neq 0, b \neq 0\} \cup \{[1 : 0 : 0]\}$$

of $\mathbb{P}^2$ locally closed?

Solution: In other words, we wish to verify that every point in the subset has an open neighborhood U in $\mathbb{P}^2$ such that $W \cap U$ is closed in U . This is just checking if W is a quasi-projective subvariety of $\mathbb{P}^2$. $W = D_{\mathbb{P}^2}(X, Y) \cup \{[1 : 0 : 0]\}$. By similar methods to the above, we have that the projective closure of W is $\mathbb{P}^2$. We show that W is not open

in $\mathbb{P}^2$, by showing that the point P admits no open neighborhood. If it did, then there exists some open U of $\mathbb{P}^2$ containing P such that $U \cap W$ was closed in U . Now, if we pass to the affine chart U_0 , then $W \cap U_0$ is the union of the $\mathbb{A}^2 \setminus \{X\text{-axis}\} \cup \{(0, 0)\}$. The open neighborhoods U of P are attained by removing a finite number of curves in $\mathbb{A}^2$ not passing through P . We note that it’s not possible to do this in such a way that $U \cap W$ is closed in U , since $U \cap W$ has dimension 2, and so it can’t be a proper closed subset of U ; we can’t have $U \cap W = U$ because any curve containing all nonzero points on X -axis also contains the origin. More formally, if S was open, there would be some distinguished open $D(f)$ such that $f(1, 0, 0) \neq 0$. Then, the polynomial $f(1, 0, t)$ is not identically zero and does not vanish when $t = 0$. Therefore, for some nonzero t , $f(1, 0, t) \neq 0$, so $[1 : 0 : t] \in D(f)$. However, points of this form are not contained in W . $\square$

§2.4 Regular Maps Between Quasi Projective Varieties

We first define quasi-projective algebraic varieties. This will unify both affine and projective algebraic varieties, saving us from having to prove a (analogous) lemma for affine algebraic varieties and projective algebraic varieties.

Definition 2.4.1. A **quasi-projective algebraic variety** is the intersection between an open subset and a closed subset of $\mathbb{P}^n$ in the Zariski topology.

Example 2.4.2

Every projective variety is quasi-projective (just take the open subset to be $\mathbb{P}^n$ itself).

Example 2.4.3

Open subsets of affine/ projective varieties are quasi-projective varieties.

Example 2.4.4

Any affine variety is quasi-projective: it is the intersection between $\mathbb{A}^n$ (which is open in $\mathbb{P}^n$) and a projective algebraic variety V .

Example 2.4.5

$\mathbb{A}^1 \setminus \{0\}$ is a quasi-projective algebraic variety that is not projective nor affine.

Exercise 2.4.6. Prove that any quasiprojective variety is open in its projective closure.

Solution: Let $V = U \cap W$, where U is open in $\mathbb{P}^n$ and W is closed in $\mathbb{P}^n$. Since W is closed, we find that $\bar{V} \subseteq W$. Then, we have that $\bar{V} \cap U \subseteq W \cap U = V$, as well as $V \subseteq \bar{V}$ and $V \subseteq U$. Thus, $V \subseteq \bar{V} \cap U$ as well, so we attain equality: $\bar{V} \cap U = V$. Since U is open, it follows that V is open in $\bar{V}$. $\square$

We move on to define regular maps between quasi-projective algebraic subvarieties. Let $V \subset \mathbb{P}^m$ and $W \subset \mathbb{P}^n$ be quasi-projective algebraic subvarieties. We expect a regular map

$\varphi: V \rightarrow W$ to be a function which can be expressed as polynomials in the homogeneous coordinates:

$$\varphi([x_0 : \cdots : x_m]) = [f_0(x_0, \dots, x_m) : \cdots : f_n(x_0, \dots, x_m)].$$

In order for this to be a well-defined function, all the f_i must be homogeneous polynomials of the same degree, so that

$$[f_0(\lambda x_0, \dots, \lambda x_m) : \cdots : f_n(\lambda x_0, \dots, \lambda x_m)] = [\lambda^d f_0(x_0, \dots, x_m) : \cdots : \lambda^d f_n(x_0, \dots, x_m)].$$

All the coordinates are multiplied by λ^d , so this is the same point in $\mathbb{P}^m$ as

$$[f_0(x_0, \dots, x_m) : \cdots : f_n(x_0, \dots, x_m)].$$

There is another condition which must be imposed to get a well-defined function $V \rightarrow \mathbb{P}^m$: we must never have

$$f_0(x_0, \dots, x_n) = \cdots = f_m(x_0, \dots, x_n) = 0$$

because $[0 : \cdots : 0]$ is not the homogeneous coordinates of a point in $\mathbb{P}^m$.

This is a very strong condition and there are too few lists of polynomials which satisfy it. However, as for the case of regular maps on open subsets, we can turn the notion of regularity into a local notion, by asking for the existence of local representations by homogeneous polynomials near each point. The homogeneous nature of the coordinates allows us to do this in such a way that the different lists of polynomials define the same map wherever they overlap. We provide a complete definition.

Definition 2.4.7. A **regular map** $\varphi: V \rightarrow W$ between quasi-projective varieties, $V \subseteq \mathbb{P}^n$, $W \subseteq \mathbb{P}^m$, is a function φ such that for every $x \in V$, there exists an open neighborhood $U \subseteq V$ of x and polynomials $f_0, \dots, f_m \in k[X_0, \dots, X_n]$ such that

1. the $f_0, \dots, f_m$ are all homogeneous of the same degree,
2. for every $y \in U$, $f_0, \dots, f_m$ are not all zero at y ,
3. $\varphi(x_0 : \cdots : x_n) = [f_0(x_0 : \cdots : x_n) : \cdots : f_m(x_0 : \cdots : x_n)]$, for all $x = [x_0 : \cdots : x_n] \in U$.

Definition 2.4.8. A **regular function** from a projective algebraic variety V is a regular map $V \rightarrow \mathbb{A}^1$.

Remark 2.4.9 — We note that this immediately gives us a notion for regular maps between projective varieties. Further, it matches with our notion of regular maps between affine varieties.

Remark 2.4.10 — Note that we also get a notion of **isomorphism** for quasi-projective varieties, which is a regular map with an inverse (which is also a regular map).

Remark 2.4.11 — We also note that it suffices to use finitely many opens when covering V , by the Hilbert basis theorem.

Remark 2.4.12 — Later, we will prove that there are no non-constant regular functions on $\mathbb{P}^n$; that is, a map from $\mathbb{P}^n \rightarrow \mathbb{A}^1$. This can be viewed as an analogue of Liouville’s theorem in complex analysis.

To help explain this, we consider an example.

Example 2.4.13

Let V be the projective closure of the parabola $V = \{[w : x : y] \in \mathbb{P}^2 : wy = x^2\}$. Let

$$V' = V \cap \{[w : x : y] : w \neq 0\} = \{(x, y) \in \mathbb{A}^2 : y = x^2\}.$$

There is a regular map $\varphi' : V' \rightarrow \mathbb{A}^1$ given by $\varphi'(x, y) = x$. Does this extend to a regular map $\varphi : V \rightarrow \mathbb{P}^1$? The idea is that the point at infinity $[0 : 0 : 1] \in V$ maps to the point at infinity $[0 : 1] \in \mathbb{P}^1$.

To attempt to construct such a map, write φ' in homogeneous coordinates using the embedding $\mathbb{A}^2 \subset \mathbb{P}^2$: $[1 : x : y] \mapsto [1 : x]$. Now we homogenize to get the map $[w : x : y] \mapsto [w : x]$ for $w \neq 0$. Under the same presentation, this maps $[0 : 0 : 1] \in V$ to $[0 : 0]$, so this is not well defined for $w = 0$. What we do is fix the map by expressing the same map differently. Using the homogeneous nature of the coordinates, and the equation $x^2 = wy$ defining V , we have

$$[w : x] = [wx : x^2] = [wx : wy] = [x : y]$$

whenever the values multiplied/divided by w and x are non-zero.

The expression $[x : y]$ is well-defined at $[0 : 0 : 1]$, with value $[0 : 1]$. On the other hand, $[x : y]$ gives $[0 : 0]$ at the point $[1 : 0 : 0] \in V$, so we cannot use $[x : y]$ alone to define a map $V \rightarrow \mathbb{P}^1$. At least one of these two expressions is defined everywhere on V , and they agree where they overlap, so the two expressions together give a well-defined regular map $\varphi : V \rightarrow \mathbb{P}^1$ (which extends φ') defined by

$$\varphi([w : x : y]) = \begin{cases} [w : x] & \text{if } w \neq 0, \\ [x : y] & \text{if } y \neq 0. \end{cases}$$

Note that each expression is defined on a Zariski open subset of V . This is important because it is how we ensure that the value of φ at each point is polynomially related to its value at nearby points. (Open sets are the natural way to talk about “nearby points” in a topological space. This still applies in the Zariski topology, even though open sets are very big.)

Note that questions 5 and 6 on problem sheet 2 give examples of regular maps defined everywhere except at a single point of an affine algebraic subvariety, where there is an obvious value the map ‘should’ take at the missing point, but the map is not regular at that point because there is no way to extend it to that point using polynomials. This is why we are not allowed just to write down polynomials on arbitrary (non-open) subsets of V and claim they define a regular map.

Remark 2.4.14 — In practice, every regular map can be written down by specifying lists of polynomials on just finitely many open sets, like φ (this follows ultimately from the Hilbert Basis Theorem). To check that a purported definition like φ in the previous example really does define a regular map $V \rightarrow W$, you have to check:

1. each set on which an expression is defined is Zariski open;
2. an expression never gives $[0 : \cdots : 0]$ on its associated set;
3. two expressions agree wherever they are both defined;
4. the image of the map is contained in W .

Example 2.4.15

As another example, let's try to extend the inverse of φ (from the previous example) from affine to projective algebraic varieties. On affine algebraic varieties, the inverse of φ' is

$$\psi': \mathbb{A}^1 \rightarrow V'$$

given by

$$\psi'(t) = (t, t^2).$$

In projective coordinates, this is

$$[1 : t] \mapsto [1 : t : t^2].$$

Homogenizing (inserting powers of s to make all the polynomials on the right-hand side degree 2), we get

$$[s : t] \mapsto [s^2 : st : t^2].$$

Now s^2, st, t^2 are never simultaneously zero for $[s : t] \in \mathbb{P}^1$, so in this case the single expression $[s^2 : st : t^2]$ is enough to define a regular map

$$\varphi: \mathbb{P}^1 \rightarrow V$$

(note that the image of φ is indeed contained in V).

The two maps $\varphi: V \rightarrow \mathbb{P}^1$ (again, from before) and $\psi: \mathbb{P}^1 \rightarrow V$ are inverses, so we conclude that the projective parabola V is isomorphic to $\mathbb{P}^1$.

Remark 2.4.16 — Note that this homogenization procedure does not always work. There are regular maps between affine algebraic subvarieties which it is impossible to extend to regular maps between their projective closures (there are points for which it is impossible to avoid sending them to $[0 : \cdots : 0]$). Such maps extend to regular maps.

§2.4.i Regular Maps Equal on a Dense Subset

We already proved the following lemma for regular maps between affine algebraic varieties:

Lemma 2.4.17

Let $\varphi, \psi: V \rightarrow W$ be regular maps between quasi-projective varieties V, W . If there exists a Zariski dense subset $A \subseteq V$ such that $\varphi|_A = \psi|_A$, then $\varphi = \psi$.

Proof. Let $Z = \{x \in V: \varphi(x) = \psi(x)\}$. By hypothesis, Z contains a dense subset of V . Hence, in order to show that $Z = V$, it suffices to show that Z is closed in V .

We will use the following topological fact: Let S be any topological space and $\{U_\alpha\}$ be an open cover of S . If $Z \cap U_\alpha$ is closed in the subspace topology on U_α for every α , then Z is closed as a subset of S .

From the definition of regular maps, we can cover V by Zariski open sets U_α such that on each U_α , both φ and ψ are defined by sequences of homogeneous polynomials:

$$\varphi|_{U_\alpha} = [f_{\alpha,0}: \dots: f_{\alpha,m}], \quad \psi|_{U_\alpha} = [g_{\alpha,0}: \dots: g_{\alpha,m}].$$

(Note that we can do this for any finite number of regular functions φ_i .) Now,

$$Z \cap U_\alpha = \{x \in U_\alpha: [f_{\alpha,0}(x): \dots: f_{\alpha,m}(x)] = [g_{\alpha,0}(x): \dots: g_{\alpha,m}(x)]\}.$$

This is the same as the set of $x \in U_\alpha$ where the vectors $(f_{\alpha,0}(x), \dots, f_{\alpha,m}(x))$ and $(g_{\alpha,0}(x), \dots, g_{\alpha,m}(x))$ are proportional (for any choice of homogeneous coordinates for x). This condition is equivalent to

$$f_{\alpha,i}(x)g_{\alpha,j}(x) - f_{\alpha,j}(x)g_{\alpha,i}(x) = 0 \quad \text{for all } i, j \in \{0, \dots, m\}.$$

Thus,

$$Z \cap U_\alpha = U_\alpha \cap \tilde{V}(f_{\alpha,i}(x)g_{\alpha,j}(x) - f_{\alpha,j}(x)g_{\alpha,i}(x)), \quad \text{for all } i, j \in \{0, \dots, m\}.$$

This last condition is given by homogeneous polynomials, and therefore defines a closed subset in the subspace topology on U_α . $\square$

Corollary 2.4.18

If $\varphi, \psi: V \rightarrow W$ are regular maps with V irreducible and there is an open subset A such that $\varphi|_A = \psi|_A$, then $\varphi = \psi$.

Proof. This is because Zariski opens of V are dense. $\square$

Remark 2.4.19 — The lemma therefore tells us that, given a list of polynomials on a Zariski open subset of V , there is at most one regular map which is given by that list of polynomials on that set. That is, we can think of regular maps as polynomials.

§2.4.ii Isomorphisms of Quasi-Projective Varieties

Example 2.4.20

We have in particular an induced notion of isomorphism for quasi-projective algebraic varieties. We can now make rigorous the claim that $\mathbb{A}^1 \setminus \{0\}$ is isomorphic to the affine hyperbola

$$H = \{(x, y) \in \mathbb{A}^2 : xy = 1\}.$$

Proof. The set

$$\mathbb{A}^1 \setminus \{0\} = \mathbb{P}^1 \setminus \{[1 : 0], [0 : 1]\}$$

is a Zariski open subset of $\mathbb{P}^1$, because its complement is finite. Hence, $\mathbb{A}^1 \setminus \{0\}$ is a quasi-projective algebraic variety. The map

$$\varphi : \mathbb{A}^1 \setminus \{0\} \rightarrow H$$

given by $\varphi(t) = (t, 1/t)$ can be written in homogeneous coordinates as

$$\varphi([1 : t]) = [1 : t : 1/t] = [t : t^2 : 1].$$

So homogenizing, we get

$$\varphi([s : t]) = [st : t^2 : s^2].$$

So long as $[s : t] \in \mathbb{P}^1 \setminus \{[1 : 0], [0 : 1]\}$, this does give a point in

$$H = \{[w : x : y] \in \mathbb{P}^2 : xy = w^2\} \cap \mathbb{A}^2.$$

So φ is a regular map

$$\mathbb{A}^1 \setminus \{0\} \rightarrow H.$$

The projection $(x, y) \mapsto x$ is a regular inverse to φ . Hence $\mathbb{A}^1 \setminus \{0\}$ and H are isomorphic as quasi-projective algebraic varieties. We note that the map φ does not extend to a map $\mathbb{A}^1 \rightarrow \bar{H}$, the projective closure of the hyperbola: $\bar{H} = H \cup \{[1 : 0 : 0], [0 : 1 : 0]\}$. $\square$

§2.5 Rational Maps

§2.5.i Rational maps between quasi-projective algebraic varieties

Given V, W quasi-projective with V irreducible, we define rational maps $V \dashrightarrow W$ as in the affine case:

Definition 2.5.1. Given V, W quasi-projective with V irreducible, we define **rational maps** $V \dashrightarrow W$ as equivalence classes of (U, f) with $U \subseteq V$ open and $f : U \rightarrow W$ regular, identifying (U, f) with (U', f') if $f|_{U \cap U'} = f'|_{U \cap U'}$.

Using the definition of regular maps, it is equivalent to the following definition:

Lemma 2.5.2

Let $V \subseteq \mathbb{P}^m$ and $W \subseteq \mathbb{P}^n$ be quasi-projective algebraic subvarieties, with V irreducible. Let S denote the set of sequences $(f_0, \dots, f_n) \in k[X_0, \dots, X_m]^{n+1}$ such that:

1. $f_0, \dots, f_n$ are homogeneous of the same degree;
2. $f_0, \dots, f_n$ are not all identically zero on V (note that this looks a little like the $b \neq 0$ condition in defining the field of fractions);
3. there exists a non-empty Zariski open set $A \subseteq V$ such that, for all $x \in A$,

$$[f_0(x) : \dots : f_n(x)] \in W.$$

Define an equivalence relation $\sim$ on S by

$$(f_0, \dots, f_n) \sim (g_0, \dots, g_n) \quad \text{if} \quad f_i g_j = f_j g_i \text{ for all } i, j.$$

Then, **rational maps** $\varphi: V \dashrightarrow W$ are the same as equivalence classes in S for $\sim$.

Proof. Expand the definition of rational map in terms of regular maps. □

Remark 2.5.3 — We usually specify rational maps by just giving one representative $[f_0 : \dots : f_n]$ in S .

Definition 2.5.4. If $\varphi: V \dashrightarrow W$ is a rational map, we say that $x \in V$ is a **regular point** of φ when there exists a representative (U, f) for φ with $x \in U$,

Definition 2.5.5. We define the **domain**, $\text{dom}(\varphi)$, as the open set of regular points of a rational map.

Remark 2.5.6 — Note that this is an open subset of V because it’s a union of open subsets.

Lemma 2.5.7

A rational map $\varphi: V \dashrightarrow W$ is **regular** at a point $x \in V$ if there exists at least one list of polynomials $(f_0, \dots, f_n) \in S$ representing φ such that

$$(f_0(x) : \dots : f_n(x)) \neq (0, \dots, 0).$$

Proof. Unravel the definition. □

Remark 2.5.8 — Note that the domain of definition of a rational map can change if we change the codomain, i.e. the target set W .

Remark 2.5.9 — We note that a rational map is regular if and only if it is regular at every point.

Example 2.5.10

For example, consider the map $\mathbb{P}^1 \rightarrow \mathbb{P}^2$ defined by

$$[s : t] \mapsto [s^2 : st : t^2].$$

This is regular at every point. We could interpret the same formula as defining a rational map $\mathbb{P}^1 \dashrightarrow W$ where $W \subseteq \mathbb{P}^2$ is the open set

$$W = \{[w : x : y] : w \neq 0\}.$$

As a rational map $\mathbb{P}^1 \dashrightarrow W$, this is not regular at the point $[0 : 1]$ because this point maps to $[0 : 0 : 1] \notin W$.

Example 2.5.11

As another example, consider the identity map $\text{id} : \mathbb{A}^1 \rightarrow \mathbb{A}^1$, whose domain is all of $\mathbb{A}^1$. We can also view the identity as a rational map $\mathbb{A}^1 \rightarrow \mathbb{A}^1 \setminus \{0\}$, which is not regular at the origin. Here, the domain must be $\mathbb{A}^1 \setminus \{0\}$.

Remark 2.5.12 — It follows immediately from the definition of regular maps between quasi-projective algebraic varieties that if a rational map is regular at every point, then it is a regular map.

Example 2.5.13

Let C denote the affine algebraic subvariety

$$C = \{(x, y) \in \mathbb{A}^2 : y = x^3\}.$$

This has projective closure

$$\overline{C} = \{[w : x : y] \in \mathbb{P}^2 : w^2 y = x^3\} = C \cup \{[0 : 0 : 1]\}.$$

Consider the regular map of affine algebraic varieties $\varphi : C \rightarrow \mathbb{A}^1$ given by

$$\varphi(x, y) = x.$$

If we try to extend this to a map of projective algebraic varieties $\overline{\varphi} : \overline{C} \dashrightarrow \mathbb{P}^1$, we would say that for points $[1 : x : y] \in C \subseteq \overline{C}$,

$$\varphi([1 : x : y]) = [1 : x],$$

and this homogenizes to

$$\overline{\varphi}([w : x : y]) = [w : x].$$

Thus $\overline{\varphi}$ is a rational map $\overline{C} \dashrightarrow \mathbb{P}^1$. The above expression for $\overline{\varphi}$ is not defined at the point $[0 : 0 : 1] \in \overline{C}$. We can prove that there is no other expression for $\overline{\varphi}$ which is defined at that point, and so $\overline{\varphi}$ is not regular at $[0 : 0 : 1]$. (we prove this in Exercise 2.5.15) However, $\overline{\varphi}$ does define a rational $\overline{C} \rightarrow \mathbb{P}^1$ whose domain is C .

Remark 2.5.14 — Thus a regular map between irreducible affine algebraic varieties extends to a rational map between their projective closures, but the extended map is not necessarily regular at the points at infinity.

Exercise 2.5.15. Consider the example from the lectures:

$$\overline{C} = \{[w : x : y] \in \mathbb{P}^2 : w^2 y = x^3\}$$

and $\phi : \overline{C} \dashrightarrow \mathbb{P}^1$ is the rational map represented by $[w : x : y] \mapsto [w : x]$. Prove that ϕ is not regular at $[0 : 0 : 1]$.

Solution: Consider the regular map $\mathbb{A}^1 \rightarrow C$ given by $\psi : t \mapsto (t, t^3)$. Homogenizing, we obtain a regular map:

$$\overline{\psi} : \mathbb{P}^1 \rightarrow \overline{C}$$

$$[s : t] \mapsto [s^3 : s^2 t : t^3].$$

Consider a local representation of ϕ by some homogeneous polynomials $f, g \in k[W, X, Y]$, then the equivalence relation between different representations of rational maps gives:

$$Xf = Wg$$

for any point of $\overline{C}$. Precomposing by $\overline{\psi}$, one obtains:

$$s^2 t f(s^3, s^2 t, t^3) = s^3 g(s^3, s^2 t, t^3)$$

This holds for all $(s, t) \neq (0, 0)$, by continuity it must hold for $(0, 0)$, hence it gives equality in $k[S, T]$:

$$\begin{aligned} S^2 T f(S^3, S^2 T, T^3) &= S^3 g(S^3, S^2 T, T^3) \\ \implies T f(S^3, S^2 T, T^3) &= S g(S^3, S^2 T, T^3) \end{aligned}$$

Then S divides $f(S^3, S^2 T, T^3)$, but the smallest power of S appearing here is S^2 , hence S^2 divides $f(S^3, S^2 T, T^3)$, and then S divides $g(S^3, S^2 T, T^3)$. Then, taking $s = 0$ and $t = 1$, one has that $g(0, 0, 1) = 0$, and then $f(0, 0, 1) = 0$, hence the map is not defined at $(0, 0, 1)$. $\square$

Exercise 2.5.16.

1. Let $\phi: \mathbb{P}^n \rightarrow \mathbb{P}^m$ be a regular map. Prove that there exist polynomials $f_0, \dots, f_m \in k[X_0, \dots, X_n]$ such that the expressions

$$[f_0(x_0, \dots, x_n) : \dots : f_m(x_0, \dots, x_n)]$$

defines ϕ at every point of $\mathbb{P}^n$. (use the fact that $k[X_0, \dots, X_n]$ is a UFD).

2. Prove that, if $\phi: \mathbb{P}^n \rightarrow \mathbb{P}^1$ is a non-constant regular map, then the image of ϕ is not contained in $\mathbb{A}^1$.
3. Prove that every rational map $\mathbb{P}^1 \dashrightarrow \mathbb{P}^n$ is regular.

Solution: 1. First, consider some cover of $\mathbb{P}^n$ defined by $U_j = D(h_j)$. Then, by quasi-compactness, we take finitely many of these. Since ϕ is regular, it must be that there exists an open set U and polynomials $f_0, \dots, f_m$ such that they're all homogeneous of the same degree, for every $y \in U$, not all of the f_i are zero, and ϕ can be expressed as $[f_1(x_0, \dots, x_n) : \dots : f_m(x_0, \dots, x_n)]$ for all $[x_0 : \dots : x_n] \in U$. Suppose that we have another expression of ϕ on a different open set V given by $g_0, \dots, g_m$ such that $U \cap V \neq \emptyset$. Then, on the open set $U \cap V$, the expressions involving the f_i and that of g_i must coincide. Now, define

$$u = \gcd(f_0, \dots, f_m), \quad v = \gcd(g_0, \dots, g_m),$$

and define the rational functions

$$F_i = \frac{f_i}{u}, \quad G_i = \frac{g_i}{v}.$$

Given that $f_i g_j = f_j g_i$, this implies that $F_i G_j = F_j G_i$ and so these expressions coincide as well. Thus, we can find some rational function h such that $F_i = h G_i$, for all i . Writing $h = \frac{a}{b}$, we then get that

$$b F_i = a G_i.$$

Then, $b \nmid a$, so by UFD property, $b \mid G_i$. However, then b divides all of G_i , so $b \mid \gcd(G_0, \dots, G_m) = 1$, so b is a unit. The analogous argument can be done for a , so $h = \frac{a}{b}$ is a scalar. Therefore, F_i and G_i are different up to a scalar, say c , and this choice of F_i then expresses ϕ .

2. Using the previous question, ϕ can be globally written using two homogeneous polynomials f, g . If its image is contained in $\mathbb{A}^1$, then $f \in k[X_0, \dots, X_n]$ never vanishes on $\mathbb{A}^{n+1} \setminus \{0\}$. From the Nullstellensatz, one must have $(f) = k[X_0 : \dots : X_n]$. That is, f is of degree 0 and then ϕ is constant, or $(f) = (X_0, \dots, X_n)$, which is impossible because $n \geq 1$.

3. It suffices to show that the restriction to the two affine spaces $\mathbb{A}^1$ covering $\mathbb{P}^1$ is regular, hence to show that any rational map $\phi: \mathbb{A}^1 \dashrightarrow \mathbb{P}^n$ is regular. Such a map is

defined polynomials $f_0, \dots, f_m \in k[X]$, and maps x to $[f_0(x) : \dots : f_m(x)]$. Now, consider the gcd of the f_i . Dividing out by this gcd, we can define ϕ as $x \mapsto [f'_0(x) : \dots : f'_m(x)]$, and the f'_i have no common zero on $\mathbb{A}^1$, hence ϕ is regular because it is well defined at all points. $\square$

Remark 2.5.17 — Another way to do 2, would be to use the fact that $\mathbb{P}^n$ is projective, so complete, and it is clearly connected. Therefore, a regular map from $\mathbb{P}^n$ to $\mathbb{A}^1$ would be constant. However, this is a contradiction to the assumption that ϕ is nonconstant.

Remark 2.5.18 — The above is not true for rational maps $\mathbb{P}^m \dashrightarrow \mathbb{P}^n$ for $m > 1$. For example, consider the rational map

$$\psi: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$$

given by

$$[x : y : z] \mapsto [yz : zx : xy],$$

which is not regular at $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]$.

Exercise 2.5.19.

1. Prove that, if $\phi: \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ is a rational function, then there exist polynomials $f, g \in k[X, Y]$ such that $\phi = f/g$ and such that $g(x, y) \neq 0$ on all of $\text{dom}(\phi)$.
2. Deduce that, if a rational function $\phi: \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ is regular on $\mathbb{A}^2 \setminus \{(0, 0)\}$, then it is regular at $(0, 0)$ too.
3. Consider the quasi-projective variety $V = \mathbb{A}^2 \setminus \{(0, 0)\}$. Prove that the ring of regular functions on V is the polynomial ring $k[X, Y]$.
4. Prove that V is not isomorphic to any affine variety. (Use the fact that, in an affine variety W , every proper ideal in $k[W]$ defines a non-empty subset.)

Solution: 1. Dividing by the gcd, we can take a fraction f/g representing ϕ such that f and g are coprime. Consider $(x, y) \in \text{dom}(\phi)$: there is f', g' such that $\phi = f'/g'$ and $g'(x, y) \neq 0$. In particular $g'f = f'g$: because f and g are prime, one has that g divides g' , hence $g(x, y) \neq 0$.

2. From the last question, $\phi = f/g$, with $g \neq 0$ on $\mathbb{A}^2 \setminus \{(0, 0)\}$. If $g(0, 0) = 0$, then from the Nullstellensatz $\sqrt{(g)} = (X, Y)$. Then, $X^n \mid g$ and $Y^m \mid g$ for some n, m , which isn't possible unless g is a constant, which is eliminated. Thus, $g(0, 0) \neq 0$, and ϕ is regular at $(0, 0)$.

3. Consider the homomorphism of rings obtained by restriction:

$$k[X, Y] = k[\mathbb{A}^2] \rightarrow \mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2 \setminus \{(0, 0)\})$$

We claim that this is injective. Note that $V = \mathbb{A}^2 \setminus \{(0, 0)\}$ is Zariski dense in $\mathbb{A}^2$. Now, if $f \in k[X, Y]$ such that $f|_V = 0$, then $V \subseteq V(f)$. Since V is dense, $V(f) = \mathbb{A}^2$, so f is the zero polynomial. Surjectivity from the last question, hence it is an isomorphism.

4. If V were affine, then the ideal (X, Y) in the coordinate ring $k[X, Y]$ would correspond to some non-empty subset of V , but there is no corresponding point in V . $\square$

Exercise 2.5.20. Prove that the rational maps $\phi: [X : Y : Z] \mapsto [X : Y]$ and $\psi: [X : Y : Z] \mapsto [XZ : YZ]$ from $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ are the same map.

Solution: We have that $\text{dom}\phi = \mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$ and $\text{dom}\psi = D(Z) \cap \text{dom}\phi$. Now, since ϕ and ψ agree on $D(Z) \cap \text{dom}\phi$, which is open in $\mathbb{P}^2$, they agree on a dense open, so are the same map. $\square$

§2.5.ii Birational maps

Just as in the affine case, if we have irreducible quasi-projective varieties V, W, T and rational maps

$$\varphi: V \dashrightarrow W \quad \text{and} \quad \psi: W \dashrightarrow T,$$

if the image of φ is dense in W , then the composition $\psi \circ \varphi$ is a rational map $V \dashrightarrow T$. The following definitions are the same as in the affine case:

Definition 2.5.21. A rational map $\varphi: V \dashrightarrow W$ is **dominant** if its image is dense in W .

Definition 2.5.22. A rational map $\varphi: V \dashrightarrow W$ is a **birational equivalence** if it is dominant and there exists a rational map $\psi: W \dashrightarrow V$ such that

$$\psi \circ \varphi = \text{id}_V \quad \text{and} \quad \varphi \circ \psi = \text{id}_W$$

(where these composite rational maps are defined). Irreducible quasi-projective varieties V and W are **birational** if there exists a birational equivalence $V \dashrightarrow W$.

Notice that the codomain of a dominant rational map is automatically irreducible, because of this lemma:

Lemma 2.5.23

Let V, W be quasi-projective varieties, and let $f: V \rightarrow W$ be a dominant rational map. If V is irreducible, then W is irreducible.

Proof. The idea is to reduce to f regular and apply some topological manipulations.

Since f is rational, it's defined on some open set $U \subseteq V$. Now, note that open subsets of quasiprojective varieties are quasiprojective varieties. Further, open subsets of irreducible spaces are irreducible. This implies that $f: U \rightarrow W$ is a regular map of quasiprojective varieties with U irreducible. If we take U large enough, then we can also assume that f is dominant.

Suppose that W is reducible: consider a decomposition $W = W_1 \cup W_2$, with W_1, W_2 proper closed subsets. Then

$$U = f^{-1}(W_1) \cup f^{-1}(W_2),$$

which are both closed subsets of U . We wish to attain a contradiction, so we want to show that each $f^{-1}(U_i)$ is a proper subset. The image of f is dense, so if $\text{Im}(f) \subseteq W_1$, then $W_1 = W$, similarly for W_2 ; this is a contradiction to W_i being proper. Therefore, $f^{-1}(W \setminus W_i) \neq \emptyset$, for both i , so $f^{-1}(W_i) \subsetneq U$ is a proper closed subset, which contradicts the irreducibility of U . $\square$

Example 2.5.24 ($\mathbb{A}^n$ is birational to $\mathbb{P}^n$)

Note that $\mathbb{A}^n$ is birational to $\mathbb{P}^n$: consider the regular map

$$\varphi: \mathbb{A}^n \rightarrow \mathbb{P}^n: (x_1, \dots, x_n) \mapsto [1: x_1: \dots: x_n]$$

and the rational map

$$\psi: \mathbb{P}^n \dashrightarrow \mathbb{A}^n: [x_0: \dots: x_n] \mapsto (x_1/x_0, \dots, x_n/x_0).$$

Each of these is dominant and composing them in either direction gives the identity, so these are birational equivalences. Note that φ is dominant because its image is $\{x_0 \neq 0\}$, which is open in $\mathbb{P}^n$. $\mathbb{P}^n$ is irreducible, so $\text{Im}\varphi$ is dense in $\mathbb{P}^n$.

Observe that φ is an isomorphism from $\mathbb{A}^n$ to an open subset of $\mathbb{P}^n$. We can generalize this to the following stronger result, which makes precise the intuition that varieties are birational if and only if they are the same ‘almost everywhere’.

Lemma 2.5.25

Irreducible quasi-projective varieties V and W are birational if and only if there exist non-empty Zariski open subsets $A \subseteq V$ and $B \subseteq W$ such that A is isomorphic to B (as quasi-projective varieties).

Proof. If we’re given an isomorphism $f: A \xrightarrow{\sim} B$ with inverse $g: B \rightarrow A$, then (A, f) and (B, g) are inverse rational maps.

Conversely, let $\varphi: V \dashrightarrow W$ and $\psi: W \dashrightarrow V$ be an inverse pair of rational maps. Let $A_1 = \text{dom}(\varphi)$ and $B_1 = \text{dom}(\psi)$. Note that A_1 is a non-empty open subset of V and B_1 is a non-empty open subset of W . Since φ induces a continuous map $A_1 \rightarrow W$, set $A = \varphi^{-1}|_{A_1}(B_1) \subseteq A_1$ is open. Furthermore, since φ is dominant, its image intersects the open set $B_1 \subseteq W$. Therefore A is non-empty. Similarly let $B = \psi^{-1}(A_1)$ is a non-empty open subset of W . One can now check that $\varphi|_A$ and $\psi|_B$ form an inverse pair of isomorphisms between A and B , since the original maps φ and ψ are inverses where they are defined. $\square$

Recall that if V is a quasi-projective algebraic variety, we define a rational function on V to be a rational map $\varphi: V \dashrightarrow \mathbb{A}^1$. By definition, this is the same as a rational map $\varphi': V \dashrightarrow \mathbb{P}^1$ except that we declare φ to be non-regular at points where $\varphi'(x) = \infty = [0: 1] \in \mathbb{P}^1$. We can therefore say

$$\varphi(x) = [f(x): g(x)] = [1: \frac{g(x)}{f(x)}] = \frac{g(x)}{f(x)} \in \mathbb{A}^1$$

whenever $f(x) \neq 0$, for suitable polynomials f, g . Of course, as always with rational maps, we might need to use different polynomials to evaluate it at different points.

Remark 2.5.26 — The rational functions on V form a field $k(V)$, the fraction field of $k[V]$.

Corollary 2.5.27

Let U be an open subset of an irreducible quasi projective variety V . Then U is birational to V .

Proof. Note that U is irreducible, since it's an open subset of an irreducible V . Thus, we apply the lemma above with $W = U$, and $A, B = U$. $\square$

Theorem 2.5.28

Two quasiprojective varieties V, W are birational if and only if $k(V)$ is isomorphic to $k(W)$ as k -algebras.

Proof. This is analogous to the affine case. $\square$

Exercise 2.5.29. Which of the following expressions define rational maps $\varphi: \mathbb{P}^n \dashrightarrow \mathbb{P}^m$ with $n, m = 1$ or 2 between projective spaces of the appropriate dimensions? In each case, determine $\text{dom}\varphi$, say if φ is birational, and if so describe the inverse map.

1. $(x, y, z) \mapsto (x, y)$;
2. $(x, y) \mapsto (x, y, 1)$;
3. $(x, y) \mapsto (x, y, 0)$;
4. $(x, y, z) \mapsto (1/x, 1/y, 1/z)$;
5. $(x, y, z) \mapsto ((x^3 + y^3)/z^3, y^2/z^2, 1)$;
6. $(x, y, z) \mapsto (x^2 + y^2, y^2, y^2)$.

Solution: 1. This is rational because it is regular on $\text{dom}\varphi = \mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$. This is dominant because it's surjective from the domain. However, there is no dominant rational inverse because the fiber of a generic point is a line $[sx : sy : z]$.

2. This is not even rational, because the map 1 is of different degree, compared to x and y , and a rational map requires all present polynomials to be of equal homogeneous degree.

3. This is rational because it's regular everywhere. $\text{dom}\varphi = \mathbb{P}^1$. This is not dominant because it misses a whole copy of $\mathbb{A}^2$.

4. This map is defined whenever x, y, z are nonzero. To make this homogeneous, we can multiply by xyz to see that $\varphi: [x : y : z] \mapsto [yz : xz : xy]$. Thus, $\text{dom}\varphi = \mathbb{P}^2 \setminus \{[0 : 0 : 1], [0 : 1 : 0], [1 : 0 : 0]\}$. This is dominant, missing the points with only one entry 0. It's also birational, having itself as the inverse.

5. We first homogenize: $\varphi: [x : y : z] \mapsto [x^3 + y^3 : y^2z : z^3]$. This is well defined at all points in $\mathbb{P}^2$ except the points $[x : y : 0]$, where $x^3 + y^3 = 0$, so $\text{dom}\varphi = \mathbb{P}^2 \setminus \{[x : y : 0] \in \mathbb{P}^2 : x^3 + y^3 = 0\}$. It is not possible to form a birational inverse from the image to the domain. One way to see this isn't possible is by restricting to the affine open $z = 1$. The map becomes $\varphi: (x, y) \mapsto (x^3 + y^3, y^2)$, so it's not possible to determine x uniquely, starting from the right. We note that

$\text{Im}\varphi = \{[x : y : 1] : [x : y] \in \mathbb{A}^2\} \cup \{[1 : 0 : 0]\}$. This first set is isomorphic to $\mathbb{A}^2$, which is an open subset of $\mathbb{P}^2$. Therefore, this is dense, and so φ is dominant.

6. $\text{dom}\varphi = \mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$. This is not dominant because it misses a whole copy of $\mathbb{P}^1$ in the second and third coordinates. Therefore, it can’t be birational. $\square$

Exercise 2.5.30. Consider the curve $C = V(Y^2 - X^3) \subseteq \mathbb{A}^2$, and the map $\phi: C \rightarrow \mathbb{P}^1$ defined by $(x, y) \mapsto [x : y]$. Is it a regular map?

Solution: Firstly, this is clearly a rational map. The only apparent issue seems to be $(0, 0)$, so we check regularity there. Motivated by the isomorphism $C \setminus \{(0, 0)\}$ to $\mathbb{A}^1 \setminus \{0\}$ defined by $(x, y) \mapsto \frac{y}{x}$ with inverse $t \mapsto (t^2, t^3)$, we use the inverse as a parametrization of the curve. If ϕ is regular at $(0, 0)$, we would have $\phi = [f : g]$, so we would have that $Yf = Xg$ around $(0, 0)$. Using the parametrization, we have that $t^3 f(t^2, t^3) = t^2 g(t^2, t^3)$. This holds for all $t \neq 0$, so by continuity, it must hold for $t = 0$ as well. Therefore, we have the following equality:

$$Tf(T^2, T^3) = g(T^2, T^3).$$

$T \mid g(T^2, T^3)$, but the smallest power of T appearing is T^2 , so $T^2 \mid g(T^2, T^3)$. However, then $T \mid f(T^2, T^3)$, but then $g(0, 0) = 0$ and $f(0, 0) = 0$, so this is not well-defined at $(0, 0)$. $\square$

Exercise 2.5.31. Consider $C = V(XY - Z^2) \subseteq \mathbb{P}^2$, and the map $\phi: C \dashrightarrow \mathbb{P}^1$ defined by $[X : Y : Z] \mapsto [X : Z]$. Is it rational? Is it regular?

Solution: First, ϕ is clearly rational. If not at the point $[0 : 1 : 0]$, this map is well-defined, so we check regularity at $[0 : 1 : 0]$. We can actually extend ϕ at this point by defining $\phi([x : y : z]) = [z : y]$. Then, we have that

$$\phi([x : y : z]) = \begin{cases} [x : z] & \text{if } x \neq 0, \\ [z : y] & \text{if } y \neq 0. \end{cases}$$

On the intersection, we have that $x, y \neq 0$, so $[x : z] = [xy : zy] = [z^2 : zy] = [z : y]$, so this is regular! $\square$

Exercise 2.5.32. Show that if X is a variety, then giving a rational map from X to $\mathbb{P}_k^1$ is equivalent to giving a rational function on X .

Solution: Let $X \dashrightarrow \mathbb{P}_k^1$ be a rational map represented by a pair (U, f) . Then let us denote by ∞ to be the point at infinity $[1 : 0]$. $f^{-1}(\infty)$ is closed in U , and its complement $V = U \setminus f^{-1}(\infty)$ is an open subset of U . The restriction of f to V takes values in $\mathbb{A}_k^1$, and can thus be regarded as a regular function on V ; then the pair (V, f) is a rational function on X . Conversely, given a rational function on X represented by a pair (V, f) , we can regard f as a map $V \rightarrow \mathbb{A}_k^1$ and compose with the inclusion into $\mathbb{P}_k^1$ to obtain a rational map from X to $\mathbb{P}_k^1$. These two constructions are inverse to each other. $\square$

Exercise 2.5.33. Let $f: \mathbb{P}_k^2 \dashrightarrow \mathbb{P}_k^1$ be a rational map. Show that there exist homogeneous polynomials f_0, f_1 in $k[x, y, z]$, of the same degree, such that f is represented by the pair

(U, g) , where U is the complement of $\tilde{Z}(f_0, f_1)$ in $\mathbb{P}_k^2$, and g is the map $U \rightarrow \mathbb{P}_k^1$ defined by

$$g([x : y : z]) = [f_0(x, y, z) : f_1(x, y, z)].$$

Solution. By the exercise above, a rational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ can be represented by a pair (V, f) , where V is a distinguished open subset $D(h)$ of $\mathbb{A}_k^2 \subset \mathbb{P}_k^2$ and f is a regular function on V . Then, we note that Proposition 1.6.10 tells us that on distinguished opens, we can write rational functions as

$$f = \frac{g}{h^r},$$

where g and h lie in $k[x, y]$. Let f_0 be the homogenization of g with respect to z and f_1 be the homogenization of h^r . Then, multiplying either f_0 or f_1 by a suitable power of z we can arrange for f_0 and f_1 to have the same degree. Then

$$\frac{f_0(x, y, 1)}{f_1(x, y, 1)} = \frac{g(x, y)}{h(x, y)^r} = f(x, y),$$

so our rational function f is represented by the map

$$[x : y : z] \mapsto [f_0(x, y, z) : f_1(x, y, z)]$$

on the complement of the common zero locus of f_0 and f_1 . $\square$

§2.6 Projections from a Point to a Hyperplane

§2.6.i Linear Spaces in $\mathbb{P}^n$

We want to define a fundamental example of a rational map: projection from a point to a hyperplane. First, we need to make a few other definitions.

Definition 2.6.1. A **hyperplane** in $\mathbb{P}^n$ is a projective algebraic subvariety defined by single homogeneous linear equation:

$$H = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : h_0x_0 + \dots + h_nx_n = 0\}$$

for some $h_0, \dots, h_n \in k$, not all zero.

Definition 2.6.2. More generally, a **linear subspace** of $\mathbb{P}^n$ is a subset defined by any set of homogeneous linear equations.

Example 2.6.3

Some example of linear subspaces are $\mathbb{P}^n$ itself (empty set of equations), $\emptyset$ (too many equations), and singletons. We can't define the singleton $\{[p_0 : \dots : p_n]\}$ by the equations $x_0 = p_0, \dots, x_n = p_n$ because these are not homogeneous. Instead, we use homogeneous equations asserting that the ratios between pairs of coordinates are correct:

$$\{[p_0 : \dots : p_n]\} = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : p_i x_j = p_j x_i \text{ for all } i, j\}.$$

Clearly, the point is contained in the set on the right. Conversely, if we have some point in the vanishing locus on the right, then $x_i \neq 0$ for some i , which gives a formula $x_j = \frac{p_j}{p_i} x_i$ for all j .

If Λ is a linear subspace of $\mathbb{P}^n$, then the **affine cone** $C(\Lambda)$ (the set of points in $\mathbb{A}^{n+1}$ satisfying the same equations as Λ) is a vector subspace of k^{n+1} . As a vector space, we know what is meant by $\dim_k C(\Lambda)$. We define

$$\dim \Lambda := \dim_k C(\Lambda) - 1.$$

Remark 2.6.4 — Note that we have not yet defined the dimension of an arbitrary algebraic variety; this definition is only for linear subspaces of projective space. The -1 is because $C(\Lambda)$ contains a line for each point in Λ .

Example 2.6.5

For example, $\mathbb{P}^n$ has dimension n , a hyperplane has dimension $n - 1$, and a point has dimension 0.

Lemma 2.6.6

If $\dim \Lambda = d$, then $\Lambda \cong \mathbb{P}^d$.

Proof. We identify Λ as

$$\Lambda = (C(\Lambda) \setminus \{0\}) / (\text{multiplying by scalars}).$$

This is just $(k^{d+1} \setminus \{0\}) / k^\times \cong \mathbb{P}^d$. □

§2.6.ii Lines in $\mathbb{P}^n$

Definition 2.6.7. A **line** in $\mathbb{P}^n$ is a linear subspace of dimension 1.

Lemma 2.6.8

For any two distinct points $p, q \in \mathbb{P}^n$, there exists a unique line L_{pq} through p and q .

Proof. One could prove this by saying: $\mathbb{P}^n$ can be written as a union $\mathbb{A}^n \cup \mathbb{P}^{n-1}$, and going through the cases $p, q \in \mathbb{A}^n$, $p, q \in \mathbb{P}^{n-1}$, $p \in \mathbb{A}^n$ and $q \in \mathbb{P}^{n-1}$. In order to make this into a full proof, we would need to check that a line in $\mathbb{P}^n$, when intersected with $\mathbb{A}^n$, is the same as the ordinary definition of a line in $\mathbb{A}^n$ (which is indeed true).

Instead we shall give a proof using linear algebra. A benefit of this proof is that it gives a description of the homogeneous coordinates of points in the line L_{pq} . Let $p = [p_0 : \dots : p_n]$ and $q = [q_0 : \dots : q_n]$. The affine cones $C(p)$ and $C(q)$ are one-dimensional vector spaces generated by $(p_0, \dots, p_n)$ and $(q_0, \dots, q_n)$ respectively. Since $p \neq q$, these vector spaces $C(p)$ and $C(q)$ are linearly independent; thus, there is a unique 2-dimensional vector subspace $W \subseteq k^{n+1}$ which contains $C(p)$ and $C(q)$. The image of $W \setminus \{0\}$ in $\mathbb{P}^n$ is the unique line through p and q . Explicitly: W consists of all linear combinations of the vectors $(p_0, \dots, p_n)$ and $(q_0, \dots, q_n)$. It follows that

$$L_{pq} = \{[p_0s + q_0t : \dots : p_ns + q_nt] \in \mathbb{P}^n : [s : t] \in \mathbb{P}^1\}.$$

It is clear that this is isomorphic to $\mathbb{P}^1$, so it is indeed a line. □

§2.6.iii Projections

A fundamental example of a rational map is projection from a point to a hyperplane. Let $p = [p_0 : \dots : p_n] \in \mathbb{P}^n$ and let $H \subseteq \mathbb{P}^n$ be a hyperplane such that $p \notin H$. To simplify the calculations, we shall assume that

$$H = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : x_n = 0\};$$

note that $p_n \neq 0$.

Claim 2.6.9. If $x \neq p$, then the line between p and x meets H at exactly one point.

Geometrically, this is clear; one can prove it algebraically via linear algebra using the affine cones, or by the following calculation: let $x \in \mathbb{P}^n \setminus \{p\}$. Then

$$L_{px} = \{[p_0s + x_0t : \dots : p_ns + x_nt] \in \mathbb{P}^n : [s : t] \in \mathbb{P}^1\}. \quad (2.1)$$

Hence, to find $L_{px} \cap H$, we need to choose $[s : t]$ such that $p_ns + x_nt = 0$; we can choose $[s : t] = [x_n : -p_n]$ (note that $p_n \neq 0$ because $p \notin H$, so we do not get $[0 : 0]$). Substituting into (2.1), the unique point of $L_{px} \cap H$ is

$$[p_0x_n - x_0p_n : \dots : p_{n-1}x_n - x_{n-1}p_n : 0].$$

The final $0 = p_nx_n - x_np_n$ is what we expect for a point in H . Note that, if $p \neq x$, then this is not $[0 : \dots : 0]$ so it is well-defined. Thus, for $x \in \mathbb{P}^n \setminus \{p\}$, it makes sense to define $\pi(x)$ to be the unique point of $L_{px} \cap H$:

$$\pi(x) = [p_0x_n - x_0p_n : \dots : p_{n-1}x_n - x_{n-1}p_n : 0]$$

Definition 2.6.10. The map $\pi(x)$ is called the **projection from p to H** .

The above calculation shows that π is a rational map $\mathbb{P}^n \dashrightarrow H$, regular on $\mathbb{P}^n \setminus \{p\}$. We show below that π is not regular at p .

Lemma 2.6.11

Let $n \geq 2$. The projection of $\mathbb{P}^n$ from p to H is not regular at p .

Proof. Intuitively: there are many lines passing through p and p (not a typo), so the projection would have to map p to ‘everywhere at once’.

We can make this rigorous: pick a point $s \in H$ and consider the line L_{ps} . For any $x \in L_{ps} \setminus \{p\}$, the geometric description of π shows that $\pi(x) = s$. If we assume that π is regular at p , then it restricts to a regular map $L_{ps} \rightarrow H$. We have just shown that this map is constant on $L_{ps} \setminus \{p\}$ and therefore it is constant on L_{ps} . Hence $\pi(p) = s$.

However, since $n \geq 2$, we can pick some other point $t \in H$ with $t \neq s$. This is because $H \cong \mathbb{P}^{n-1}$. Repeating the same argument using L_{pt} , if π is regular at p , then we also have that $\pi(p) = t$. This is a contradiction. $\square$

Remark 2.6.12 — When $n = 1$, then $H\{h\} \in \mathbb{P}^1$, a constant point. Thus, $\pi(x) = h$, which is regular.

Remark 2.6.13 — One could replace this particular fixed H by any hyperplane not containing p , and carry out the same recipe. That is, a useful recipe to show that a rational map ϕ is not regular at p is by restricting ϕ to a simpler geometry; for instance, lines.

Exercise 2.6.14. Let:

$$f(X_0, \dots, X_n) = \sum_{0 \leq i, j \leq n} a_{ij} X_i X_j \in k[X_0, \dots, X_n]$$

be a homogeneous polynomial of degree 2 (where a_{ij} are not all zero). The projective algebraic variety $V \subset \mathbb{P}^n$ defined by the equation $f = 0$ is called a quadric. We aim to show that, if V is irreducible, then it is birational to $\mathbb{P}^{n-1}$.

1. Prove that V is irreducible if and only if V is the power of an irreducible polynomial.
2. Let $p = [0 : \dots : 0 : 1] \in \mathbb{P}^n$. What condition on the coefficients a_{ij} is equivalent to $p \in V$?

From now, we assume that V is nondegenerate, i.e. is defined by an irreducible polynomial, and that $p \in V$.

3. Let H be the hyperplane defined by the equation $X_n = 0$. For each point $x \in H$, let L_x denote the line through x and p . Prove that either $L_x \subset V$ or $L_x \cap V$ contains 1 or 2 points.
4. Let

$$U = \{x \in H : \#(L_x \cap V) = 2\}.$$

By finding equations for $H \setminus U$, prove that U is a Zariski open subset of H .

5. What condition on the coefficients a_{ij} is equivalent to $U \neq \emptyset$? (You may find it helpful to use the fact that H is irreducible.)

From now on, we assume that $U \neq \emptyset$.

6. For $x \in U$, write $L_x \cap V = \{p, \psi(x)\}$. Find the coordinates of the points $\psi(x)$. Conclude that ψ is a rational map $H \dashrightarrow V$.
7. By considering the projection from p to H , deduce that V is birational to H .
8. Determine the domain of ψ . What is $\psi(x)$ for $x \in \text{dom}(\psi) \setminus U$?
9. Apply this to the quadric defined by the equation $XY = WZ$ in $\mathbb{P}^3$: write down the rational map $\psi: H \dashrightarrow V$ and its domain of definition for this case.
10. Let $\pi: V \dashrightarrow H$ denote the projection from p . What is $\text{dom}(\pi)$? The answer is different for $n = 2$ and $n \geq 3$. For $n = 2$, give an algebraic proof and an informal geometric explanation of what is going on. For $n \geq 3$, give either a geometric or an algebraic proof.

Solution: 1. Consider the decomposition of f as a product of powers of irreducible polynomials f_i : because f is homogeneous, each of these polynomials is homogeneous, and then in the primary decomposition

$$(f) = \bigcap_i (f_i),$$

each ideal is homogeneous and not of the form $(X_0, \dots, X_n)$, hence from the projective Nullstellensatz this decomposition corresponds to the decomposition of V into irreducible components.

2. It is $a_{nn} = 0$.

3. The line L_x is the union of the points $[x_0 : \dots : x_{n-1} : t]$ and p . The point parametrized by t is in V if and only if

$$\sum_{i,j=0}^{n-1} a_{ij}x_i x_j + t \sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i = 0. \quad (2.2)$$

It is a polynomial of degree at most 1 in t , hence either all t are solutions, so $L_x \subset V$, or it has zero solution, so $L_x \cap V$ is only the point p , or it has one solution, so $L_x \cap V$ is p plus another solution.

4. By the above reasoning, $H \setminus U$ is then the set of $[x_0 : \dots : x_{n-1}]$ such that (2.2) is of degree zero in t , hence it is given by the equation

$$\sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i = 0,$$

hence U is open in H .

5. $H \setminus U$ is defined by a single linear equation, hence U is nonempty if and only if this equation is not trivial, i.e. if

$$(a_{0n} + a_{n0}, \dots, a_{n-1,n} + a_{n,n-1}) \neq 0.$$

6. Using (2.2), we obtain that $\psi(x)$ is given in coordinates by

$$t = -\frac{\sum_{i,j=0}^{n-1} a_{ij}x_i x_j}{\sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i}.$$

Hence one obtains $\psi: H \dashrightarrow V$ given by

$$[x_0 : \dots : x_{n-1}] \mapsto \left[x_0 \sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i : \dots : x_{n-1} \sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i : - \sum_{i,j=0}^{n-1} a_{ij}x_i x_j \right].$$

This is a rational map.

7. The projection from p to H defines a dominant rational map

$$\pi: V \dashrightarrow H$$

regular on $V \setminus \{p\}$. For $x \in U$, considering that $p, x, \psi(x)$ are aligned, one has that

$$L_{\psi(x)} \cap H = \{x\},$$

hence $\pi \circ \psi(x) = x$. For $x \in V \setminus \{p\}$, $p, x, \pi(x)$ are aligned, hence

$$V \cap L_{\pi(x)} = \{p, x\},$$

hence $\psi \circ \pi(x) = x$. Hence ψ and π are birational inverse of each other.

8. By definition,

$$U \subseteq \text{dom}(\psi).$$

Consider $W \subset H$ defined by the homogeneous equations

$$W = \left\{ [x_0 : \cdots : x_{n-1}] \in H : \sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i = \sum_{i,j=0}^{n-1} a_{ij}x_i x_j = 0 \right\}. \quad (2.3)$$

Then we claim that $\text{dom}(\psi) = H \setminus W$. Indeed, ψ is regular on $H \setminus W$, given by the polynomial expression (2.3). On $\text{dom}(\psi) - U$, $\psi(x) = p$, the idea being that the two solutions p and $\psi(x)$ collide at p on this closed subset.

Now, given the reasoning of 3., the locus W is the locus where $L_x \subset V$. If $n = 2$, this is not possible because V is not degenerate, hence $\text{dom}(\psi) \neq H$ only when $n \geq 3$. On the locus where

$$\sum_{i,j=0}^{n-1} a_{ij}x_i x_j = 0$$

but

$$\sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i \neq 0,$$

one has $\psi(x) = x \in H$, and on the locus where

$$\sum_{i=0}^{n-1} (a_{in} + a_{ni})x_i = 0$$

but

$$\sum_{i,j=0}^{n-1} a_{ij}x_i x_j \neq 0,$$

$\psi(x) = p \notin H$, hence by continuity ψ cannot be regular on W .

9. In this case, H is defined by the equation $Z = 0$, and U is defined in H by the equation $W \neq 0$. Then the map ψ is defined by

$$\psi: H \dashrightarrow V$$

$$[w : x : y] \mapsto [w^2 : wx : wy : xy].$$

10. We have that π , as a projection, is regular on $V - \{p\}$, the only question remaining is if it is regular at p .

For $n = 2$, H is isomorphic to $\mathbb{P}^1$, and $H - U$ is defined by a single nontrivial linear equation, hence is a single point that we denote by y . If we extend π by $\pi(p) = y$, then it is regular. Indeed, the quadratic polynomial can be written

$$F(X, Y, Z) = aX^2 + bY^2 + dXY + eXZ + fYZ.$$

By changing coordinates on the coordinates X, Y , we can always assume that $a = 0$ and $e \neq 0$. Hence, on the open set $\mathbb{A}^2$, one has $[x : y : 1] \in V$ if and only if

$$by^2 + dxy + ex + fy = 0,$$

and then, on the open subset $\mathbb{A}^2 \cap V - \{p\}$, if $y \neq 0$, and

$$\pi([x : y : 1]) = [x : y] = [ex : ey] = [-by^2 - dxy - fy : ey] = [-f - by - dx : e].$$

And, in these coordinates, $\pi(p) = \pi([0 : 0 : 1]) = y = [-f : e]$, hence the above expression gives a polynomial expression of π on the open neighborhood $\mathbb{A}^2 \cap V$ of p , i.e. π is regular at p . In particular, π and ψ provide an isomorphism between V and $\mathbb{P}^1$, hence any nondegenerate conic is isomorphic to $\mathbb{P}^1$.

Suppose now that $n \geq 3$. We show now that π is not regular at p . In particular, it means that any nondegenerate quadric is birational, but not a priori isomorphic, to $\mathbb{P}^{n-1}$. Now, for any $x \in W$, we have that $L_x \subset V$, and contains x and p : it means that for any $y \in L_x - p$, $\pi(y) = x$. Then, by continuity, if π were regular at p , one would have $\pi(p) = y$. But W is the intersection of a hyperplane with a quadric in H , hence is a quadric in $\mathbb{P}^{n-2}$ for $n-2 \geq 1$, hence must contain at least 2 points, hence the value of $\pi(p)$ cannot be well-defined. $\square$

§2.7 Products of Quasi Projective Varieties

Recall that if V, W are affine varieties, then $V \times W$ is also an affine variety. The key here is the isomorphism $\mathbb{A}^m \times \mathbb{A}^n \cong \mathbb{A}^{m+n}$. We want to do a similar construction for projective or quasi projective varieties.

For projective algebraic varieties, things are harder because $\mathbb{P}^m \times \mathbb{P}^n \not\cong \mathbb{P}^{m+n}$. To see informally why $\mathbb{P}^1 \times \mathbb{P}^1 \not\cong \mathbb{P}^2$, recall that $\mathbb{P}^1 \cong \mathbb{A}^1 \cup \{\text{point}\}$ so

$$\begin{aligned} \mathbb{P}^1 \times \mathbb{P}^1 &\cong (\mathbb{A}^1 \times \mathbb{A}^1) \cup (\mathbb{A}^1 \times \{\text{point}\}) \cup (\{\text{point}\} \times \mathbb{A}^1) \cup (\{\text{point}\} \times \{\text{point}\}) \\ &\cong \mathbb{A}^2 \cup \mathbb{A}^1 \cup \mathbb{A}^1 \cup \{\text{point}\}. \end{aligned}$$

Meanwhile

$$\mathbb{P}^2 = \mathbb{A}^2 \cup \mathbb{P}^1 = \mathbb{A}^2 \cup \mathbb{A}^1 \cup \{\text{point}\}.$$

Thus, informally, $\mathbb{P}^1 \times \mathbb{P}^1$ contains an extra copy of $\mathbb{A}^1$ compared to $\mathbb{P}^2$.

We could try giving an ad hoc definition for $\mathbb{P}^m \times \mathbb{P}^n$. It is fairly clear what algebraic subsets of $\mathbb{P}^m \times \mathbb{P}^n$ should mean (sets defined by polynomials in the two sets of homogeneous coordinates $[x_0 : \dots : x_m]$, $[y_0 : \dots : y_n]$; in order for the zero set of such a polynomial to be well-defined, it must be **bihomogeneous**, that is, homogeneous in the x variables and homogeneous in the y variables, but potentially of different degrees in x and y). Similarly, we could give a definition of regular maps between subvarieties of $\mathbb{P}^m \times \mathbb{P}^n$ involving bihomogeneous polynomials. But it would be annoying to have just defined quasi-projective varieties, unifying affine and projective varieties, and then immediately have to introduce ad hoc definitions for another different kind of variety. So we aim to construct the product in a way which makes it a (quasi-)projective variety, and then we can just reuse the definitions from before.

§2.7.i The Segre Embedding

To construct the product $\mathbb{P}^m \times \mathbb{P}^n$ as a projective algebraic subvariety, we embed it inside some larger $\mathbb{P}^N$. The homogeneous coordinates of a point in $\mathbb{P}^m \times \mathbb{P}^n$ will be given by an $(m+1) \times (n+1)$ matrix, so we need

$$N = (m+1)(n+1) - 1 = mn + m + n.$$

Label the homogeneous coordinates of a point in $\mathbb{P}^N$ as if they were entries of a matrix

$$[(z_{ij} : 0 \leq i \leq m, 0 \leq j \leq n)],$$

rather than the usual $[z_0 : \cdots : z_N]$.

Define a map $\sigma_{m,n} : \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^N$ defined by

$$([x_0 : \cdots : x_m], [y_0 : \cdots : y_n]) \mapsto (z_{ij}),$$

where $z_{ij} = x_i y_j$, for each pair of indices i, j . In other words, the homogeneous coordinates of

$$\sigma_{m,n}([x_0 : \cdots : x_m], [y_0 : \cdots : y_n])$$

are given by the product matrix

$$\begin{pmatrix} x_0 \\ \vdots \\ x_m \end{pmatrix} \begin{pmatrix} y_0 & \cdots & y_n \end{pmatrix}.$$

We shall note that this matrix has rank 1, since all of its columns are a multiple of one, and at least one of these columns is nonzero. In fact, we can attain all rank 1 matrices in this manner, since to define such a matrix, we need one nonzero column (and scalar multiply it). Define

$$\Sigma_{m,n} := \{[z_{00} : \cdots : z_{mn}] \in \mathbb{P}^N : \text{the matrix } (z_{ij}) \text{ has rank } 1\}.$$

Recall that a matrix has rank r if and only if it has a nonzero $r \times r$ -minor and all $(r+1) \times (r+1)$ -minors are zero (to clarify, when we say ‘the minor’, we are talking about the determinant of the minor). For $\Sigma_{m,n}$, the conditions transfers to be that all 2×2 minors to be zero. Therefore, we can describe $\Sigma_{m,n}$ as the vanishing locus of

$$z_{ij} z_{kl} = z_{kj} z_{il},$$

for all $0 \leq i, k \leq m, 0 \leq j, l \leq n$, so $\Sigma_{m,n}$ is a variety.

Lemma 2.7.1

$\sigma_{m,n}$ is a bijection from $\mathbb{P}^m \times \mathbb{P}^n$ to $\Sigma_{m,n}$.

Proof. We can define an inverse to $\sigma_{m,n}$ as follows. Let $a \in \Sigma_{m,n}$, and let A be a matrix giving homogeneous coordinates for a . A is not the zero matrix (because it is a set of homogeneous coordinates), so we can pick j such that the j -th column of A contains a non-zero entry. Define

$$\pi_1(a) \in \mathbb{P}^m$$

to be the point with homogeneous coordinates given by the j -th column of A , that is,

$$\pi_1(a) = [A_{0j} : \cdots : A_{mj}].$$

This is independent of the choice of j because the matrix has rank 1 (every non-zero column is a multiple of every other non-zero column).

Similarly we can pick i such that the i -th row of A contains a non-zero entry, and define

$$\pi_2(a) \in \mathbb{P}^n$$

to be the point with homogeneous coordinates given by the i -th row of A . Again this is independent of the choice of i .

Now $(\pi_1, \pi_2) : \Sigma_{m,n} \rightarrow \mathbb{P}^m \times \mathbb{P}^n$ is an inverse to $\sigma_{m,n}$. (proof is nonexaminiabile) $\square$

Remark 2.7.2 — Note that the maps π_i are regular. This depends locally on sets of the form $\{[z] \in \mathbb{P}^n : \text{column/ row of } z \text{ is nonzero}\}$. For instance, projection onto the first column is an expression of the closed subset $\{\text{first column is zero}\}$.

Corollary 2.7.3

$\mathbb{P}^n \times \mathbb{P}^m$ can be identified with a projective subvariety $\Sigma_{m,n}$ of $\mathbb{P}^{(m+1)(n+1)-1}$.

Definition 2.7.4. The map

$$\sigma_{m,n} : \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^N$$

is called the **Segre embedding**, and its image $\Sigma_{m,n} \subset \mathbb{P}^N$ is called the **Segre variety**.

The Zariski topology on $\mathbb{P}^N$ induces a subspace topology on $\Sigma_{m,n}$. One can check that this topology is what we expect, namely:

Lemma 2.7.5

Let $V \subset \mathbb{P}^m \times \mathbb{P}^n$. Then $\sigma_{m,n}(V) \subset \Sigma_{m,n}$ is closed if and only if

$$V = \{([x_0 : \cdots : x_m], [y_0 : \cdots : y_n]) : f_i(x_0, \dots, x_m, y_0, \dots, y_n) = 0 \text{ for } 1 \leq i \leq s\},$$

where $f_1, \dots, f_s \in k[X_0, \dots, X_m, Y_0, \dots, Y_n]$ are bihomogeneous polynomials; that is, homogeneous in x and y separately.

Proof. Suppose that $\sigma_{m,n}(V)$ is Zariski closed in $\mathbb{P}^N$. Then it is defined by some homogeneous polynomials $g_r(z_{00}, \dots, z_{mn})$. Making the substitutions $z_{ij} = x_i y_j$ (as in the definition of $\sigma_{m,n}$), we get a finite set of polynomials which define V . If g_r is homogeneous in z_{ij} of degree d_r , then $g_r \circ \sigma_{m,n}$ is bihomogeneous of degree (d_r, d_r) .

It is easy to see that if V is defined by polynomials f_r , where f_r is bihomogeneous of degree (d_r, d_r) , then we can reverse this process to get homogeneous polynomials in z_{ij} which define $\sigma_{m,n}(V)$.

But what if the defining polynomials for V include some f which is bihomogeneous of degree (d, e) , where $d \neq e$? Without loss of generality, suppose that $d > e$. Then $f = 0$ is equivalent to the system of equations

$$y_0^{d-e} f = 0, \quad \dots, \quad y_n^{d-e} f = 0,$$

because we cannot have all y_j to be zero. These equations are bihomogeneous of degree (d, d) , and so we have reduced to the previous case. (This proof is unexaminiabile) $\square$

If $V \subseteq \mathbb{P}^m$ and $W \subseteq \mathbb{P}^n$ are projective algebraic subvarieties, then $V \times W \subseteq \mathbb{P}^m \times \mathbb{P}^n$ is Zariski closed: the homogeneous polynomials defining V become bihomogeneous polynomials of bidegree $(d, 0)$, while those defining W become bihomogeneous polynomials of bidegree $(0, e)$.

Similarly, if $V \subseteq \mathbb{P}^m$ and $W \subseteq \mathbb{P}^n$ are quasi-projective algebraic subvarieties, then the product $V \times W$ is also quasi-projective in $\Sigma_{m,n} \subseteq \mathbb{P}^{(m+1)(n+1)-1}$.

Remark 2.7.6 — Note that the Zariski topology on the product (induced by the Segre embedding) is not the same as the product topology, and is much finer.

Example 2.7.7

$\mathbb{A}^n \times \mathbb{A}^m$ is open in $\mathbb{P}^n \times \mathbb{P}^m$.

Proof. We note that $\mathbb{A}^n \times \mathbb{P}^m$ is open in $\mathbb{P}^n \times \mathbb{P}^m$ because it is the complement of $V(X_0) \times \mathbb{P}^m$, which is closed. Then, $\mathbb{A}^n \times \mathbb{A}^m$ is the intersection of $\mathbb{A}^n \times \mathbb{P}^m$ and $\mathbb{P}^n \times \mathbb{A}^m$. $\square$

Example 2.7.8

When $m = n = 1$, $N = 3$. The Segre variety $\Sigma_{1,1} \subset \mathbb{P}^3$ is defined by the single equation

$$\det \begin{pmatrix} z_{00} & z_{01} \\ z_{10} & z_{11} \end{pmatrix} = z_{00}z_{11} - z_{10}z_{01} = 0.$$

The Segre embedding is given by

$$\sigma_{m,n}([x_0 : x_1], [y_0 : y_1]) = [x_0y_0 : x_0y_1 : x_1y_0 : x_1y_1].$$

We see that $\Sigma_{m,n}$ is an irreducible quadric hypersurface in $\mathbb{P}^3$. Therefore, it is birational to $\mathbb{P}^2$ (by a question in problem sheet 3). This is not surprising, because of course $\mathbb{P}^1 \times \mathbb{P}^1$ should have an open subset isomorphic to $\mathbb{A}^1 \times \mathbb{A}^1 \cong \mathbb{A}^2$, which in turn is a subset of $\mathbb{P}^2$. We note that this is a *doubly ruled surface*.

One useful example of a subvariety of a product is the graph of a regular function.

Example 2.7.9 (Graphs of Regular Functions)

Let $V \subset \mathbb{P}^n$ and $W \subset \mathbb{P}^m$ be quasi-projective varieties, and let $\varphi: V \rightarrow W$ be a regular map. The **graph** of φ is

$$\Gamma_\varphi = \{(x, y) \in V \times W : y = \varphi(x)\}.$$

We claim that Γ_φ is closed in $V \times W$; thus, a quasi-projective variety in its own right.

Proof. We prove the claim made in the above example. We observe that Γ_φ is the preimage of the diagonal $\Delta \subset \mathbb{P}^m \times \mathbb{P}^m$ under the regular map

$$(\iota \circ \varphi, \iota): V \times W \longrightarrow \mathbb{P}^m \times \mathbb{P}^m,$$

where $\iota: W \hookrightarrow \mathbb{P}^m$ denotes the inclusion map. If ψ denotes this map, then we see that

$$\psi^{-1}(\Delta) = \{(x, y): \iota\varphi(x) = \iota(y)\} = \{(x, y): \varphi(x) = y\} = \Gamma_\varphi.$$

Since $\psi := (\iota \circ \varphi, \iota)$ is a regular map, it is continuous. Therefore it suffices to check that the diagonal is a Zariski closed subset of $\mathbb{P}^m \times \mathbb{P}^m$. This is true because we can describe the diagonal by bihomogeneous equations as follows:

$$\Delta = \{([x_0 : \cdots : x_m], [y_0 : \cdots : y_m]): x_i y_j = x_j y_i \text{ for all } i, j\}.$$

□

Remark 2.7.10 — We note that Γ_f is isomorphic to V via projection onto the first factor.

Exercise 2.7.11. Let X be a quasiprojective variety and $h: X \rightarrow X$ be a regular map. Prove that the fixed point locus $\{x \in X: h(x) = x\}$ is closed.

Solution: We note that the fixed point locus is the preimage of the diagonal $\Delta \subset \mathbb{P}^n \times \mathbb{P}^n$ of the regular map $F: X \rightarrow X \times X$ defined by $(x, h(x))$. That is,

$$\{x \in X: h(x) = x\} = F^{-1}(\Delta).$$

Since F is regular, we have that $F^{-1}(\Delta)$ is closed, since the diagonal Δ is closed. □

Exercise 2.7.12. A quadric in $\mathbb{P}^2$ is called a conic. Let:

$$f(X, Y, Z) = aX^2 + bY^2 + cZ^2 + dXY + eXZ + fYZ \in k[X, Y, Z]$$

where a, b, c, d, e, f are not all zero. Let $V \subset \mathbb{P}^2$ be the conic defined by f .

1. Prove that two quadratic polynomials f and g define the same conic if and only if $f = \lambda g$ for some $\lambda \in k \setminus \{0\}$.
2. For any point $p \in \mathbb{P}^2$, prove that the set of conics which contains p corresponds to a hyperplane of $\mathbb{P}^5$.
3. Prove that, given any 5 points in $\mathbb{P}^2$, there exist a conic passing through all 5 points.
4. Find conditions on 5 points in $\mathbb{P}^2$ which are equivalent to:
 - a) There exists a nondegenerate conic passing through 5 points.
 - b) There exist a unique conic passing through the 5 points.

Solution: 1. The projective Nullstellensatz says that $V(f) = V(g)$ if and only if $(f) = (g)$, so we are done.

2. Writing $p = [x : y : z] \in \mathbb{P}^2$, we have that $p \in V$ if and only if

$$ax^2 + by^2 + cz^2 + dxy + exz + fyz = 0.$$

This last equation is a hyperplane in (a, b, c, d, e, f) .

3. From the last question, the set of conics passing through five given points in $\mathbb{P}^2$ is a projective subspace of $\mathbb{P}^5$ given by five linear equations, hence it is nontrivial.

4a. It is the case if and only if any 3 points are not aligned. Indeed, if the conic is degenerate, it is a line or a union of two lines, and each of the five points have to lie on at least one of those lines, hence at least 3 points are aligned.

Otherwise, if 3 points are aligned, consider the line $\phi: \mathbb{P}^1 \rightarrow \mathbb{P}^2$ passing by these 3 points. Then $f \circ \phi \in k[S, T]$ is an homogeneous polynomial of degree at most 2 whose zero set gives at least 3 points of $\mathbb{P}^1$, hence it vanishes, then the whole line is contained in the quadric. Denoting by g the homogeneous polynomial of degree 1 defining the line, and using the projective Nullstellensatz, one has $f \in (g)$, hence f is not irreducible, hence the conic is degenerate.

4b. The condition is that any 4 points are not aligned. Indeed, if 4 points are aligned, then the line passing through the 4 points, and any other line passing through the fifth point, is a conic passing through the 5 points, hence there are infinitely many solutions.

Otherwise, we can find 3 points which are not aligned, and consider coordinates such that these points are $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $C = [0 : 0 : 1]$. If a conic passes through these 3 points, we have $a = b = c = 0$, hence it is defined by:

$$f(X, Y, Z) = dXY + eXZ + fYZ \in k[X, Y, Z]$$

we have then a $\mathbb{P}^2$ of other solutions, and given two other points $D = [x : y : z]$ and $E = [x' : y' : z']$, they define a matrix:

$$\begin{pmatrix} xy & xz & yz \\ x'y' & x'z' & y'z' \end{pmatrix}$$

there is more than one solution if and only if this matrix is of rank ≤ 1 , which means, using 2×2 submatrices:

$$\begin{cases} xx'(yz' - y'z) = 0, \\ yy'(zx' - z'x) = 0, \\ zz'(xy' - x'y) = 0. \end{cases}$$

Then $x = 0$ means that B, C, D are aligned, $x' = 0$ means that B, C, E are aligned, and $yz' - y'z = 0$ means that A, D, E are aligned, and circular permutations; in particular, all of those constraints concern different sets of 3 points. Suppose by contradiction that there is more than one solution: hence, at least two different constraints have to hold, hence two different sets of 3 points are aligned, which means that there is at least 4 points aligned. $\square$

Exercise 2.7.13. Prove that the product of two irreducible algebraic sets is irreducible.

Solution: This follows analogously to the affine case. We shall just run through the argument again. Let V, W be irreducible quasi projective varieties. Suppose that $V \times W = Z_1 \cup Z_2$, with $Z_i \subset V \times W$ closed. For any $x \in V$, we have $\{x\} \times W$ is isomorphic to W as a topological space, hence irreducible. But

$$\{x\} \times W = (\{x\} \times W \cap Z_1) \cup (\{x\} \times W \cap Z_2)$$

is a decomposition into two closed subsets, hence by irreducibility of W , for each $x \in V$, there is $i \in \{1, 2\}$ such that $\{x\} \times W \subset Z_i$. Then

$$V = \{x \in V : \{x\} \times W \subset Z_1\} \cup \{x \in V : \{x\} \times W \subset Z_2\},$$

and the two subsets are closed. To see that these sets are closed, we use the projection map $\pi_V: V \times W \rightarrow V$. We have that

$$V \setminus \{x \in V: \{x\} \times W \subset Z_i\} = \pi_V((V \times W) \setminus Z_i).$$

Noting that $(V \times W) \setminus Z_i$ is open (because Z_i is closed) and that π_V is an open map, we have that $V \setminus \{x \in V: \{x\} \times W \subset Z_i\}$ is open, so we have proven true. To finish, the irreducibility of V tells us that

$$V = \{x \in V: \{x\} \times W \subset Z_i\}$$

for some $i \in \{1, 2\}$; thus, $V \times W = Z_i$. $\square$

Exercise 2.7.14. Let $f: X \rightarrow Y$ be a regular map between varieties. Suppose that X is quasi-projective. Let $\sigma: Y \rightarrow X$ be a regular map such that $f \circ \sigma = \text{Id}_Y$ such a map is called a **section** of f . Show that $\sigma(Y)$ is closed in X .

Solution: By construction, we have

$$\sigma(Y) = \{x \in X: (\sigma \circ f)(x) = x\}.$$

Let $\delta_X: X \rightarrow X \times X$ (respectively $\Delta_X \subseteq X \times X$) be the diagonal map (respectively the diagonal) of X . Let $\Gamma_{\sigma \circ f} \subseteq X \times X$ be the graph of $\sigma \circ f$. By the above, we have

$$\sigma(Y) = \delta_X^{-1}(\Delta_X \cap \Gamma_{\sigma \circ f})$$

and this set is closed because Δ_X and $\Gamma_{\sigma \circ f}$ are closed. Note that the fact that these two are closed come from X being quasi projective. $\square$

§2.8 Complete Varieties

We begin with a remark on compactness.

Remark 2.8.1 — Over the complex numbers, every projective algebraic variety is compact in the analytic topology. This is because they are closed subsets of $\mathbb{P}_{\mathbb{C}}^n$, which is compact.

In the Zariski topology, the notion of compactness is not very interesting: every algebraic variety is compact in the Zariski topology, even affine algebraic varieties. Affine algebraic varieties do not behave in ways matching our intuition about compactness: this intuition only works in Hausdorff spaces.

There is a converse to this, which tells us that there is a very close relationship between analytic and algebraic geometry in $\mathbb{P}_{\mathbb{C}}^n$:

Theorem 2.8.2 (Chow's theorem)

Let V be an analytic subset of $\mathbb{P}_{\mathbb{C}}^n$ which is closed in the analytic topology. Then V is a projective algebraic subvariety.

We won't define analytic subsets here, but roughly it means a set defined by zeroes of holomorphic functions. This theorem requires too much complex analytic geometry to

prove here.

One can prove analytically that every holomorphic function on a connected compact complex manifold is constant (for example, this holds on the Riemann sphere, which is equal to $\mathbb{P}_{\mathbb{C}}^1$). Polynomials are holomorphic, so every regular function on a connected projective algebraic variety over $\mathbb{C}$ is constant. Once we define regular functions on projective algebraic varieties, it will turn out that the same is true over any field.

Recall that we have seen that the image of a closed subset under a regular map need not be closed.

Definition 2.8.3. A map $f: X \rightarrow Y$ of topological spaces is **closed** if $f(Z)$ is closed for every $Z \subseteq X$ closed.

A stronger condition on maps is that of being universally closed, also known as proper:

Definition 2.8.4. A map $f: X \rightarrow Y$ of topological spaces is **proper** (or **universally closed**) if for any space W , the map

$$f \times \text{id}_W: X \times W \rightarrow Y \times W,$$

is closed.

Remark 2.8.5 — It is almost immediate that proper maps are closed; for this, just take W to be a one point set or the empty space, so that $X \times W \cong X$ and $Y \times W \cong Y$. The converse is not true at all, for being proper is much stronger than being closed.

Of course, in the case of varieties, the definitions above become regular maps with all other parts the same.

Example 2.8.6

Indeed, the *constant* regular map $f: \mathbb{A}^1 \rightarrow \{p\}$ is certainly closed (as every subset of a one-point topological space is closed), but it is not proper; for example, the product

$$f \times \text{id}: \mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \{p\} \times \mathbb{A}_k^1 \cong \mathbb{A}^1$$

is not closed. Indeed, let W be the closed subset of $\mathbb{A}^1 \times \mathbb{A}^1$ defined by the equation $XY = 1$; then

$$(f \times \text{id})(W) = \{p\} \times (\mathbb{A}_k^1 \setminus \{0\}),$$

which is not closed in $\{p\} \times \mathbb{A}_k^1$.

Having seen this example, one might reasonably wonder whether any map of varieties is proper. In fact we will see that this is rather common, and that any map of projective varieties is proper (and thus, in particular, closed). We first introduce an auxiliary notion that is rather useful.

Definition 2.8.7. A quasi-projective variety V is **complete** (or **proper**) if the projection (really the constant map)

$$V \rightarrow \{p\}$$

(where $\{p\}$ is a single point, considered as a variety over k) is proper, i.e. for any quasi-projective variety W , the second projection

$$p_2: V \times W \rightarrow W$$

is closed.

Example 2.8.8

We have seen that $\mathbb{A}^1$ is not complete.

Example 2.8.9

Analogously, $\mathbb{A}^n$ is not complete. We will prove later that affine varieties are not complete unless they are a finite union of points.

Remark 2.8.10 — We will see that completeness of varieties is a notion that is (roughly) analogous to compactness for manifolds. We note again that compactness itself is not a very useful notion for varieties as every variety is compact; thus the usual intuition of a compact object being ‘no holes’ fails in algebraic geometry. Instead, the correct intuition is that a complete variety is one with ‘no holes’.

We will eventually prove the below theorem:

Theorem 2.8.11

A quasi-projective variety is complete precisely when it’s a projective variety.

Notice that if we go beyond the world of quasi-projective varieties (we have not defined non-quasi-projective varieties at all in this course) then it is possible to find algebraic varieties which are complete but not projective. We remark that, over the complex numbers, an algebraic variety is complete if and only if it is compact for the analytic topology (this is hard to prove).

§2.8.i Formal Properties of Proper Maps

Before proving completeness of projective varieties, we develop several formal properties associated with the notions of completeness and properness. The claims here all follow more-or-less directly from the definitions.

Lemma 2.8.12

Let $i: V \rightarrow W$ be a closed embedding (i.e. an injection which is closed). Then i is proper.

Proof. If i is a closed embedding then it is clearly closed, as it is a homeomorphism onto its image and its image is closed. But for any variety Z , $i \times \text{id}: V \times Z \rightarrow W \times Z$ is also a closed embedding, hence is closed. So i is proper. $\square$

Lemma 2.8.13

Compositions of proper maps are proper.

Proof. This follows because compositions of closed maps are clearly closed. $\square$

From the previous two lemmas, we deduce

Corollary 2.8.14

A closed subvariety of a complete variety is complete.

Proof. Let W be complete and V be closed in W . Then the map $V \rightarrow \{p\}$ is the composition of the inclusion of V in W (a closed embedding) and the map $W \rightarrow \{p\}$, which is proper, hence proper by the previous lemma. $\square$

Lemma 2.8.15

Let $f: V \rightarrow W$ be a proper map of quasi projective varieties, and let y be a point of W . Then $f^{-1}(y)$ is a complete variety.

Proof. Let Z be a quasi-projective variety, and let Y be a closed subset of $f^{-1}(y) \times Z$. We must show that the image of Y under the projection to Z is closed. But this image coincides with the image of Y under the map $V \times Z \rightarrow W \times Z$ which is closed since f is proper. $\square$

Lemma 2.8.16

Let $f: V \rightarrow W$ be a proper map. Then for any Z , the map $f \times \text{id}: V \times Z \rightarrow W \times Z$ is proper.

Proof. For any variety Y , the map

$$f \times \text{id}: V \times (Z \times Y) \rightarrow W \times (Z \times Y)$$

is closed. The result follows by shifting around the parenthesis:

$$(f \times \text{id}_Z) \times \text{id}_Y = f \times (\text{id}_{Z \times Y}).$$

This is closed since f was proper. $\square$

Corollary 2.8.17

Any product of complete varieties is complete.

Proof. If V is complete, then $V \rightarrow \{p\}$ is proper, hence from the last lemma $V \times W \rightarrow W$ is proper. Composing this with the proper map $W \rightarrow \{p\}$, we find that $V \times W$ is complete. $\square$

Proposition 2.8.18

Let V be a complete variety and $f: V \rightarrow W$ a surjective regular map. Then W is complete.

Proof. Let Z be a variety, and Y a closed subset of $W \times Z$. We must show that its image under the projection to Z is closed. On the other hand, let $Y' = (f \times \text{id}_Z)^{-1}(Y) \subseteq V \times Z$ be the preimage of Y in $V \times Z$. Then the projection of Y' to Z is closed since V is complete. But since f is surjective, the projection of Y' to Z coincides with the projection of Y to Z , which is hence closed. $\square$

§2.8.ii Images of Complete Varieties

Remark 2.8.19 — Every holomorphic function on a connected compact complex manifold is constant (for example $\mathbb{P}_{\mathbb{C}}^n$). Polynomials are holomorphic, so every regular function on a connected projective algebraic variety over $\mathbb{C}$ is constant. The same is true over arbitrary fields. This is true more generally for complete varieties. We note that there are more general varieties than the quasi-projective ones; these contain examples of things that are complete but not projective

Proposition 2.8.20

If V is complete, and $f: V \rightarrow W$ is any regular map of quasiprojective varieties, then f is proper. In particular, the image of V under any regular map is closed.

Proof. Consider the graph

$$\Gamma_f := \{(x, y) : y = f(x)\};$$

we have shown that it is a closed subvariety of $V \times W$, isomorphic to V . The map

$$V \rightarrow V \times W, \quad x \mapsto (x, f(x))$$

is an isomorphism on its closed image Γ_f (with inverse given by projection onto the first factor), hence it is a closed embedding, so it is proper. On the other hand the projection

$$V \times W \rightarrow W$$

is proper because by Lemma 2.8.16, since V is complete. The composition of these two maps is the map $f: V \rightarrow W$, so f is proper. $\square$

Remark 2.8.21 — We note that this is a useful proof technique, to use a regular map to embed inside a product, then project onto a factor.

A Hausdorff topological space is compact if and only if its image under any continuous map is closed, so the above proposition shows that completeness is a good replacement for compactness in the (very non Hausdorff) Zariski topology. We can now prove half of Theorem 2.8.11:

Corollary 2.8.22

A complete quasi-projective variety is projective.

Proof. Any quasi-projective variety V comes with an embedding $i: V \rightarrow \mathbb{P}^n$ for some n . If V is complete, then the previous proposition tells us that the image of i is closed, so $i(V)$ is closed in $\mathbb{P}^n$. Thus, V is a projective variety. $\square$

Proving that a projective variety is complete is harder and we will return to it later.

Corollary 2.8.23

Every regular function on a connected complete variety is constant.

Proof. Let V be a connected complete variety and $\varphi: V \rightarrow \mathbb{A}^1$ a regular function. Let $\iota: \mathbb{A}^1 \rightarrow \mathbb{P}^1$ be the natural inclusion. Then $\iota \circ \varphi: V \rightarrow \mathbb{P}^1$ is a regular map, so by Proposition 2.8.20, its image is a closed subset of $\mathbb{P}^1$. But the image of $\iota \circ \varphi$ is contained in $\mathbb{A}^1$, so it cannot be all of $\mathbb{P}^1$. Therefore the image of φ is finite (say by the fundamental theorem of algebra). Since V is connected, its image is also connected and therefore consists of a single point. $\square$

Remark 2.8.24 — In the above, we used that closed subsets of $\mathbb{P}^1$ are finite subsets.

Thus complete (and then, using the theorem, projective) algebraic varieties are essentially ‘opposite’ to affine ones, since an affine algebraic variety is determined by its ring of regular functions while a projective algebraic variety has no regular functions except constants.

Corollary 2.8.25

The image of a regular map from a connected complete variety to an affine variety is a point.

Proof. Suppose we have a regular map $\varphi: V \rightarrow W$, where V is projective and connected and W is affine. We can suppose that $W \subseteq \mathbb{A}^m$, and let $X_1, \dots, X_m$ denote the coordinate functions on W . Then $X_1 \circ \varphi, \dots, X_m \circ \varphi$ are all regular functions on a connected complete variety; thus, constant by Corollary 2.8.23. Therefore, φ is constant. $\square$

Lemma 2.8.26

Let $V \subseteq \mathbb{P}^n$ be an infinite complete variety and let $H \subseteq \mathbb{P}^n$ be a hyperplane. Then the intersection $V \cap H$ is non-empty.

Proof. Suppose for contradiction that $V \cap H = \emptyset$. Then $V \subseteq \mathbb{P}^n \setminus H \cong \mathbb{A}^n$. Hence we get an injective regular map $\iota: V \rightarrow \mathbb{A}^n$. By Corollary 2.8.25, ι is constant on each irreducible component of V . Since ι is injective, each irreducible component of V is a point. But V has only finitely many irreducible components, so this contradicts the hypothesis that V is infinite. $\square$

§2.8.iii Affine Open Covers

By definition, every quasi-projective algebraic variety is contained in a projective algebraic variety. We can use this to reduce some proofs for quasi-projective algebraic varieties to the projective case (proving from the outside in). On the other hand, it is often useful to know that we can find affine varieties as open sets inside each quasi-projective algebraic variety. This can be used to reduce some proofs to the affine case (proving from the inside out).

Lemma 2.8.27

Let V be a quasi-projective variety. For every point $x \in V$, there exists an open set $U \subseteq V$ which contains x and is (isomorphic to) an affine variety.

Proof. Write $V = V_0 \cap U_0$ where $V_0 \subseteq \mathbb{P}^n$ is closed and $U_0 \subseteq \mathbb{P}^n$ is open. Given a point $x \in V$, we may assume that x is in $\mathbb{A}^n \subseteq \mathbb{P}^n$ (embedded by setting $X_0 = 1$; we can always achieve this by changing the coordinate system if necessary). Since $\mathbb{P}^n \setminus U_0$ is a projective algebraic variety which does not contain x , there is some homogeneous polynomial f which vanishes on $\mathbb{P}^n \setminus U_0$ but not at x . Then x is contained in the set

$$U := V_0 \cap D(f) = V \cap D(f),$$

where

$$D(f) = \{y \in \mathbb{A}^n : f(1, y_1, \dots, y_n) \neq 0\}.$$

Note that we have $V_0 \cap D(f) = V \cap D(f)$ because $D(f) \subseteq U_0$. U is an open subset of V because $D(f)$ is an open subset of $\mathbb{P}^n$. U is a closed subset of $D(f)$, so in order to show that U is an affine variety, it suffices to show that $D(f)$ is an affine variety. We can prove this using the ‘hyperbola trick’: consider the set

$$E(f) = \{(y_1, \dots, y_n, z) \in \mathbb{A}^{n+1} : z \cdot f(1, y_1, \dots, y_n) = 1\}.$$

The claim is that $E(f) \cong D(f)$. The way we prove this is by considering the projection map $\pi: E(f) \rightarrow D(f)$ defined by $(y_1, \dots, y_n, z) \mapsto (y_1, \dots, y_n)$. π is clearly regular. The inverse is ϕ , and is given by $\phi: D(f) \rightarrow E(f)$ defined by

$$(y_1, \dots, y_n) \mapsto \left(y_1, \dots, y_n, \frac{1}{f(1, y_1, \dots, y_n)} \right).$$

ϕ is clearly regular. Clearly, these are inverses to each other so we have the desired isomorphism. $\square$

§2.8.iv Proof of Completeness

We will now prove the completeness of projective varieties (Theorem 2.8.11, which we recall here for convenience):

Theorem 2.8.28

Let V be a projective variety. For any quasiprojective variety W , the projection map $p_2: V \times W \rightarrow W$ is closed. Thus, V is complete.

Proof. We can cover W by affine varieties U_α . Take $Z \subseteq V \times W$ closed. To show that $p_2(Z) \subseteq W$ is closed, it suffices to show that each $p_2(Z) \cap U_\alpha$ is closed in U_α , for every α (by the topological fact from the proof of Lemma 2.4.17). In other words (replacing W by U_α), it suffices to prove Theorem 2.8.28 for the case where W is affine.

Then we can replace $V \subseteq \mathbb{P}^m$ by $\mathbb{P}^m$ and $W \subseteq \mathbb{A}^n$ by $\mathbb{A}^n$, so $Z \subseteq V \times W$ is closed in $\mathbb{P}^m \times \mathbb{A}^n$. Then, $p_2(Z) \subseteq \mathbb{A}^n$ is closed if and only if $p_2(Z) \subseteq W$ is closed. Therefore, it's enough to prove that $p_2: \mathbb{P}^m \times \mathbb{A}^n \rightarrow \mathbb{A}^n$ is closed.

Claim 2.8.29. The projection map

$$p_2: \mathbb{P}^m \times \mathbb{A}^n \rightarrow \mathbb{A}^n$$

is closed.

Proof. We can describe a Zariski closed subset $Z \subseteq \mathbb{P}^m \times \mathbb{A}^n$ as the zero set of some polynomials

$$f_1, \dots, f_r \in k[X_0, \dots, X_m, Y_1, \dots, Y_n]$$

which are homogeneous with respect to $X_0, \dots, X_m$. ($Y_1, \dots, Y_n$ are affine coordinates, so there is no homogeneity condition with respect to them.) For each point $(y_1, \dots, y_n) \in \mathbb{A}^n$, we can substitute the values $(y_1, \dots, y_n)$ into these polynomials and get a projective algebraic subvariety

$$Z_y = \{[x_0 : \dots : x_m] \in \mathbb{P}^m : f_i(x, y) = 0 \text{ for all } i\}.$$

Observe that $y \in p_2(Z)$ if and only if Z_y is non-empty.

Let I_y denote the ideal in $k[X_0, \dots, X_m]$ generated by the polynomials $f_i(X_0, \dots, X_m, y)$. By the Projective Nullstellensatz, Z_y is non-empty if and only if $\sqrt{I_y}$ is not equal to either the full ring $k[X_0, \dots, X_m]$ or to the irrelevant ideal $(X_0, \dots, X_m)$. We claim that $\sqrt{I_y}$ is not $k[X_0, \dots, X_m]$ or $(X_0, \dots, X_m)$ precisely when I_y does not contain S_d for any positive integer d , where S_d denotes the set of all homogeneous polynomials of degree d in $k[X_0, \dots, X_m]$.

For each positive integer d , write

$$W_d = \{(y_1, \dots, y_n) \in \mathbb{A}^n : I_y \not\supseteq S_d\}.$$

We have shown that

$$p_2(Z) = \bigcap_{d \in \mathbb{N}} W_d.$$

Let the polynomials $f_0, \dots, f_r$ have degrees $d_0, \dots, d_r$ with respect to the X variables. We shall show that W_d is closed for $d \geq \max(d_0, \dots, d_r)$. Since the W_d are a descending chain of sets, the intersection $\bigcap_{d \geq 0} W_d$ is a nested intersection. Therefore,

$$\bigcap_{d \in \mathbb{N}} W_d = \bigcap_{d \geq \max(d_i)} W_d.$$

This is sufficient to show that $p_2(Z)$ is closed, since then $p_2(Z)$ is an intersection of closed subsets.

Fix $d \geq \max(d_i)$. If $g \in S_d$, then $g \in I_y$ if and only if we can write

$$g(X_0, \dots, X_m) = \sum_{i=1}^r f_i(X_0, \dots, X_m, y) h_i(X_0, \dots, X_m)$$

for some homogeneous polynomials $h_1, \dots, h_r$, where $\deg h_i = d - d_i$. Hence $S_d \cap I_y$ is the image of the linear map

$$\alpha_{d,y}: \bigoplus_{i=1}^r S_{d-d_i} \rightarrow S_d$$

given by

$$\alpha_{d,y}(h_1, \dots, h_r) = \sum_{i=1}^r f_i(X_0, \dots, X_m, y) h_i(X_0, \dots, X_m).$$

Therefore,

$$\begin{aligned} W_d &= \{y \in \mathbb{A}^n: I_y \not\supseteq S_d\} \\ &= \{y \in \mathbb{A}^n: \alpha_{d,y} \text{ is not surjective}\} \\ &= \{y \in \mathbb{A}^n: \text{rank } \alpha_{d,y} < \dim S_d\} \\ &= \{y \in \mathbb{A}^n: \text{all } (\dim S_d \times \dim S_d)\text{-minors of } \alpha_{d,y} \text{ vanish}\}. \end{aligned}$$

Here, we fix bases for S_d and $\bigoplus_i S_{d-d_i}$ and use these to write $\alpha_{d,y}$ as a matrix. The determinants of these submatrices are polynomials in $y_1, \dots, y_n$, proving that W_d is Zariski closed in $\mathbb{A}^n$. Therefore, this is an affine variety. $\square$

$\square$

Exercise 2.8.30. Let $p \in \mathbb{P}^2$. Prove that $\mathbb{P}^2 \setminus \{p\}$ is irreducible. Further, show that $\mathbb{P}^2 \setminus \{p\}$ is not isomorphic to a projective variety.

Solution. $\mathbb{P}^2 \setminus \{p\}$ is an open subset of $\mathbb{P}^2$, which is irreducible. Thus, irreducible. As for the second statement, if it were isomorphic, then the inclusion from $\mathbb{P}^2 \setminus \{p\} \hookrightarrow \mathbb{P}^2$ would be a closed subset of $\mathbb{P}^2$. However, this is not true. $\square$

3 Dimension

§3.1 Different Approaches

We want to define the dimension of algebraic varieties. There are several different definitions, all equivalent but each being useful in different situations. Over the course of the next few lectures we will see four different (equivalent) notions of dimension:

- An axiomatic approach: define some axioms that dimension ought to follow, and prove that these assign every variety a well-defined dimension.
- Birational/field-theoretic: $\dim V$ as an invariant of $k(V)$.
- Topological: $\dim V$ as an invariant of the space V .
- (For affine varieties only) Algebraic: $\dim R$ as an invariant of a commutative ring R , and $\dim V$ as $\dim k[V]$ (Krull Dimension).

§3.1.i An Axiomatic Approach

None of the definitions above are particularly obvious, so we begin with the axiomatic approach—by listing some properties that the ‘dimension’ of an irreducible quasi-projective variety V ought to have. (We only consider irreducible varieties here, because a reducible variety might have components of different dimensions, so it is harder to be confident about what properties the dimension of a reducible variety should have.)

1. $\dim \mathbb{A}^n = n$.
2. If U is an open subset of V , then $\dim U = \dim V$ (note that this holds for manifolds in differential geometry).

Remark 3.1.1 — In particular, because $\mathbb{A}^n$ is open in $\mathbb{P}^n$, Axioms 1 and 2 implies that $\dim \mathbb{P}^n = \dim \mathbb{A}^n = n$, as expected. Moreover, if V and W are birational, then some open subset of V is isomorphic to some open subset of W , then Axiom 2 implies that $\dim V = \dim W$, hence dimension is a birational invariant. In particular, the dimension of V is expected to depend only on $k(V)$.

Not every quasi-projective variety is isomorphic to some $\mathbb{A}^n$, so the two preceding axioms do not suffice to attribute a dimension to each quasi-projective variety. But we can add:

3. If $f: V \rightarrow W$ is finite-to-one (i.e. each fiber of f is finite) and dominant, then $\dim V = \dim W$.

This axiom is natural, because, the fibers of f being points means intuitively that f ‘doesn’t add any dimension’ to V : geometrically, f is a kind of ramified covering. It turns out that these properties are enough to compute the dimension of every irreducible quasi-projective variety, thanks to the following lemma.

Lemma 3.1.2 (Projective Noether Normalization)

Let V be an irreducible projective variety. Then there exists a finite-to-one surjective regular map $V \rightarrow \mathbb{P}^d$ for some d .

Proof. Embed V in $\mathbb{P}^N$ as a closed subset for some sufficiently large N ; this is possible because V is projective. If $V = \mathbb{P}^N$ then we are done. Otherwise, choose a point p not in V . Consider a hyperplane $H \cong \mathbb{P}^{N-1}$ in $\mathbb{P}^N$ not containing p , and the projection $\pi: \mathbb{P}^N \setminus \{p\} \rightarrow \mathbb{P}^{N-1}$ from p to H . Because $V \subseteq \mathbb{P}^N \setminus \{p\}$, this restricts to a regular map $\pi: V \rightarrow \mathbb{P}^{N-1}$. We want to show this map has finite fibers.

The fiber of this map at $x \in H$ is $L_{px} \cap V$, which is a closed subset of $L_{px} \cong \mathbb{P}^1$ (because V is closed) not containing p , hence is a finite set of points, hence π is finite-to-one. Denote by $V_1 = \pi(V) \subseteq \mathbb{P}^{N-1}$, V_1 is closed because V is projective, hence complete. If $V_1 = \mathbb{P}^{N-1}$ then we are done, otherwise we can continue as before to obtain a sequence of finite-to-one surjective maps:

$$V = V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r = \mathbb{P}^{N-r},$$

with each $V_i \subseteq \mathbb{P}^{N-i}$ closed. The composition of all these arrows is then a finite-to-one and surjective map $V \rightarrow \mathbb{P}^{N-r}$. $\square$

Then, if V is quasi-projective and irreducible, by Lemma 2.8.27, V has a non-empty affine open subset $U \subseteq V$. Using an embedding $U \subseteq \mathbb{A}^n \subseteq \mathbb{P}^n$ and taking the projectivization, one obtains an irreducible projective variety $\bar{U}$ birational to V : then Axiom 2 gives $\dim V = \dim \bar{U}$. This lemma gives a finite-to-one and surjective map $f: \bar{U} \rightarrow \mathbb{P}^d$, then by Axiom 3 $\dim \bar{U} = \dim \mathbb{P}^d = d$.

§3.1.ii An Algebraic Approach

There is one problem with the axiomatic definition given above: there were a lot of choices made in the procedure used to attribute a dimension to a quasi-projective variety (choice of open affine subset, of embedding in a $\mathbb{P}^N$, of points and hyperplane), and we are not sure a priori that different choices would have led to the same dimension. In fact this is the case, and we will prove this by finding an algebraic invariant corresponding to the geometric notion of dimension introduced here.

Theorem 3.1.3

Let V and W be irreducible varieties over k , and let $f: V \rightarrow W$ be a dominant, finite-to-one regular map. Then the induced map $f^*: k(W) \rightarrow k(V)$ makes $k(V)$ into an algebraic extension of $k(W)$.

We note that this theorem indicates that the dimension should involve ‘algebraic’ properties of $k(V)$. We first establish a lemma:

Lemma 3.1.4

Let W be an irreducible affine variety over k , and let $V \subseteq \mathbb{A}^1 \times W$ be an irreducible closed subvariety. Suppose that the projection map $\pi: V \rightarrow W$ is dominant and finite-to-one. Then the induced map $\pi^*: k(W) \rightarrow k(V)$ is an algebraic extension of fields.

Proof. Let t be the coordinate on $\mathbb{A}^1$, so that the coordinate ring of $\mathbb{A}^1 \times W$ is $k[W][t]$. Let I be the ideal of $k[W][t]$ consisting of the functions that vanish on V , so that $k[V] \cong k[W][t]/I$. Note that $\pi^*: k[W] \rightarrow k[W][t]/I$ is injective, since π is dominant. Then, $k(V)$ is generated by $k(W)$ and t , so $k(V)$ is generated over $k(W)$ by t by dominance of π . We just now need to show that t is algebraic over $k(W)$. If I were the zero ideal, then $V \cong \mathbb{A}^1 \times W$ by the Nullstellensatz, and so the projection of V onto W is not finite-to-one. Hence I is not the zero ideal. Thus I contains a nonconstant polynomial $P(t)$ (which exists by injectivity of π^*) with coefficients in $k[W]$. Moreover, $P(t) = 0$ in $k(V)$. Thus, t is algebraic over $k(W)$ as desired. $\square$

Proof of Theorem 3.1.3. Replacing V and W by suitable open subsets, we may assume that both V and W are affine. Embed V in $\mathbb{A}^n$, and consider the closed subset $\Gamma_f \subseteq \mathbb{A}^n \times W$ given by $\Gamma_f = \{(x, y) \in \mathbb{A}^n \times W : x \in V, y = f(x)\}$. Then V is isomorphic to Γ_f defined by $x \mapsto (x, f(x))$, and the composition of this isomorphism with the projection $\Gamma_f \rightarrow W$ is precisely the map f . We can thus factor f as a series of projections:

$$V \cong \Gamma_f = V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_n = W$$

where $V_i \subseteq \mathbb{A}^{n-i} \times W$ is the closure of the image of V_{i-1} under the projection $\mathbb{A}^{n-i+1} \times W \rightarrow \mathbb{A}^{n-i} \times W$. By the lemma, the induced extension $k(V_{i+1})/k(V_i)$ is algebraic, hence $k(W) \hookrightarrow k(V)$ is algebraic. $\square$

This theorem leads us to consider naturally the transcendence degree:

Definition 3.1.5. Let $k \subseteq K$ be fields. A subset $S \subseteq K$ is **algebraically independent** over k if the natural map $k[S] \rightarrow K$ defined by $s \mapsto s$ is an injection. In other words, there is no nonzero polynomials with coefficients in k with $|S|$ variables such that $P(S) = 0$ in K .

Definition 3.1.6. Let $k \subseteq K$ be fields. A subset $S \subseteq K$ is a **transcendence basis** of K over k if S is algebraically independent over k and the extension $k(S) \hookrightarrow K$ is algebraic.

Remark 3.1.7 — It is a theorem of field theory that transcendence bases exist, and moreover that any two transcendence bases have the same cardinality. We call this cardinality the **transcendence degree** of K over k . We write this as $\text{trdeg}(K : k)$.

Remark 3.1.8 — We remark that if $k \rightarrow K$ is a field extension and $S \subseteq K$ is algebraically independent over k , then $k(S)$ is the field of fractions of a polynomial ring in $|S|$ variables. Such extensions of k are called **purely transcendental**. Here, S is a transcendence basis for $K(S)$. The fact that transcendence bases exist shows that every field extension can be decomposed into a purely transcendental extension followed by an algebraic extension.

Definition 3.1.9. The **dimension** of an irreducible quasi-projective variety V is the transcendence degree (over k) of the field of rational functions $k(V)$.

Notice the formal similarity between this definition of dimension and the one for vector spaces: one replace here bases of vector spaces by transcendence bases of fields of rational functions.

We see that this algebraic definition satisfies the three axioms:

1. We have $k(\mathbb{A}^n) = k(X_1, \dots, X_n)$, which is a purely transcendental extension of transcendence degree n .
2. The definition of dimension depends only on $k(V)$, and it is a birational invariant, as expected.
3. If $f: V \rightarrow W$ is finite-to-one and dominant, then from Theorem 3.1.3 (or just the corollary below), $k(V)$ is algebraic over $k(W)$ and hence they have the same transcendence degree.

As a proof of the third statement above, we may consider $k \rightarrow k(W) \rightarrow k(V)$, and note that

$$\text{trdeg}(k(V) : k) = \text{trdeg}(k(V) : k(W)) + \text{trdeg}(k(W) : k),$$

and $\text{trdeg}(k(V) : k(W)) = 0$, because it is algebraic.

Remark 3.1.10 (Axiomatic $\implies$ Algebraic) — As for the converse (that the axiomatic definition satisfies the algebraic one), this requires the projective Noether normalization. Let V be an irreducible quasi projective variety, and choose a nonempty affine open subset $U \subseteq V$. Taking the projective closure of U , we have that $\bar{U}$ is a projective variety. Then, the projective Noether normalization gives a finite to one surjective regular map $\bar{U} \rightarrow \mathbb{P}^d$, for some d . Thus, $\dim V = \dim \bar{U} = d$. Now, since $\bar{U} \rightarrow \mathbb{P}^d$ is finite to one and surjective (so dominant), it induces an algebraic extension of field $k(\mathbb{P}^d) \hookrightarrow k(\bar{U})$, and so $\text{trdeg}_k k(\bar{U}) = \text{trdeg}_k k(\mathbb{P}^d) = d$. Since U is a nonempty open of $\bar{U}$ and V , these two are birational, meaning that $k(V) = k(U) = k(\bar{U})$. Therefore, we arrive at $\text{trdeg}_k k(V) = d$.

§3.1.iii Basic Properties of Dimension

We establish here basic properties of dimension. Notice that it is useful to know how to use more algebraic or more geometric proofs, depending on the context.

Lemma 3.1.11

If V, W are irreducible quasi-projective varieties, and $f: V \rightarrow W$ is dominant, then $\dim W \leq \dim V$.

Proof. Because f is dominant, it induces a field extension $f^*: k(W) \rightarrow k(V)$. Then the transcendence degree of $k(V)$ over k is the transcendence degree of $k(W)$ over k plus the transcendence degree of $k(V)$ over $k(W)$, and hence $\dim W \leq \dim V$. $\square$

Lemma 3.1.12

Let V, W be irreducible quasi-projective varieties. Then we have an equality $\dim(V \times W) = \dim V + \dim W$.

Proof. Replacing V and W first with affine open subsets, and then with their projective closures, which does not change the dimension, we can assume that V and W are projective. If $d = \dim V$ and $e = \dim W$, then we have surjective, finite-to-one regular

maps $f: V \rightarrow \mathbb{P}^d$ and $g: W \rightarrow \mathbb{P}^e$. Then $f \times g: V \times W \rightarrow \mathbb{P}^d \times \mathbb{P}^e$ is a surjective, finite-to-one regular map and hence:

$$\dim(V \times W) = \dim(\mathbb{P}^d \times \mathbb{P}^e)$$

But $\mathbb{P}^d \times \mathbb{P}^e$ has an open subset isomorphic to $\mathbb{A}^d \times \mathbb{A}^e \cong \mathbb{A}^{d+e}$, and is then of dimension $d + e$. $\square$

Recall that, when $V \subseteq \mathbb{P}^n$ is a nonempty projective variety, the projective cone $\widehat{V} \subseteq \mathbb{A}^{n+1}$ is defined by:

$$\widehat{V} = \{0\} \cup \{(x_0, \dots, x_n) : [x_0 : \dots : x_n] \in V\}.$$

It is a closed subset given by the zero locus of the homogeneous ideal defining V . V is obtained by quotienting $\widehat{V} - \{0\}$ by the scaling, so, intuitively, it must have one dimension less than $\widehat{V}$. This is indeed the case:

Corollary 3.1.13

Let $V \subseteq \mathbb{P}^n$ be irreducible and projective, and let $\widehat{V} \subseteq \mathbb{A}^{n+1}$ be the affine cone of V . Then $\dim \widehat{V} = 1 + \dim V$.

Proof. If $V = \mathbb{P}^n$ then $\widehat{V} = \mathbb{A}^{n+1}$ and we are done. If not, then V is not contained in some hyperplane $H \subseteq \mathbb{P}^n$, and by applying an automorphism of $\mathbb{P}^n$ we may assume that H is the hyperplane $\{x_0 = 0\}$. Then $\dim V = \dim U$, where U is the complement of H in V . The cone $\widehat{U}$ of U is the set of all points of the form $(t, tx_1, \dots, tx_n)$, where $[1 : x_1 : \dots : x_n]$ is a point of U . On one hand, $\widehat{U}$ is dense in $\widehat{V}$ so that $\dim \widehat{U} = \dim \widehat{V}$; on the other hand we have an isomorphism

$$(\mathbb{A}^1 \setminus \{0\}) \times U \rightarrow \widehat{U} \setminus \{(0, \dots, 0)\}$$

$$(t, [1 : x_1, \dots, x_n]) \rightarrow (t, tx_1, \dots, tx_n)$$

This is certainly well-defined. Further, this is regular. The inverse is given by the natural projection to the first coordinate. Thus:

$$\dim \widehat{U} = \dim((\mathbb{A}^1 \setminus \{0\}) \times U) = \dim(\mathbb{A}^1 \setminus \{0\}) + \dim U = 1 + \dim U$$

where we have used the previous lemma, and the fact that $\mathbb{A}^1 \setminus \{0\}$ is open in $\mathbb{A}^1$, hence is of dimension 1. $\square$

Remark 3.1.14 — Note that in the above, the affine cone is defined for $t = 0$, but to restrict to an isomorphism, we neglect $t = 0$ because injectivity is not guaranteed here.

Exercise 3.1.15. Let $H \subseteq \mathbb{P}^n$ be a hyperplane. Let $V \subset H$ be an irreducible projective algebraic variety (note: $V \subset H$, rather than $V \subset \mathbb{P}^n$). Let x be a point of $\mathbb{P}^n \setminus H$. Let C be the union of all lines joining V to x (this is called the cone over V with vertex x). Prove that C is a projective algebraic variety, that it is irreducible and that $\dim(C) = \dim(V) + 1$.

Solution: By choosing appropriate coordinates, we can assume that $p = [0 : \cdots : 0 : 1]$ and

$$H = \{[x_0 : \cdots : x_{n-1} : 0]\} \simeq \mathbb{P}^{n-1}.$$

Let $V = V(f_1, \dots, f_r)$, for homogeneous polynomials $f_1, \dots, f_r \in k[X_0, \dots, X_{n-1}]$. Then, C is the set of points of the form $[sx_0 : \cdots : sx_{n-1} : t]$ for $[s : t] \in \mathbb{P}^1$ and $[x_0 : \cdots : x_{n-1}] \in V$. We claim that is is equal to the zero set of the homogeneous polynomials

$$g_i(X_0, \dots, X_n) := f_i(X_0, \dots, X_{n-1}) \in k[X_0, \dots, X_n].$$

To see this, we note that if $x = [sx_0 : \cdots : sx_{n-1} : t] \in C$, then

$$g_i(sx_0, \dots, sx_{n-1}, t) = s^d f_i(x_0, \dots, x_{n-1}) = 0,$$

so $C \subseteq V(g_1, \dots, g_r)$. Conversely, if $p = [x_0, \dots, x_n] \in V(g_1, \dots, g_r)$, then firstly, $f_i(x_0, \dots, x_{n-1}) = 0$ for all i . If $[x_0, \dots, x_{n-1}]$ is not zero, we therefore have that $[x_0 : \cdots : x_{n-1} : 0] \in V$. Then, the original point lies on a line adjoining this point in V to $[0 : \cdots : 0 : 1]$, so $p \in C$. In the case that $[x_0, \dots, x_{n-1}]$ is zero, we find that our original point was just the point x , so is certainly in C .

Hence C is a projective variety. V is irreducible by assumption, so $\sqrt{(f_1, \dots, f_r)}$ is prime. It follows by definition of the g_i that $\sqrt{(g_1, \dots, g_r)}$ is prime as well, implying that C is irreducible. The open subset $C \setminus \{p\}$ is isomorphic with $V \times \mathbb{A}^1$ from the map $C \setminus \{p\} \rightarrow V \times \mathbb{A}^1$ defined by

$$[sx_0 : \cdots : sx_{n-1} : t] \mapsto ([x_0 : \cdots : x_{n-1} : \frac{t}{s}].$$

Therefore, we can compute the dimension easily:

$$\dim(C) = \dim(C - \{p\}) = \dim(V \times \mathbb{A}^1) = \dim(V) + 1.$$

□

Remark 3.1.16 — We could’ve justified slightly more abstractly, and using the fiber dimension theorem (to come). Continue with the notation above. We note that C is the image of the map $F: V \times \mathbb{P}^1 \rightarrow \mathbb{P}^n$, defined by

$$([x_0 : \cdots : x_n], [s : t]) \mapsto [sx_0 : \cdots : sx_n : t].$$

This is clearly regular. Also, $V \times \mathbb{P}^1$ is a product of irreducible projective varieties, so is irreducible and projective (thus complete) in its own right. Since $C = \text{Im} F$, we find that C is closed in $\mathbb{P}^n$ and irreducible, so is an irreducible quasi-projective variety. Now, given a point $c \in C$, the fiber $F^{-1}(c)$ is generally 0-dimensional because it’s usually just a point. Therefore, $\dim C - \dim(V \times \mathbb{P}^1) = 0$, so $\dim C = \dim V + 1$.

§3.2 Dimension and Intersections

Before digressing into the algebraic geometry, let’s recall two facts about dimensions of vector spaces:

Proposition 3.2.1

Let $E \subseteq k^n$ a sub-vector space, and $H \subseteq k^n$ a hyperplane not containing E , then $\dim(E \cap H) = \dim E$.

Proposition 3.2.2

Let $E, F \subseteq k^n$ two sub-vector spaces of dimension d, e , such that $d + e \geq n$. Then $\dim(E \cap F) \geq d + e - n$.

The goal of this section will be to provide analogous results for quasi-projective varieties.

§3.2.i Dimension of Intersections**Theorem 3.2.3**

Let $V \subseteq \mathbb{P}^n$ be an irreducible quasi-projective variety of dimension ≥ 1 , and let H be a hyperplane of $\mathbb{P}^n$ not containing V . Then either $V \cap H$ is empty or every irreducible component of $V \cap H$ has dimension $\dim V - 1$. Moreover, if V is projective, $V \cap H$ is never empty.

Proof. Firstly we reduce to the case where V is projective. Let $\bar{V}$ be the closure of V in $\mathbb{P}^n$, then it is projective, and does not contain H . $V \cap H$ is open in $\bar{V} \cap H$: in particular, either $V \cap H$ is empty, or each irreducible component of $V \cap H$ is open in an irreducible component of $\bar{V} \cap H$. The claim for $\bar{V}$ implies then the claim for V .

We assume now that V is projective, and we argue by induction on n . If $n = 1$ the result is clear. Indeed, V must be all of $\mathbb{P}^1$, and H is a point. Therefore, $V \cap H = H$, so the intersection is of dimension 0.

Assume the result is true for $n - 1$. If V is all of $\mathbb{P}^n$, the claim is clear, so assume this is not the case. We first show that V can't contain H . If so, then $I(V) \subseteq I(H)$, so $I(V)$ is a homogeneous prime ideal contained in the ideal generated by the linear polynomial L defining H . Let $P(X)$ be an irreducible homogeneous polynomial contained in $I(V)$; then $L(X)$ divides $P(X)$ so $L(X) = P(X)$. Thus $I(V) = I(H)$, which is impossible since then $V = H$.

Now fix a point $p \in H$ but not in V , and a hyperplane, say K , $\mathbb{P}^{n-1}$ not containing p . Consider the projection from p to $\mathbb{P}^{n-1}$, $\pi: \mathbb{P}^n \rightarrow \mathbb{P}^{n-1}$, which gives a regular map $\pi: V \rightarrow K$. Because $p \notin V$, this is a finite-to-one map surjective map on its image $V' := \pi(V)$ which is automatically irreducible, and closed because V is projective, and then complete, and $\dim V = \dim V' \geq 1$. On the other hand, let H' be the closure of $H \setminus \{p\}$, so that H' is also a hyperplane in $\mathbb{P}^{n-1}$, which does not contain V' because H contains p . It follows that π is a surjective finite-to-one map from $V \cap H$ to $V' \cap H'$, hence it maps each irreducible component of $V \cap H$ to an irreducible of $V' \cap H'$ of the same dimension. By the induction hypothesis on $V' \subseteq \mathbb{P}^{n-1}$, $V' \cap H'$ is not empty and each of its irreducible components is of dimension $\dim V' - 1 = \dim V - 1$, hence the same holds for $V \cap H$.

As for the final claim, if V is projective, assume that $V \cap H = \emptyset$. Suppose H is the hyperplane identified with $x_0 = 0$. Then, $V \cap H = \emptyset$ implies V has $x_0 \neq 0$, so V is contained in $\mathbb{A}^n \cong \mathbb{P}^n \setminus H$. However, V is projective, so complete, but a complete closed subvariety of $\mathbb{A}^n$ is a finite set of points. Thus, $\dim V = 0$, which is a contradiction. $\square$

Corollary 3.2.4

Let $V, W \subseteq \mathbb{A}^n$ be irreducible and quasi-projective of dimensions d and e , respectively, and suppose that $d + e \geq n$. Then either $V \cap W$ is empty or every irreducible component of $V \cap W$ has dimension at least $d + e - n$.

Proof. We first note that $V \cap W = (V \times W) \cap \Delta$, where $\Delta \subseteq \mathbb{A}^n \times \mathbb{A}^n$ is the diagonal. Note that $\Delta = V(x_1 - y_1, \dots, x_n - y_n)$, where x_i are the coordinates on the first factor and the y_i are coordinates on the second factor. If we let H_i be the hyperplane $V(x_i - y_i)$ then intersecting with Δ is the same as successively intersecting with each H_i . By the previous result, when we intersect with one of the H_i , the dimension of each irreducible component drops by at most one or the whole intersection becomes empty. At the end of this process the dimension of each irreducible component is at least $\dim(X \times Y) - n = d + e - n$. $\square$

Remark 3.2.5 — We note that equality is achieved when no $(V \times W) \cap (\cap_{j \in J} H_j) \subseteq H_i$, for $i \notin J$. This can be observed by examining the proof.

Theorem 3.2.6

Let $V, W \subseteq \mathbb{P}^n$ be irreducible projective varieties of dimensions d and e , respectively, and suppose that $d + e \geq n$. Then $V \cap W$ is nonempty, and every irreducible component of $V \cap W$ has dimension at least $d + e - n$.

Proof. Let $\widehat{V}$ and $\widehat{W}$ be the affine cones of V and W respectively: they are respectively of dimension $d + 1$ and $e + 1$ from Corollary 3.1.13. Their intersection is nonempty since both contain the origin; thus by the previous Corollary every irreducible component of $\widehat{V} \cap \widehat{W}$ has dimension at least $(d + 1) + (e + 1) - (n + 1) = d + e - n + 1$. In particular this is at least one, so this intersection contains a point other than the origin. Since $\widehat{V} \cap \widehat{W}$ is the affine cone of $V \cap W$, it follows that $V \cap W$ is nonempty. Explicitly, we may write $\widehat{V} \cap \widehat{W} \cap (\mathbb{A}^{n+1} \setminus \{0\}) \neq \emptyset$, and $V \cap W$ is just the quotient of this by scaling.

Now if Z is an irreducible component of $V \cap W$ then its projective cone $\widehat{Z}$ is an irreducible component of $\widehat{V} \cap \widehat{W}$, which has dimension at least $d + e - n + 1$. It follows from Corollary 3.1.13 that Z has dimension at least $d + e - n$. $\square$

Veronese Embedding

In order to generalize Theorem 3.2.3 from intersections with hyperplanes to intersections with hypersurfaces, we use the Veronese embedding. This is defined as follows.

Let d and n be positive integers and let $N = \binom{n+d}{d} - 1$. There are $N + 1$ monomials of degree d in variables $X_0, \dots, X_n$; that is, expressions of the form $X_0^{a_0} X_1^{a_1} \dots X_n^{a_n}$ where $a_0, \dots, a_n \in \mathbb{Z}_{\geq 0}$ with $a_0 + \dots + a_n = d$.

Definition 3.2.7. Let N be as above. We define a regular map $\nu_{n,d}: \mathbb{P}^n \rightarrow \mathbb{P}^N$ by writing down all these monomials of degree d as coordinates (in some order). This is called the **degree d Veronese embedding** of $\mathbb{P}^n$.

Remark 3.2.8 — The order doesn't really matter, but one usually writes something sensible (maybe lexicographic).

Example 3.2.9

For example, for $n = d = 2$ we get $N = 5$ and

$$\nu_{2,2}([X_0 : X_1 : X_2]) = [X_0^2 : X_1^2 : X_2^2 : X_0X_1 : X_1X_2 : X_0X_2].$$

Remark 3.2.10 — Note that $\nu_{n,d}$ is a regular map. Further, it's a surjection onto its image $V_{n,d} \subseteq \mathbb{P}^N$. The image is a projective algebraic variety by completeness of $\mathbb{P}^n$.

Remark 3.2.11 — One can write down explicit polynomials defining this algebraic subvariety (they are determinants of 2×2 matrices).

Remark 3.2.12 ($\nu_{n,d}$ is an isomorphism) — The most important fact (which we will not prove) is that $\nu_{n,d}$ is indeed a closed embedding, so $\nu_{n,d}$ is injective as well. Therefore, $\nu_{n,d}$ is an isomorphism $\mathbb{P}^n \rightarrow V_{n,d}$ is an isomorphism.

The benefit of doing all this is that applying $\nu_{n,d}$ turns intersections with degree d hypersurfaces into intersections with hyperplanes. That is, let $V \subseteq \mathbb{P}^n$ be a projective variety and $H \subseteq \mathbb{P}^n$ is a hypersurface defined by some homogeneous polynomial $f = \sum_I a_I X^I$ of degree d . Then because the monomials of degree d become individual homogeneous coordinates via the Veronese embedding, the equation for $\nu_{n,d}(H)$ is a linear equation $\sum_I a_I Z_I = 0$. Thus $\nu_{n,d}(H) = V_{n,d} \cap Z$, where $Z \subseteq \mathbb{P}^N$ is the hyperplane mapped from H through the Veronese embedding. In particular,

$$\nu_{n,d}(V \cap H) = \nu_{n,d}(V) \cap Z.$$

Therefore, instead of studying the intersection between $V \subseteq \mathbb{P}^n$ and a hypersurface $H \subseteq \mathbb{P}^n$, we can instead study the intersection between $\nu_{n,d}(V) \subseteq V_{n,d} \subseteq \mathbb{P}^N$ and a hyperplane $Z \subseteq \mathbb{P}^N$.

Example 3.2.13

Consider the hyperplane $H = V(X^2 + 3XY - 5Z^2) \subseteq \mathbb{P}^2$. The corresponding hyperplane through the Veronese embedding is $Z = V(Z_{X^2} + 3Z_{XY} - 5Z_{Z^2}) \subseteq \mathbb{P}^5$. Indeed,

$$\nu_{n,d}(\mathbb{P}^2 \cap H) = V_{2,2} \cap Z.$$

In particular, $\nu_{n,d}$ is an isomorphism, so we can use Theorem 3.2.3 to deduce the same result for intersections with hypersurfaces:

Theorem 3.2.14

Let $V \subseteq \mathbb{P}^n$ be irreducible and quasi-projective variety of dimension ≥ 1 , and let H be a hypersurface of $\mathbb{P}^n$ not containing V . Then either $V \cap H$ is empty or every irreducible component of $V \cap H$ has dimension $\dim V - 1$. Moreover, if V is projective, $V \cap H$ is never empty.

Proof. Let the degree of H be d . The irreducible components of $V \cap H$ are isomorphic to the irreducible components of $\nu_{n,d}(V \cap H) = \nu_{n,d}(V) \cap H'$, where $H' \subseteq \mathbb{P}^N$ is a hyperplane (Here, $N = \binom{n+d}{d} - 1$). Then, we apply Theorem 3.2.3 to $\nu_{n,d}(V) \cap H'$. $\square$

Example 3.2.15

The theorem above tells us that the dimension of an irreducible hypersurface V of $\mathbb{P}^n$ has dimension $n - 1$. V certainly doesn't contain $\mathbb{P}^n$ and its intersection is nonempty, given that V is nonempty. $\mathbb{P}^n \cap V = V$ is irreducible, so has dimension $n - 1$.

Exercise 3.2.16. Let $V \subseteq \mathbb{P}^n$ be a projective algebraic variety in which all irreducible components have dimension $n - 1$. Prove that V is a hypersurface.

Solution: Denote by V_i the irreducible components of V . There is some nonzero polynomial f_i vanishing on V_i . Since each V_i is irreducible, each irreducible part of f_i vanishes on V_i , hence we can take f_i to be irreducible. The zero set W_i of f_i is then an irreducible hypersurface, so has dimension $n - 1$. We also have that $V_i \subset W_i$ with V_i and W_i both irreducible of dimension $n - 1$, so we actually achieve equality: $V_i = W_i$. Therefore, V is the hypersurface defined by the homogeneous polynomial $f := \prod_i f_i$. $\square$

Exercise 3.2.17. Prove that the variety $\mathbb{P}^2 \setminus X$, where X is a plane conic, is affine.

Solution: We use the Veronese embedding. First, note that $\mathbb{P}^2 \setminus X$ is open, and so we want to show that this is affine. Under the Veronese embedding $\nu_{2,2}: \mathbb{P}^2 \rightarrow \mathbb{P}^5$, X gets mapped to the intersection of $V_{2,2}$ with a hyperplane in $\mathbb{P}^5$. Thus, the image of $\mathbb{P}^2 \setminus X$ under the Veronese embedding is

$$\nu_{2,2}(\mathbb{P}^2) \setminus \nu_{2,2}(X) = V_{2,2} \setminus (V_{2,2} \cap H) = V_{2,2} \cap (\mathbb{P}^5 \setminus H).$$

Now, we know that $\mathbb{P}^5 \setminus H \cong \mathbb{A}^5$, so we have that $\nu_{2,2}(\mathbb{P}^2) \setminus \nu_{2,2}(X) \cong V_{2,2} \cap \mathbb{A}^5$, so it's a closed subvariety of $\mathbb{A}^5$, which is affine. Therefore, $\nu_{2,2}(\mathbb{P}^2) \setminus \nu_{2,2}(X)$ is affine, but $\mathbb{P}^2 \setminus X$ is isomorphic to its image in the Veronese embedding, so we are done. $\square$

Exercise 3.2.18. Let $\alpha: \mathbb{P}^n \rightarrow \mathbb{P}^m$ be regular; if $m < n$, then α is constant.

Solution: Let $\pi: \mathbb{A}^{n+1} \setminus \{\text{origin}\} \rightarrow \mathbb{P}^n$ be the map $(a_0, \dots, a_n) \mapsto (a_0 : \dots : a_n)$. Then $\alpha \circ \pi$ is regular, and there exist polynomials $F_0, \dots, F_m \in k[X_0, \dots, X_n]$ such that $\alpha \circ \pi$ is the map

$$(a_0, \dots, a_n) \mapsto (F_0(a) : \dots : F_m(a)).$$

As $\alpha \circ \pi$ factors through $\mathbb{P}^m$, the F_i must be homogeneous of the same degree. Note that

$$\alpha(a_0 : \dots : a_n) = (F_0(a) : \dots : F_m(a)).$$

If $m < n$ and the F_i are nonconstant, then Theorem 3.2.14 shows that each intersection of the hypersurface cut out by F_i is nonempty. Explicitly, we can cut out $\mathbb{P}^n$ by a hypersurface $m + 1$ times, and do this n times. At each step, the dimension of the irreducible components drops by one, so we have that $\dim V(F_0, \dots, F_m) \geq n - m - 1 \geq 0$. This means that we have a common zero for the polynomials, so α is not defined on all of $\mathbb{P}^n$. Hence the F_i 's must be constant. $\square$

§3.3 Topological Definition of Dimension

Lemma 3.3.1

Let V be an irreducible quasi-projective variety and let W be a irreducible closed subset of V . Then $\dim W \leq \dim V$, with equality if and only if $W = V$.

Proof. For the first statement, replacing V with an affine open subset, we may embed V in $\mathbb{A}^n \subseteq \mathbb{P}^n$, and taking their projective closure, we can assume that V and W are projective. Then V admits a surjective, finite-to-one map $f: V \rightarrow \mathbb{P}^d$, where $d = \dim V$. Let $W' := f(W)$, which is closed in $\mathbb{P}^d$ by completeness of W , and such that $\dim W' = \dim W$ because $f|_W: W \rightarrow W'$ is finite-to-one and surjective. Because W' is a closed subvariety of $\mathbb{P}^d$, by considering successive projections as in the proof of the projective Noether normalization Lemma (3.1.2), it admits a finite-to-one map to $\mathbb{P}^{d-r}$ for $r \geq 0$, hence $\dim W = \dim W' = d - r \leq d = \dim V$, which finishes the proof of the first part.

For the second part, let $V \subseteq \mathbb{P}^n$ be an irreducible quasi-projective variety and let W be a irreducible closed subset of V . Suppose W is a proper closed subset of V . Then $I(W)$ strictly contains $I(V)$, so there exists a polynomial f that vanishes on W but not on V . Then $W \subseteq V \cap V(f)$ and $V(f) \subseteq \mathbb{P}^n$ is a hypersurface not containing V . From Theorem 3.2.14 each irreducible component of $V \cap V(f)$ have dimension $\dim V - 1$, hence from the first part of the proof $\dim W \leq \dim V - 1$. $\square$

Lemma 3.3.2

Let $V \subseteq \mathbb{A}^n$ be an irreducible affine variety containing at least two points. Then there exists a hyperplane H , not containing V , such that $H \cap V \neq \emptyset$.

Proof. Let x, y be distinct points of V . Via an affine coordinate change we may first assume that x is the origin, and then via a linear change of coordinates we may assume that y is the point $(1, 0, 0, \dots, 0)$, so that the hyperplane $H = V(x_1)$ will do. $\square$

Corollary 3.3.3

Let V be an irreducible quasi-projective variety of dimension $d > 0$. Then there exists an irreducible closed subvariety $W \subseteq V$ of dimension exactly $d - 1$.

Proof. Let $U \subseteq V$ be an open dense affine and $U \hookrightarrow \mathbb{A}^n$ any embedding. Since U also has dimension $d > 0$, it has at least two points, so that by the previous lemma we may find a hyperplane $H \subseteq \mathbb{A}^n$, not containing U , with $H \cap U \neq \emptyset$. Let W' be any irreducible component of $U \cap H$, so that by Theorem 3.2.3, W' has dimension $d - 1$. Letting W be the closure of W' in V we obtain the desired subvariety. $\square$

Our previous definition of dimension was algebraic. We can also describe the dimension of a variety in terms of its topology.

Theorem 3.3.4

Let V be an irreducible quasi-projective variety. The (topological) **dimension** of V is the maximum integer d such that there exists a chain of irreducible closed subsets

$$V = V_d \supsetneq V_{d-1} \supsetneq \cdots \supsetneq V_0 \supsetneq \emptyset.$$

Some care is required in the statement of this theorem to get the numbering right. The point is that $\dim V_i = i$, so V_0 is still non-empty. In Theorem 3.3.2, it is essential to require all the V_i to be irreducible. Otherwise we could make the chain arbitrarily long by inserting reducible sets with more and more components, all of dimension i , in between V_i and V_{i+1} .

Proof. First we prove that such a sequence with $d = \dim V$ exists. First put $V_d = V$. By the previous corollary, we may find an irreducible closed subvariety $V_{d-1} \subsetneq V_d$ of dimension exactly $d - 1$. Continuing this inductively, we obtain $V_i \subsetneq V_{i+1}$ with $\dim V_i = i$ until we get to V_0 with $\dim V_0 = 0$, where this process must stop.

In the other direction, to show that there is no such sequence with $d > \dim V$ terms, this follows immediately from the fact that $\dim V_i < \dim V_{i+1}$ (Lemma 3.3.1). $\square$

Remark 3.3.5 — We note that Theorem 3.3.2 is not really strong enough to be useful; that is, we haven’t shown that all maximal chains have the same length. For example in $\mathbb{P}^n$, we can write down a chain of closed subsets

$$\mathbb{P}^n \supsetneq \mathbb{P}^{n-1} \supsetneq \mathbb{P}^{n-2} \supsetneq \cdots \supsetneq \mathbb{P}^1 \supsetneq \{pt\} \supsetneq \emptyset.$$

This chain is maximal; that is, we cannot insert another irreducible closed subset anywhere in the middle of it. But just exhibiting this chain is not enough to prove that $\dim \mathbb{P}^n = n$ – maybe there is a completely different chain which is longer.

It turns out that that it is true that all maximal chains of irreducible closed subsets in an irreducible quasi-projective variety V has length equal to $\dim V$. This is another hard theorem, using ideas similar to *Hilbert’s Hauptidealsatz* or *Krull’s Height Theorem*.

§3.3.i Krull Dimension of Rings

Definition 3.3.6. The **Krull dimension** of a commutative ring R is the supremum of the set of integers d for which there exists a chain of prime ideals

$$I_d \subsetneq I_{d-1} \subsetneq \cdots \subsetneq I_0 \subsetneq R.$$

When V is an irreducible affine variety, these corresponds to prime ideals in $k[V]$. Therefore, we find that $\dim V$ is equal to the Krull dimension of the integral domain $k[V]$ in this case.

Remark 3.3.7 — If R is an integral domain, then we can take $I_d = (0)$.

Example 3.3.8

If $R = k[V]$ is the coordinate ring of an affine variety, then $\dim(R) = \dim(V)$ by the correspondence between prime ideals and irreducible subvarieties. In particular $\dim(k[V])$ is always finite.

Remark 3.3.9 — We note that if V is a not necessarily irreducible affine variety, then $\dim k[V]$ is the maximum of the dimension of the irreducible components of V . Note that each component contributes to a chain of ideals which gives its dimension.

The fact that the Krull dimension of an integral domain is equal to the transcendence degree of its field of fractions is an algebraic version of the geometric arguments that we used (with finite-to-one projections corresponding to algebraic extensions, and intersections with a hyperplane corresponding to adding a transcendental element).

§3.4 The Fiber Dimension Theorem

We provide some motivation first. Let $f: V \rightarrow W$ be a regular map of varieties, with W irreducible and $p \in W$ a point. We ask what the dimension of the fiber of $f^{-1}(p)$ should be. The fiber of f over p is the subset $f^{-1}(p)$; it is a closed subvariety of V because $\{p\} \in W$ is closed and f is continuous. If f is the projection $V \times W \rightarrow W$, where V and W are irreducible, then the fibers $f^{-1}(p)$ are of the form $V \times \{p\}$, and are isomorphic to V . Recall that in this case we have seen that

$$\dim(V \times W) = \dim V + \dim W,$$

which we can rewrite in this case by saying that the dimension of the domain is the sum of the dimension of the target and the dimension of a fiber. One can then ask whether something like this holds for general surjective maps. (Surjectivity, or at least dominance, is necessary here, since one can otherwise compose with an embedding of Y in a variety of much higher dimension without changing X or the fibers.)

There are a number of technical issues that must be addressed in order to make a statement like this work for varieties.

Example 3.4.1

The first is that the fiber of a map $f: V \rightarrow W$ need not be irreducible, and its irreducible components need not even all have the same dimension: consider for instance the fiber over 0 of the map

$$\begin{aligned} \mathbb{A}^3 &\rightarrow \mathbb{A}^2 \\ (x, y, z) &\mapsto (xy, xz) \end{aligned}$$

Over $(0, 0)$, the fiber is the union of the line $V(Y, Z)$ and of the plane $V(X)$, irreducible of dimension 1 and 2, but over any other points the fiber is irreducible of dimension 1. In this example, we see that the fibers can only have a dimension at least the expected dimension $\dim V - \dim W$, and that, generically on W , the fibers are of the expected dimensions.

Example 3.4.2

Notice that in general $f^{-1}(w)$ is not irreducible. For example, if we take the map $f: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ which map x to x^n , for any $w \in \mathbb{A}^1 - \{0\}$, $f^{-1}(w)$ is made of n points (the n -th roots of w).

The fiber dimension theorem will generalize these observations.

Theorem 3.4.3 (The Fiber Dimension Theorem)

Let V, W be irreducible quasi-projective varieties and let $f: V \rightarrow W$ be a dominant regular map. Then:

- (i) For every $w \in W$, each irreducible component of $f^{-1}(w)$ has dimension at least $\dim V - \dim W$.
- (ii) There exists a non-empty open subset $U \subset W$ such that each irreducible component of $f^{-1}(w)$ is of dimension $\dim V - \dim W$ for all $w \in U$.

Proof. First, we cover W with affine open subsets U_i , and then, we cover the open subsets $f^{-1}(U_i) \subset V$ by affine open subsets U'_{ij} . Each U_i is dense in W , so each U'_{ij} is dense in $f^{-1}(U_i)$, hence $f(U'_{ij})$ is dense in $f(f^{-1}(U_i))$, which is itself dense in U_i because f is dominant. Hence U'_{ij} and U_i are affine and irreducible, the induced map $f_{ij}: U'_{ij} \rightarrow U_i$ are dominant. If we assume that the theorem holds for V, W affine, we can apply it to f_{ij} , and the result follow for f .

We are now reduced to the case where V, W are affine, where we can work algebraically. Consider the injection of rings

$$f^*: k[W] \rightarrow k[V].$$

Take a maximal family $f_1, \dots, f_n$ of algebraically independent elements of $k[W]$: it gives a transcendence basis of $k(W)$, hence $n = \dim W$. We take a maximal family $g_1, \dots, g_m$ of element of $k[V]$ algebraically independent over $k[W]$: $f_1, \dots, f_n, g_1, \dots, g_m$ gives a transcendence basis of $k(V)$, hence $n + m = \dim V$.

We note that points in W correspond bijectively to maximal ideals of $k[W]$. Thus, let $u \in W$ and $\mathfrak{m}_u \subseteq k[W]$ be the maximal ideal corresponding to u , and $k[f^{-1}(u)] = k[V]/\mathfrak{m}_u$.

The $g_1, \dots, g_m$ are algebraically independent elements of $k[V]$ over $k[W]$, so they give algebraically independent elements of $k[V]/\mathfrak{m}_u = k[f^{-1}(u)]$ over $k[W]/\mathfrak{m}_u = k$. Hence for every irreducible component V' of $f^{-1}(u)$, $k(V')$ has transcendence degree at least $m = \dim V - \dim W$, hence we have

$$\dim V' \geq \dim V - \dim W.$$

Notice that $g_1, \dots, g_m$ is not always a transcendence basis, hence the dimension can be larger.

The f_i, g_j together generate $k(V)$ as a field, but this doesn't mean they generate $k[V]$. Thus, take $h_1, \dots, h_r \in k[V]$, such that $f_1, \dots, f_n, g_1, \dots, g_m, h_1, \dots, h_r$ generates $k[V]$: they are automatically algebraic in $f_1, \dots, f_n, g_1, \dots, g_m$. Then take polynomials

$$F_i(h_i, f_1, \dots, f_n, g_1, \dots, g_m) = 0$$

which are non-constant in h_i . In particular, write:

$$F_i(h_i) = c_d h_i^d + c_{d-1} h_i^{d-1} + \dots + c_0$$

where $c_i \in k[f_1, \dots, f_n, g_1, \dots, g_m]$, and $c_d \neq 0$. Then take $U_i \subset W$ to be the set of points $u \in W$ where c_d remain nonzero after substituting all the f_j by $f_j(u)$: it is an open subset (defined by the nonvanishing of c_d viewed as a polynomial in $f_1, \dots, f_n$ with coefficients in $k[g_1, \dots, g_m]$). Let U be the intersection of the U_i . For $u \in U$, we have that $k[f^{-1}(u)]$ is generated by $g_1, \dots, g_m, h_1, \dots, h_r$, with $F_i(h_i, f_1(u), \dots, f_n(u), g_1, \dots, g_m) = 0$ giving a non-constant polynomial equation for the h_i with coefficients in $k[g_1, \dots, g_m]$. Then the h_i are algebraic over the $g_1, \dots, g_m$ in $k[f^{-1}(u)]$, hence for any irreducible V' component of $f^{-1}(u)$, $g_1, \dots, g_m$ gives a transcendence basis of $k(V')$, hence

$$\dim V' = \dim V - \dim W.$$

□

We generally use this theorem in situations where we know the dimension of either V or W and want to work out the other. If we can work out $\dim \varphi^{-1}(w)$ for just a single $w \in W$, then we get an inequality. If we can work out $\dim \varphi^{-1}(w)$ for w in some open set then we can work out the desired dimension exactly.

Corollary 3.4.4

Let V, W, φ satisfy the assumptions in the fiber dimension theorem. If there exists $w \in W$ such that $\dim \varphi^{-1}(w) = 0$, then $\dim V = \dim W$.

Solution: Dominant regular map gives $\dim W \leq \dim V$. Conversely, this assumption tells us that $\dim \varphi^{-1}(w) \leq \dim V - \dim W$, so $\dim W \leq \dim V$. Therefore, we attain equality. □

Example 3.4.5 (Example of non-equality)

Consider $\varphi: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ given by

$$\varphi(x, y) = (x, xy).$$

Consider the vertical line

$$L_x = \{(x, y) : y \in k\}.$$

Fixing $x \neq 0$, φ restricts to an isomorphism $L_x \rightarrow L_x$ since it just scales by x . But when $x = 0$, φ maps all of L_0 down to $(0, 0)$. Hence the image of φ is

$$(\mathbb{A}^2 \setminus \{(0, y)\}) \cup \{(0, 0)\}.$$

We see that, above the open set $\{(x, y) : x \neq 0\}$, the fibers of φ are single points i.e. with dimension $2 - 2 = 0$. On the other hand, above the point $(0, 0)$, the fiber $\varphi^{-1}((0, 0))$ is a line, so has dimension $1 \geq 2 - 2$.

As an application, we give a criterion for irreducibility (recall that from Lemma 2.5.23, if $f: V \rightarrow W$ is dominant, V is irreducible implies that W is irreducible).

Proposition 3.4.6 (Irreducibility Criterion)

Let $f: V \rightarrow W$ a surjective map, and suppose that V is projective and W is irreducible, and that the fibers of f are all irreducible of the same dimension d . Then V is irreducible.

Proof. Let $V_1, \dots, V_n$ be the irreducible components of V . Because V is projective, hence complete, the $f(V_i)$ are closed, and, because f is surjective, W is the union of the $f(V_i)$. Because W is irreducible, at least one V_i must map surjectively onto W , and fix such a i . Denote

$$U_i = V_i \setminus \bigcup_{j \neq i} V_j,$$

which is a nonempty open subset of V_i . Notice that U_i is dense in V_i , and $f_i = f|_{V_i}: V_i \rightarrow W$ is surjective, so that $f(U_i)$ is dense in W . For $y \in W$,

$$f^{-1}(y) = \bigcup_j (f^{-1}(y) \cap V_j),$$

but $f^{-1}(y)$ is irreducible by hypothesis, so $f^{-1}(y) \subset V_j$ for some j . If $y \in f(U_i)$, this cannot hold for $j \neq i$; therefore, $f^{-1}(y) \subset V_i$. As $f(U_i)$ is dense in W , it intersects the nonempty open subset of (ii) of the fiber dimension theorem. For y in the intersection $U \cap f(U_i)$, the fiber

$$f^{-1}(y) = f_i^{-1}(y) \subset V_i$$

hence has dimension

$$d = \dim f^{-1}(y) = \dim V_i - \dim W.$$

Then, by (i) of the fiber dimension theorem, for each $w \in W$, each irreducible component Z of the fiber of the map $f_i: V_i \rightarrow W$ has dimension at least d . But this fiber is contained in $f^{-1}(w)$, which is irreducible of dimension d . We have then an inclusion of irreducible varieties

$$Z \subset f^{-1}(w)$$

of decreasing dimension, hence $Z = f^{-1}(w)$. We thus have $f^{-1}(w) \subset V_i$ for all $w \in W$. Because $V = \cup_w f^{-1}(w)$, it follows that $V \subseteq V_i$, so $V = V_i$ and we attain that V is irreducible. $\square$

Remark 3.4.7 — Note that this is false if V is not projective: let $V \subset \mathbb{A}^2$ be the union of the hyperbola $V(XY - 1)$ and of the origin $V(X, Y)$: V projects surjectively on $\mathbb{A}^1$, and each fiber is a single point, but V is not irreducible!

Exercise 3.4.8. Let $V \subset \mathbb{P}^n$ be an irreducible projective algebraic variety of dimension $d > 0$.

1. Let:

$$\Sigma = \{(p, q, r) \in V \times V \times \mathbb{P}^n : p \neq q \text{ and } r \in L_{pq}\}$$

and let $\bar{\Sigma}$ denote the Zariski closure of Σ . You may assume that $\bar{\Sigma}$ is irreducible, and that:

$$\bar{\Sigma} \cap \{(p, q, r) \in V \times V \times \mathbb{P}^n : p \neq q\} = \Sigma$$

Prove that the projection $\pi_{1,2}: \bar{\Sigma} \rightarrow V \times V$ onto the first two factors is surjective.

2. By applying the fiber dimension theorem to the projection $\pi_{1,2}: \bar{\Sigma} \rightarrow V \times V$, show that we have an inequality $\dim(\bar{\Sigma}) \leq 2d + 1$.

3. Let

$$S = \bigcup_{\substack{(p,q) \in V \times V \\ p \neq q}} L_{pq}$$

and let π_3 denote the projection onto the last factor $\bar{\Sigma} \rightarrow \mathbb{P}^n$. Prove that $\pi_3(\Sigma) = S$ and deduce that S is contained in a closed subset of $\mathbb{P}^n$ of dimension at most $2d + 1$.

4. Deduce that if $2d + 1 < n$, then there exist a point $r \in \mathbb{P}^n$ such that every line through r intersects V in at most one point.

5. Of the previous four parts, which remain true and which fail if we have $d = 0$?

Solution: 1. $\bar{\Sigma}$ is closed in the projective variety $V \times V \times \mathbb{P}^n$. Therefore, it is a projective variety in its own right, and so it is complete. Since the projection $\pi_{1,2}$ is clearly regular, the image is closed. Let $(p, q) \in V \times V$. If $p \neq q$, then $(p, q, p) \in \Sigma \subset \bar{\Sigma}$. Thus, (p, q) is in the image of $\pi_{1,2}: \bar{\Sigma} \rightarrow V \times V$. That is, its image contains the open dense subset $\{(p, q) \in V \times V : p \neq q\}$ and is closed, hence this map is surjective. Note that this set is open dense because it is open; it is the complement of the diagonal map.

2. If $(p, q) \in V \times V$, then $\pi_{1,2}^{-1}(p, q)$ is (isomorphic to) the line $L_{p,q}$, which is of dimension 1. Therefore, the fiber dimension theorem tells us that $\dim L_{p,q} \geq \dim \bar{\Sigma} - \dim(V \times V)$. In other words,

$$\dim \bar{\Sigma} \leq 2d + 1.$$

Note that if we consider the open subset $(V \times V) \setminus \Delta$ and consider the fiber of points inside that, these have 1-dimensional fibers as they are lines. Therefore, we actually attain equality. Thus, $\dim \Sigma = 2d + 1$.

3. The surjectivity of π_3 is almost immediate. We have that $r \in S$ if and only if there is $(p, q) \in V \times V$ with $p \neq q$ such that $r \in L_{p,q}$. This means that we have a preimage of r : $r = \pi_3(p, q, r)$ with $(p, q, r) \in \Sigma$. Thus, $\pi_3(\Sigma) = S$. In particular, $S \subset \pi_3(\bar{\Sigma})$; we show this is the closed subset of $\mathbb{P}^n$ we were looking for. As for the bound on the dimension,

$\pi_3: \bar{\Sigma} \rightarrow \pi_3(\bar{\Sigma})$ is a dominant map between irreducible quasi projective varieties, so the image has dimension at most $\dim \bar{\Sigma}$, which has dimension at most $2d + 1$.

4. If $2d + 1 < n$, it follows that S is a proper subset of $\mathbb{P}^n$. Therefore, there exists a point $r \in \mathbb{P}^n \setminus S$. Now, by definition of S , every line through r intersects V in at most one point. To prove this, suppose not; that there was a line such that it intersected V in points p, q . Then, $r \in L_{p,q}$, contradicting that $r \notin S$.

5. If $d = 0$, then by irreducibility of V , we have that V is just a point. $\Sigma = \bar{\Sigma}$ is empty, so part 1. fails. It is still true that $\dim(\bar{\Sigma}) \leq 1$, so 2. remains true. Similarly, S is also empty, so 3. remains true. Finally, every $r \in \mathbb{P}^n \setminus V$ has the desired property of part 4. $\square$

Exercise 3.4.9. Let V, W be irreducible quasi projective varieties. Prove that $\dim(V \times W) = \dim V + \dim W$ using the fiber dimension theorem.

Solution: Consider the projection $\pi: V \times W \rightarrow V$. We note that products of irreducible varieties is irreducible, and π is a surjective regular map, so is dominant. For all $v \in V$, $\pi^{-1}(v) = \{v\} \times W \cong W$. Then, $\dim \pi^{-1}(v) = \dim(V \times W) - \dim(V)$, so we find that

$$\dim(V \times W) = \dim V + \dim W.$$

$\square$

Exercise 3.4.10.

1. Show that all the regular maps from $\mathbb{P}^2$ to $\mathbb{P}^1$ are constant.
2. Using (1) or using another method, show that the regular maps from $\mathbb{P}^n$ to $\mathbb{P}^1$ are constant.

Solution: 1. Let ϕ be our map, and suppose it is nonconstant. $\mathbb{P}^2$ is complete, so $\text{Im} \phi$ is closed in $\mathbb{P}^1$, so is a projective subvariety of $\mathbb{P}^1$. Since ϕ is nonconstant and is closed in $\mathbb{P}^1$ it is either a finite set of points of $\mathbb{P}^1$. However, by irreducibility of $\mathbb{P}^2$, it follows that $\text{Im} \phi$ cannot be a finite set of points because it would be constant then, contradicting that ϕ is nonconstant. Thus, ϕ is actually surjective. Thus, $\phi: \mathbb{P}^2 \rightarrow \mathbb{P}^1$ is dominant. Now take some point in $\mathbb{P}^1$ and consider the fiber of it. Every irreducible component of that fiber will have dimension at least 1. Therefore, if we take distinct $a, b \in \mathbb{P}^1$, then their fibers should be disjoint. However, because these are two curves such that $1 + 1 \geq 2$, it follows that by Theorem 3.2.6, the intersection is nonempty. This is a contradiction.

2. We can generalize this argument for $\mathbb{P}^n \rightarrow \mathbb{P}^1$. Suppose we have a nonconstant regular map from $\mathbb{P}^n \rightarrow \mathbb{P}^1$. By completeness of $\mathbb{P}^n$ and irreducibility, we find that the regular map is surjective, thus dominant. Further any irreducible component of a fiber of a point in $\mathbb{P}^1$ has dimension at least $n - 1$. Now, if take two distinct points in $\mathbb{P}^1$, then their fibers should be disjoint. However, $(n - 1) + (n - 1) = 2n - 2 \geq n$ for $n \geq 2$, so these irreducible components have nonempty intersection. This is a contradiction. $\square$

Exercise 3.4.11. Prove that every regular map from $\mathbb{P}^2 \rightarrow \mathbb{P}^2$ is constant or surjective.

Solution: Let ϕ be our map, and suppose it's not surjective. Let $p \notin \text{Im} \phi$ then. We may assume that $p = [0 : 0 : 1]$. Then, the rational map $\pi: \mathbb{P}^2 \rightarrow \mathbb{P}^1$ defined by $[x : y : z] \mapsto [x : y]$ is defined everywhere except p , so $\pi \circ \phi$ is a regular map from $\mathbb{P}^2 \rightarrow \mathbb{P}^1$. Thus, $\pi \circ \phi$ is constant, so let q be in the image of this composition. Then, $\text{Im} \phi \subseteq \pi^{-1}(q) \cup \{p\}$, and the latter is isomorphic to a line, or $\mathbb{P}^1$. Then, ϕ is a regular map from $\mathbb{P}^2$ to $\mathbb{P}^1$ again, so is constant. $\square$

§3.5 Parameter Spaces

§3.5.i Universal Family of Hypersurfaces

The fiber dimension theorem is particularly useful when applied to *families of algebraic varieties* and *parameter spaces*. These are a powerful feature of algebraic geometry: often we can consider some collection of algebraic varieties, and construct another algebraic variety which has one point for each variety in the collection. We may also be able to fit all the varieties of the collection together into a single big algebraic variety. This is different from other forms of geometry, where a family of objects rarely forms an object of the same type.

Definition 3.5.1. Let B be a quasi-projective variety. A **family of projective algebraic subvarieties over B** is a Zariski closed subset $\mathcal{V} \subseteq B \times \mathbb{P}^n$. For each $b \in B$, we write $\mathcal{V}_b = \{x \in \mathbb{P}^n : (b, x) \in \mathcal{V}\}$ and call this a **fiber** of $\mathcal{V}$. The set B is called the **base** or **parameter space** of the family.

Remark 3.5.2 — Note that the fiber of $\mathcal{V}$ is just the fiber $\pi^{-1}(b)$, where $\pi: \mathcal{V} \rightarrow B$ is the projection.

An Example with A Hypersurface

This definition might seem rather abstract; to give some idea of what is going on, we will look at a simple example (we will see some more complex examples later).

A hypersurface of degree d in $\mathbb{P}^n$ means the zero set of a non-zero homogeneous polynomial in $k[X_0, \dots, X_n]$ of degree d . These polynomials form a vector space of dimension $\binom{n+d}{d}$, which we denote as $P_{n,d}$. One basis is given by

$$X_0^{i_1} X_1^{i_1} \dots X_n^{i_n},$$

where $\sum_j i_j = d$. Alternatively, we say that a basis is given by multi-indices $I = (i_0, i_1, \dots, i_n)$, with $|I| = \sum_j i_j = d$.

Now, note that if one homogeneous polynomial is a scalar multiple of another, then they define the same hypersurface. Hence we get a hypersurface H_a associated with each point $a \in \mathbb{P}^N$, where $N = \binom{n+d}{d} - 1$. (The homogeneous coordinates of a form the coefficients of the polynomial defining H_a .) Then, we have a map

$$\mathbb{P}(P_{n,d}) \rightarrow \{\text{hypersurfaces of degree } d \text{ in } \mathbb{P}^n\},$$

where $N = \binom{n+d}{d} - 1$.

Remark 3.5.3 — We remark two advantages of doing this:

1. This map is surjective, but it's not quite injective (however, it is injective if we restrict to irreducible polynomials).
2. By using the projective base $P_{n,d}$ instead of the affine base $V_{n,d}$, we can take advantage of properties like completeness.

We now ask why the N from the Veronese embedding appear. We note that

$$\mathbb{P}^N \cong \{\text{hyperplanes in } \mathbb{P}^N\} \hookrightarrow \{\text{degree } d \text{ hypersurfaces in } \mathbb{P}^n\}.$$

We note that the first isomorphism just sends coordinates to the coefficients of the hyperplane, and the inclusion is by intersecting with the Veronese variety.

Claim 3.5.4. The hypersurfaces form a family in $\mathbb{P}^N \times \mathbb{P}^n$.

The space we are interested in is

$$\{(a, x) \in \mathbb{P}^N \times \mathbb{P}^n : x \in H_a\}.$$

The fiber above $a \in \mathbb{P}^N$ is the hypersurface $H_a \subseteq \mathbb{P}^n$.

Claim 3.5.5. This set is defined by the vanishing of a degree $(1, d)$ bihomogeneous equation. Namely,

$$\sum_I a_I X^I = 0.$$

This is necessarily the equation we require because for $x \in H_a$, since H_a is the vanishing locus of $\sum_I a_I X^I$. The bihomogeneous degree is clearly $(1, d)$ as we can calculate just by rescaling in the coefficients or X . We have that

$$\mathcal{H} := V\left(\sum_I a_I x^I\right) \subseteq \mathbb{P}^N \times \mathbb{P}^n.$$

We call this the **universal family of degree d hypersurfaces** in $\mathbb{P}^n$, and is denoted $\mathcal{H}$.

Remark 3.5.6 — We can think of $P_{n,d} \cong \mathbb{P}^N$ as the parameter space for hypersurfaces of degree d in $\mathbb{P}^n$.

Remark 3.5.7 (Aside) — The word ‘universal’ here is related to the fact that every hypersurface of degree d appears as a fiber in this family, and most of them only appear once (if we work with schemes instead of varieties, then each hypersurface will really appear exactly once). However a rigorous definition of what it means for a family to be ‘universal’ is more subtle than this, and too complicated to define in this course (it involves the notion of a ‘flat family of schemes’).

One of the benefits of parameter spaces and families of varieties is that they give us a way of talking about all varieties with some particular property at once. If we take a family

$$\mathcal{V} \subseteq B \times \mathbb{P}^n$$

and consider the subset of fibers which satisfy an interesting geometric condition, then very often the corresponding set of points in the parameter space

$$\{b \in B : \mathcal{V}_b \text{ satisfies given condition}\}$$

is an open or closed subset of B .

Remark 3.5.8 — We may also write For a polynomial $f \in \mathbb{P}(P_{n,d})$, let's write $[f]$ for the corresponding point in $P_{n,d}$. Using the basis for $\mathbb{P}(P_{n,d})$ which consists of the monomials X^I with $|I| = d$, we see that the homogeneous coordinates of $[f] \in \mathbb{P}(P_{n,d})$ are given by the coefficients of f .

Example 3.5.9

As a simple example, if we fix a point $x \in \mathbb{P}^n$, then the set

$$\{b \in B : x \in \mathcal{V}_b\}$$

is a closed subset. This is the image of the closed set

$$(B \times \{x\}) \cap \mathcal{V} \subseteq B \times \mathbb{P}^n$$

under the projection

$$p_1 : B \times \mathbb{P}^n \rightarrow B,$$

so it is closed because $\mathbb{P}^n$ is complete.

Example 3.5.10

The set

$$\{[f] \in P_{n,d} : f \text{ is irreducible}\}.$$

is an open set, and so is the set

$$\{[f] \in P_{n,d} : f \text{ generates a radical ideal}\}.$$

This is in the below exercise.

§3.5.ii Dimension Counting

An important use of families of varieties, and the fact that the family is itself a variety, is that we can calculate the dimension of the parameter space, or of interesting subsets of it, using the fiber dimension theorem. By doing this, we can show that certain sets are empty/non-empty/finite/infinite/equal or not equal to the entire parameter space.

A Worked Example

Consider the intersection of $n + 1$ hypersurfaces in $\mathbb{P}^n$. From our earlier discussions of dimension, we expect that usually such an intersection should be empty (because $n + 1 > n$), but of course sometimes it will be non-empty. By counting dimensions of parameter spaces, we can be more specific about how often ‘sometimes non-empty’ occurs: we will prove that the subset of the parameter space where this intersection is non-empty is a closed subset, and then we will compare its dimension with the dimension of the entire parameter space.

What is the appropriate parameter space? In order to get parameter spaces for hypersurfaces, we have to fix the degree of the defining polynomial (the dimension of $P_{n,d}$ increases as d grows, so if there was a single parameter space for all hypersurfaces in degree

n it would have to be infinite-dimensional, which doesn't fit within our notion of varieties).

For simplicity, we will assume that all of the $n + 1$ hypersurfaces we are intersecting have the same degree d . We are looking at sequences of $n + 1$ hypersurfaces, so the parameter space we need is $(P_{n,d})^{n+1}$. (We could more generally pick a sequence of positive integers $d_0, \dots, d_n$ and look at intersections of the form $H_0 \cap \dots \cap H_n$ where H_0 has degree d_0 , H_1 has degree d_1 , etc. Because we have fixed the degree of each H_i , we can still form a parameter space for such sequences: it would be the product $P_{n,d_0} \times P_{n,d_1} \times \dots \times P_{n,d_n}$. The example below would still work, but would sometimes get a little more complicated.) The subset of the parameter space which we want to study is

$$S = \{(a_0, \dots, a_n) \in (P_{n,d})^{n+1} : \bigcap_{i=0}^n H_{a_i} \neq \emptyset\}.$$

The algebraic varieties we are interested in (intersections of $n + 1$ hypersurfaces) should form a family over $(P_{n,d})^{n+1}$. More precisely, we want a family of algebraic varieties over $(P_{n,d})^{n+1}$ such that the fiber above $(a_0, \dots, a_n)$ is $\bigcap_{i=0}^n H_{a_i}$. We can define this family by

$$\Sigma := \{(a_0, \dots, a_n, x) \in (P_{n,d})^{n+1} \times \mathbb{P}^n : x \in H_{a_i} \text{ for all } i\}.$$

For each i , the condition $x \in H_{a_i}$ is given by a polynomial, so Σ is an intersection of closed subsets, so Σ is a closed subset of $(P_{n,d})^{n+1} \times \mathbb{P}^n$. Thus it is a family of algebraic varieties as defined above.

The idea is now to study $\dim S$ using the fiber dimension theorem.

Let π_1 denote the projection

$$\Sigma \subseteq (P_{n,d})^{n+1} \times \mathbb{P}^n \rightarrow (P_{n,d})^{n+1}.$$

By definition,

$$\pi_1^{-1}(a_0, \dots, a_n) \cap \Sigma \neq \emptyset$$

if and only if $(a_0, \dots, a_n) \in S$. In other words, $S = \pi_1(\Sigma)$. Therefore, because $\mathbb{P}^n$ is complete and $P_{n,d} \cong \mathbb{P}^N$ (so complete), we have that the product $(P_{n,d})^{n+1} \times \mathbb{P}^n$ is complete, and Σ is a closed subset of a complete variety, so complete; therefore, $S = \pi_1(\Sigma)$ is a closed subset of $(P_{n,d})^{n+1}$.

Remark 3.5.11 — Why are we focusing on the set where $\bigcap_{i=0}^n H_{a_i}$ is non-empty rather than the set where it is empty? Because closed sets are usually more interesting than open sets, e.g. it makes sense to ask what is the dimension of a closed subset, while the dimension of an open set is always the same as the dimension of the space it is contained in.

We can work this out by two applications of the fiber dimension theorem: first we apply it to the projection $\Sigma \rightarrow \mathbb{P}^n$ to find $\dim \Sigma$, then we apply it to the projection $\Sigma \rightarrow S$ to find $\dim S$. The reason we can do this is that we know the dimension of $\mathbb{P}^n$ and we can work out the dimensions of the fibers of both projections from Σ .

To compute $\dim \Sigma$, we will apply the fiber dimension theorem to the projection $p: \Sigma \rightarrow \mathbb{P}^n$. This map is surjective: for any $x \in \mathbb{P}^n$, we can pick $a \in P_{n,d}$ such that

$x \in H_a$ and then $(a, \dots, a, x) \in p^{-1}(x) \subseteq \Sigma$. The fibers are

$$\begin{aligned} p^{-1}(x) &= \{(a_0, \dots, a_n) \in (P_{n,d})^{n+1} : x \in H_{a_i} \text{ for all } i\} \\ &= (\{a \in P_{n,d} : x \in H_a\})^{n+1}. \end{aligned}$$

Thus

$$\dim p^{-1}(x) = (n+1) \dim \{a \in P_{n,d} : x \in H_a\}.$$

In order to calculate $\dim \{a \in P_{n,d} : x \in H_a\}$, make a linear change of coordinates so that $x = [0 : \dots : 0 : 1]$. This change of coordinates won't change the dimension of $\{a \in P_{n,d} : x \in H_a\}$, so it suffices to work out the dimension for the special case of $[0 : \dots : 0 : 1]$.

Now $[0 : \dots : 0 : 1] \in H_a$ if and only if the homogeneous polynomial f vanishes at $[0 : \dots : 0 : 1]$, where $a = [f]$. The value of f at $[0 : \dots : 0 : 1]$ is just the X_n^d coefficient of f . Thus

$$[0 : \dots : 0 : 1] \in H_{[f]} \iff \text{the } X_n^d \text{ coefficient of } f \text{ is zero.}$$

In other words, $\{a \in P_{n,d} : x \in H_a\}$ is a subspace of $P_{n,d}$ defined by one linear equation, so

$$\dim \{a \in P_{n,d} : x \in H_a\} = \dim P_{n,d} - 1 = N - 1$$

where $N = \binom{n+d}{d} - 1$. (Alternatively, we could have seen this without reducing to the case $x = [0 : \dots : 0 : 1]$ by observing that the condition $x \in H_{[f]}$ is a single linear condition on the coefficients of f , as you see just by expanding out $f(x_0, \dots, x_n) = 0$.) Therefore, for every $x \in \mathbb{P}^n$,

$$\dim p^{-1}(x) = (n+1)(N-1).$$

In particular $\mathbb{P}^n$ is irreducible, p is surjective, and the fibers of p are all irreducible of the same dimension: by Proposition 3.4.6, Σ is irreducible. We can apply the fiber dimension theorem (Theorem 3.4.3) to get

$$\dim \Sigma = \dim \mathbb{P}^n + \dim p^{-1}(x) = n + (n+1)(N-1) = N(n+1) - 1.$$

By part (ii) of the fiber dimension theorem, this holds for all x in some non-empty open subset of $\mathbb{P}^n$. It doesn't matter which x we choose because we showed that all the fibers have the same dimension.

Now to compute $\dim S$, we will apply the fiber dimension theorem to the projection

$$q: \Sigma \rightarrow S.$$

This map is surjective by construction. This time, the fibers do not all have the same dimension, but the minimum dimension of the fibers is zero – to see that there are fibers of dimension zero, observe that for suitable choices of $n+1$ homogeneous polynomials of degree d , the intersection of the corresponding hypersurfaces is finite and non-empty for example:

$$f_i = X_i^d \quad \text{for } 0 \leq i \leq n-1, \quad f_n = X_0^d$$

(repeating f_0) have the unique common solution

$$[0 : \dots : 0 : 1].$$

Therefore the fiber dimension theorem implies that

$$\dim S = \dim \Sigma - 0 = N(n+1) - 1.$$

We recall how this follows from the fiber dimension theorem: by part (i), the fact that there is just a single fiber of dimension 0 implies that

$$0 \geq \dim \Sigma - \dim S.$$

By part (ii) of the fiber dimension theorem, there exists an open subset $U \subseteq S$ on which

$$\dim q^{-1}(s) = \dim \Sigma - \dim S.$$

But since $\dim q^{-1}(s)$ can never be negative, this forces

$$0 \leq \dim q^{-1}(s) = \dim \Sigma - \dim S.$$

Combining these gives $\dim S = \dim \Sigma$ as we claimed. In particular, we have

$$\dim S = \dim(P_{n,d})^{n+1} - 1.$$

This means that it is only slightly unusual for $n + 1$ hypersurfaces to have non-empty intersection: this subset of the parameter space has dimension only 1 less than the entire parameter space.

As an application of this calculation, we see that if S is a hypersurface in $(P_{n,d})^{n+1}$, and therefore it is defined by a single polynomial

$$F \in k[X_{iI} : 0 \leq i \leq n, 0 \leq I \leq N].$$

In other words, there exists some polynomial F such that, when we evaluate it at the coefficients of $n + 1$ homogeneous polynomials $f_0, \dots, f_n$ of degree d , we get zero if and only if the intersection $\bigcap_{i=0}^n H_{[f_i]}$ is non-empty.

Exercise 3.5.12. Let $V_{n,d}$ denote the vector space of homogeneous polynomials of degree d in $k[X_0, \dots, X_n]$ and let $P_{n,d}$ denote the projective space attached to $V_{n,d}$.

1. Show that, for each e such that $0 \leq e \leq d$, multiplication of polynomials $V_{n,e} \times V_{n,d-e} \rightarrow V_{n,d}$ induces a regular map:

$$\mu_{n,d,e} : P_{n,e} \times P_{n,d-e} \rightarrow P_{n,d}.$$

2. By applying completeness to $\mu_{n,d,e}$, show that the set:

$$P_{n,d,\text{irr}} = \{[f] \in P_{n,d} : f \text{ is irreducible}\}$$

is an open subset of $P_{n,d}$.

3. Let $n \geq 2$. Show that $P_{n,d,\text{irr}}$ is non-empty. (Easy approach: write down an irreducible homogeneous polynomial of degree d . Harder approach, making use of dimension: calculate the dimension of $P_{n,e} \times P_{n,d-e}$ for each e and compare with $\dim(P_{n,d})$. What goes wrong when $n = 1$ and $d \geq 2$?)
4. By considering maps of the form:

$$P_{n,a} \times P_{n,b} \rightarrow P_{n,d} : ([f], [g]) \mapsto [f^r g]$$

for all triples a, b, r and $ar + b = d$ and $r \geq 2$, show that the set:

$$P_{n,d,\text{rad}} = \{[f] \in P_{n,d} : f \text{ generates a radical ideal}\}$$

is an open subset of $P_{n,d}$.

5. Give an example, for some n, d , of two polynomials f, g such that $[f], [g] \in P_{n,d} \setminus P_{n,d,\text{rad}}$ and f and g define the same hypersurface in $\mathbb{P}^n$, but f is not a scalar multiple of g .

Solution: 1. Multiplication of polynomials is a bilinear map $V_{n,e} \times V_{n,d} \rightarrow V_{n,d}$, so it is bihomogeneous of degree 1 in the variables corresponding to $V_{n,d}$ and $V_{n,e}$. Therefore, it defines a regular map at the level of the projective spaces.

2. $P_{n,e}$ and $P_{n,d-e}$ are both isomorphic to projective spaces determined by their degree, so these are projective varieties. Thus, they are complete and so their product is complete. It follows that the image of $\mu_{n,d,e}$ is closed in $P_{n,d}$. Now, we note that factors of homogeneous polynomials are homogeneous themselves. Therefore, a homogeneous polynomial of degree d is reducible if and only if it is in the image of $\mu_{n,d,e}$ for $1 \leq e \leq d-1$. Focusing on the complement of $P_{n,d,\text{irr}}$ in $P_{n,d}$, we find that it is a finite union of closed subsets. Therefore, it is closed, so $P_{n,d,\text{irr}}$ is open.

3. $x_0 x_1^{d-1} - x_2^d$ is irreducible.

4. We begin by noting that a homogeneous polynomial h generates a radical ideal if and only if each of its irreducible components appears exactly once. This occurs if and only if one cannot write $h = f^r g$ for $r \geq 2$ and f is nonconstant. However, since h is homogeneous, f and g are homogeneous. If they have respective degrees a and b , we must have that $ar + b = d$. Therefore, $[h] \in P_{n,d}$ is in $P_{n,d,\text{rad}}$ if and only if it is not in the image of any map of the form given, for any $a \geq 1, b, r \geq 2$ such that $ar + b = d$. This map is evidently regular, and we know that the domain is complete. Therefore, the image is closed. Since $P_{n,d,\text{rad}}$ is just the complement of a finite union of closed subsets, this is open.

5. For any n and $d \geq 3$, we can take two different linear polynomials u, v . Then, $f = uv^{d-1}$ and $g = u^{d-1}v$ have the same radical, and so they define the same hypersurface in $\mathbb{P}^n$. $\square$

Exercise 3.5.13. What is the parameter space of a hyperplane and a line on it in $\mathbb{P}^3$? Find the dimension of it. More generally, what about the parameter space of a hyperplane and a line on it in $\mathbb{P}^n$?

Solution: Our parameter space is as follows:

$$X := \{(L, H) : H \text{ hyperplane in } \mathbb{P}^3 \text{ and } L \subseteq H \text{ a line}\}.$$

Let us compute $\dim X$ using the fibers. Consider the projection to the second factor. Now the space of hyperplanes in $\mathbb{P}^3$ is determined by $[a_0 : a_1 : a_2 : a_3]$, which is of dimension 3. To determine the dimension of X , we shall determine the dimension of irreducible components of a fiber of a fixed hyperplane H . We note that a hyperplane in $\mathbb{P}^n$ is isomorphic to $\mathbb{P}^{n-1}$, so we want to find the dimension of a line in $\mathbb{P}^{n-1}$. This is just $(n-1)$. In our case, the dimension of a line through H would be 2. Therefore, all fibers have dimension 2. If all fibers are of dimension 2, then certainly on a dense open in the space of hyperplanes, the fiber at each hyperplane is of dimension 2. Therefore, we have that $2 = \dim X - 3$, so $\dim X = 5$.

To generalize what we've done, the space of hyperplanes in $\mathbb{P}^n$ are of dimension n . Now, we must determine the dimension of the space of lines going through a hyperplane. For this we use the fiber dimension theorem. Consider the set

$$Y = \{(p, q) \in H \times H : p \neq q\}.$$

Then, Y is equal to $(H \times H) \setminus \Delta$, where Δ denotes the diagonal, so it is open. Since H is a hyperplane, it is irreducible, so $H \times H$ is irreducible, and Y is open in $H \times H$, so Y is irreducible. Therefore, $\dim Y = \dim H \times H = 2(n-1)$. Then, there is a natural map $\phi: Y \rightarrow \{\text{lines } L \subseteq H\}$. Note that this is a surjective regular map, so the image is irreducible. Then,

$$\phi^{-1}(L) = \{(p, q) \in L \times L : p \neq q\},$$

but this is isomorphic to $(\mathbb{P}^1 \times \mathbb{P}^1) \setminus \Delta$, since L is isomorphic to $\mathbb{P}^1$, so this fiber is irreducible. Therefore, the dimension of this fiber is 2. Since we have constant fibers over the space of lines L contained in H , we may apply the fiber dimension theorem:

$$\dim Y - \dim(\{\text{lines } L \subseteq H\}) = 2,$$

so

$$\dim(\{\text{lines } L \subseteq H\}) = 2(n-1) - 2 = 2n-4.$$

Returning, we have that

$$\dim X - n = \dim(\{\text{lines } L \subseteq H\}) \implies \dim X = 3n - 4.$$

□

4 Tangent Space and Singularities

In the mastery material, we will define the singular points of algebraic varieties. These are points where the variety is not smooth, for example the origin in the curves $y^2 = x^3$ or $y^2 = x(x + 1)$.

To determine whether a point of a variety is singular, we use the tangent space of the variety at that point. This is a generalisation of the tangent line to a smooth curve (i.e. the line whose gradient is equal to the gradient of the curve). We say that the point is singular if the dimension of the tangent space at that point is different to the dimension of the variety (for an irreducible variety – there is a slight complication for reducible varieties).

So first we have to define tangent spaces. We start with the case of affine algebraic varieties, then generalize to quasi-projective varieties. We will finish with the proof that most points of a variety are not singular – more precisely, the singular points form a proper closed subset of the variety.

§4.1 Tangent Space in Coordinates

Let $V \subseteq \mathbb{A}^n$ be an affine algebraic subvariety. We consider V equipped with an embedding in $\mathbb{A}^n$ because we will use coordinates in our first definition of the tangent space, before turning to a coordinate-free definition.

Choose a point $x \in V$. For each polynomial $f \in k[X_1, \dots, X_n]$, let df_x denote the linear map $k^n \rightarrow k$ given by

$$df_x(a_1, \dots, a_n) = \sum_{i=1}^n a_i \frac{\partial f}{\partial X_i} \Big|_x$$

Informally, df_x sends a vector $a \in k^n$ to the ‘directional derivative’ of f at x along that vector. Thus $df_x(a) = 0$ precisely for those directions in which f is stationary at x . Since the polynomials in $I(V)$ are zero on V , we should expect polynomials in $I(V)$ to be stationary along “tangent directions” to V . This motivates the following definition.

Definition 4.1.1. Let $V \subseteq \mathbb{A}^n$ be an affine algebraic subvariety and let $x \in V$. The **tangent space** to V at x is

$$T_x V = \bigcap_{f \in I(V)} \ker df_x \subseteq k^n.$$

In our definition of df_x , we used partial derivatives. Because we are only differentiating polynomials, these can be defined purely algebraically and therefore make sense over any field, even in positive characteristic where there is no analysis.

Remark 4.1.2 — Derivatives of polynomials can behave surprisingly in positive

characteristic: over a field of characteristic p we have

$$\frac{\partial X^p}{\partial X} = pX^{p-1} = 0,$$

so it is possible for a non-constant polynomial to have derivative equal to zero.

Similarly the informal motivation for the definition relied on our intuition from analysis about what happens over $\mathbb{C}$. Even over $\mathbb{C}$, our analytic intuition only works correctly if the variety is non-singular at x .

Proposition 4.1.3

Let $f_1, \dots, f_n$ be a list of polynomials which generate $I(V)$. Then,

$$T_x V = \bigcap_{i=1}^n \ker d(f_i)_x \subseteq k^n.$$

Proof. By definition:

$$T_x V = \bigcap_{f \in I(V)} \ker df_x \subseteq k^n$$

Suppose that $f_1, \dots, f_r$ generates $I(V)$. Clearly, we first have:

$$T_x V \subseteq \bigcap_{i=1}^r \ker d(f_i)_x \subseteq k^n$$

Consider $a \in \bigcap_{i=1}^r \ker d(f_i)_x$. Consider $f \in I(V)$: because $f_1, \dots, f_r$ generates the ideal, there are $g_1, \dots, g_r \in k[X_1, \dots, X_n]$ such that $f = \sum_{i=1}^r g_i f_i$. Then:

$$df_x(a) = \sum_{i=1}^r d(g_i f_i)_x(a) = \sum_{i=1}^r (g_i(x) d(f_i)_x(a) + f_i(x) d(g_i)_x(a)) = 0$$

the last equality holding because $f_i \in I(V)$, hence $f_i(x) = 0$, and $a \in \ker d(f_i)_x$. Hence $a \in \ker df_x$ for any $f \in I(V)$, hence $a \in T_x V$, which shows the reverse inclusion. $\square$

Thus it is straightforward to calculate tangent spaces in practice, from a list of generators for $I(V)$.

Remark 4.1.4 — There is just one thing to be careful of: it is not enough to take a list of functions which define V as an algebraic subvariety but do not generate the ideal (there is an issue with radical closure). For instance, consider $I = x^2$. Then, we should calculate $T_x V$ using $f = x$, not $f = x^2$.

Example 4.1.5

As a very simple example, consider the line L in $\mathbb{A}^2$ defined by the polynomial X . At the point $(0, 0)$, we have:

$$dX_{(0,0)}(a_1, a_2) = a_1 \frac{\partial X}{\partial X} \Big|_{(0,0)} + a_2 \frac{\partial X}{\partial Y} \Big|_{(0,0)} = a_1.$$

Since f generates $I(L)$, we get:

$$T_{(0,0)}V = \ker dX_{(0,0)} = \{(a_1, a_2) \in k^2 : a_1 = 0\}.$$

This is what we should expect: the tangent space to a line is a line in the same direction. However, if we were given the polynomial $g = X^2$, then $V(g)$ is again L . We could try to calculate:

$$d(X^2)_{(0,0)}(a_1, a_2) = a_1 \frac{\partial X^2}{\partial X} \Big|_{(0,0)} + a_2 \frac{\partial X^2}{\partial Y} \Big|_{(0,0)} = 0.$$

So $\ker dg_{(0,0)} = k^2$ which is too big. Thus using polynomials which do not generate the whole ideal of the variety may give the wrong answer for the tangent space.

Example 4.1.6

Consider the graph Γ of a polynomial function $g \in k[x]$. This is an affine algebraic subvariety in $\mathbb{A}^2$ defined by the polynomial

$$f(X, Y) = Y - g(X).$$

Note that f is irreducible because it is monic of degree 1 in Y , so f generates $I(\Gamma)$. We can calculate:

$$df_{(x,g(x))}(a_1, a_2) = -g'(x)a_1 + 1 \cdot a_2.$$

so:

$$T_{(x,g(x))}\Gamma = \ker df_{(x,g(x))} = k(1, g'(x)).$$

Thus the tangent space to Γ at the point $(x, g(x))$ is a line with gradient $g'(x)$, as we expect.

Example 4.1.7

Consider the cuspidal cubic curve defined by the polynomial

$$f(X, Y) = Y^2 - X^3.$$

We have

$$\frac{\partial f}{\partial X} = -3X^2 \quad \frac{\partial f}{\partial Y} = 2Y$$

Hence:

$$\frac{\partial f}{\partial X}\Big|_{(0,0)} = 0 \quad \frac{\partial f}{\partial Y}\Big|_{(0,0)} = 0$$

and so $df_{(0,0)}$ is the zero map. Therefore

$$T_{(0,0)}V = k^2.$$

We can’t see this looking at a picture: this curve appears to have only the X -axis as a tangent line at the origin. This demonstrates that we cannot rely on geometric intuition to calculate the tangent space at singular points: it is necessary to use the algebraic definition.

§4.2 Differentials in Coordinates

We can also consider the differential of a regular map:

Lemma 4.2.1

Let $V \subseteq \mathbb{A}^m$ and $W \subseteq \mathbb{A}^n$ be affine algebraic subvarieties. Let $\varphi: V \rightarrow W$ be a regular map. For any $x \in V$, φ induces a linear map

$$d\varphi_x: T_xV \rightarrow T_{\varphi(x)}W.$$

Proof. Choose polynomials $f_1, \dots, f_n$ such that $\varphi = (f_1, \dots, f_n)$. Define a linear map

$$k^m \rightarrow k^n$$

by the matrix

$$\left(\frac{\partial f_i}{\partial X_j} \Big|_x \right)_{ij}.$$

We define $d\varphi_x$ to be the restriction of this map to T_xV . (Recall that T_xV is a subspace of k^m and $T_{\varphi(x)}W$ is a subspace of k^n .) Using the chain rule for partial derivatives, one can check that:

1. $d\varphi_x$ maps T_xV to $T_{\varphi(x)}W$.
2. $d\varphi_x$ is independent of the choice of polynomials representing φ .

□

Corollary 4.2.2

In particular, if $\varphi: V \rightarrow W$ is an isomorphism, $d\varphi_x$ is an isomorphism of vector spaces.

Proof. This follows by the chain rule. $\square$

Remark 4.2.3 — This provides a nice way to determine if two varieties are isomorphic or not. For instance, it becomes immediate why $V(X^3 - Y^2)$ and $\mathbb{A}^1$ can't be isomorphic.

By the above, one can define $T_x V$ for any V quasi-projective, by taking an affine neighborhood of x in V : by the above arguments, $T_x V$ does not depend on this choice of affine neighborhood.

Corollary 4.2.4

The tangent space $T_x V$ does not depend on the embedding in $\mathbb{A}^n$.

§4.3 Definition of Singular Points

Intuition suggests that, at non-singular points, the dimension of the tangent space should be equal to the dimension of the algebraic variety. The above examples indicate that this breaks down at singular points. This motivates us to define

Definition 4.3.1. Let V be an irreducible quasi-projective variety. $x \in V$ is a **singular point** of V if

$$\dim T_x V \neq \dim V.$$

Remark 4.3.2 — Note that because we've only defined the dimension for irreducible quasi-projective varieties, we can only define this notion for varieties in the same case.

Remark 4.3.3 — Later, we will see that $\dim T_x V \geq \dim V$. This will allow us to say that x is singular if and only if $\dim T_x V > \dim V$.

To understand the definition in the reducible case, note that singularity and smoothness should be a local property. Hence take V a union of two irreducible components of different dimensions, say $V = V_1 \cup V_2$ where $\dim V_1 = 1$ and $\dim V_2 = 2$. For a point $x \in V_1 - V_1 \cap V_2$, whether x is a singular point of V should not care about V_2 ; we ought to compare $\dim T_x V$ against $\dim V_1 = 1$.

In order to fix this and correctly define singular points of reducible algebraic varieties, we introduce a new definition.

Definition 4.3.4. Let V be a quasi-projective variety and let x be a point of V . The **local dimension** of V at x , written $\dim_x V$, is the maximum of the dimensions of those irreducible components of V which contain x .

Example 4.3.5

In our previous example $V = V_1 \cup V_2$, $\dim_x V = 2$ if $x \in V_2$ (including if $x \in V_1 \cap V_2$) while $\dim_x V = 1$ if $x \in V_1 \setminus (V_1 \cap V_2)$.

Now we can define singular points of a quasi-projective varieties by using local dimension.

Definition 4.3.6. Let V be a quasi-projective variety and let $x \in V$. Then x is a **singular point** of V if $\dim T_x V \neq \dim_x V$. Otherwise, x is said to be a **smooth point** of V .

Remark 4.3.7 — We may also say ‘ V is singular at x ’, or conversely, ‘ V is smooth at x ’.

Remark 4.3.8 — We will prove later that $\dim T_x V \geq \dim_x V$ always, so we could equivalently state the definition as: x is a singular point of V if $\dim T_x V > \dim_x V$.

Exercise 4.3.9. Let $V = \bigcup_i V_i$ be a union of irreducible components. Show the following:

1. If x lies in only one irreducible component V_i , then $\dim_x V = \dim V_i$.
2. If $x \in V$ is contained in precisely one of the V_i , then x is singular in V if and only if x is singular in V_i .
3. If $x \in V_i \cap V_j$, for $i \neq j$, then x is a singular point. *Hint: Use Nakayama's lemma.* This is tricky.

Solution: 1. This is immediate.

2. Since x is singular in V is $\dim T_x V \neq \dim_x V$. Now, it is immediate that $\dim_x V = \dim V_i$, so we check that $\dim T_x V = \dim T_x V_i$. It is clear that $V_i \subseteq V$, so $I(V) \subseteq I(V_i)$. Therefore, $T_x V_i \subseteq T_x V$ because $I(V_i)$ is larger than $I(V)$. Now for the converse. $x \notin V_j$, for $j \neq i$, so choose a polynomial s_j for each such j such that $s_j(x) \neq 0$. Now, take $f \in I(V_i)$. Then, $g := f \prod_{j \neq i} s_j \in I(V)$. Therefore, if $u \in T_x V$, then $dg_x(u) = 0$. However, the product rule says that

$$d(f \prod_{j \neq i} s_j)_x(u) = f(x) d(\prod_{j \neq i} s_j)_x(u) + df_x(u) \left(\prod_{j \neq i} s_j \right) (x).$$

This implies that $df_x(u) \left(\prod_{j \neq i} s_j \right) (x) = 0$. Since $\left(\prod_{j \neq i} s_j \right) (x) \neq 0$, it follows that $df_x(u) = 0$, so $u \in T_x V_i$. Since $f \in I(V_i)$ was arbitrary, we are done.

3. □

Exercise 4.3.10. Let $V \subseteq \mathbb{A}^n$ be a reducible affine algebraic variety, with irreducible components V_1 and V_2 . Let $x \in V_1 \cap V_2$. Prove that:

$$T_x V_1 + T_x V_2 \subseteq T_x V$$

Is $T_x V_1 + T_x V_2$ always equal to $T_x V$? Here, $T_x V_1 + T_x V_2$ means the vector space spanned

by $T_x V_1$ and $T_x V_2$.

Solution: We have $I(V) \subseteq I(V_i)$; hence, if $a \in T_x V_i$, it is in the kernel of df_x for any $f \in I(V_i)$, hence a fortiori of $f \in I(V)$, hence $a \in T_x V$. It means that $T_x V_i \subseteq T_x V$ for $i = 1, 2$, hence $T_x V_1 + T_x V_2 \subseteq T_x V$ because $T_x V$ is a vector space.

For the second, the answer is no. Consider

$$V = V(Y(Y - X^2)) \subseteq \mathbb{A}^2,$$

which is the union of the line $V_1 = V(Y)$ and the parabola $V_2 = V(Y - X^2)$, and their intersection $x = (0, 0)$. It's clear that

$$T_x V_1 = \ker \begin{pmatrix} 0 & 1 \end{pmatrix},$$

so the kernel is spanned by $Y = 0$. Analogously,

$$T_x V_2 = \ker \begin{pmatrix} 0 & 1 \end{pmatrix},$$

so the kernel is spanned by $Y = 0$ again. Both $T_x V_1, T_x V_2 \subseteq k^2$ are the x -axis $Y = 0$. But each irreducible factor of $Y(Y - X^2)$ generates $I(V_i)$, and so we have:

$$d(Y(Y - X^2))_{(0,0)} = (Yd(Y - X^2) + (Y - X^2)dY)_{(0,0)} = 0.$$

Hence $T_x V = k^2$, which is strictly larger than $T_x V_1 + T_x V_2$, which is only the x -axis. $\square$

Exercise 4.3.11. Let V, W be affine varieties. Let $v \in V$ and $w \in W$. Prove that $V \times W$ is non-singular at (v, w) if and only if V is non-singular at v and W is non-singular at w .

Solution: Consider a closed embedding $V \subseteq \mathbb{A}^n$ and $f_1, \dots, f_r \in k[X_1, \dots, X_n]$ generating $I(V)$, and a closed embedding $W \subseteq \mathbb{A}^m$ and $g_1, \dots, g_s \in k[Y_1, \dots, Y_m]$ generating $I(W)$. Then $V \times W \subseteq \mathbb{A}^{n+m}$, and $I(V \times W)$ is generated by $\bar{f}_1, \dots, \bar{f}_r, \bar{g}_1, \dots, \bar{g}_s$, the above polynomials seen as polynomials in $k[X_1, \dots, X_n, Y_1, \dots, Y_m]$. Then, for $(x, y) \in V \times W$, $T_{(x,y)}(V \times W) \subseteq k^{n+m}$ is the intersection of the kernel of $d(\bar{f}_i)_{(x,y)}$ and $d(\bar{g}_i)_{(x,y)}$: but the $\bar{f}_i$ respectively $\bar{g}_i$ are constant in the Y respectively X variables, hence:

$$\ker d(\bar{f}_i)_{(x,y)} = \ker d(f_i)_x \times k^m$$

$$\ker d(\bar{g}_i)_{(x,y)} = k^n \times \ker d(g_i)_y$$

which means that:

$$T_{(x,y)}(V \times W) = T_x V \times T_y W.$$

In particular, we have:

$$\dim(T_{(x,y)}(V \times W)) = \dim T_x V + \dim T_y W.$$

If the V_i respectively the W_j are the irreducible components of V respectively W , then the $V_i \times W_j$ are irreducible, and these form the irreducible components of $V \times W$, and $\dim(V_i \times W_j) = \dim V_i + \dim W_j$. For $(x, y) \in V \times W$, $\dim_{(x,y)}(V \times W)$ is then the maximum of the dimension of the $V_i \times W_j$ containing (x, y) , hence such that $x \in V_i$ and $y \in W_j$, hence:

$$\dim_{(x,y)}(V \times W) = \dim_x V + \dim_y W.$$

For any variety Z and $z \in Z$, we have $\dim T_z Z \geq \dim_z Z$ with equality if and only if Z is non-singular at z , hence, from the previous two equations, $V \times W$ is non-singular at (x, y) if and only if V is non-singular at x and W is non-singular at y . $\square$

§4.3.i Singular Locus of a Variety

Definition 4.3.12. Let V be a quasi-projective variety. The set of singular points of V is called the **singular locus** of V and denoted $\text{Sing } V$.

The singular locus turns out to be a proper closed subset of V . We will prove this for irreducible quasi-projective varieties. Generalising to reducible varieties requires the fact that every point which lies in the intersection of two or more irreducible components is singular, and require more algebra (we will prove that in the next section).

Theorem 4.3.13

Let V be an irreducible quasi-projective variety. Then $\text{Sing } V$ is a proper closed subset of V .

We prove this for irreducible V , and the more general step requires Nakayama's lemma. A key intermediate step in the proof of this theorem is the fact that $\dim T_x V \geq \dim_x V$ for every point $x \in V$.

The Singular Locus of a Hypersurface

Proof of Theorem 4.3.13 for Hypersurfaces. We first prove this for affine hypersurfaces, then go to the projective version. Let $V = V(f) \subseteq \mathbb{A}^n$, for f irreducible, be a hypersurface. Since f generates $I(V)$, the tangent space $T_x V$ is just $\ker df_x$. In other words, $T_x V$ is the kernel of a linear map $k^n \rightarrow k$. We now split into cases.

1. x is nonsingular $\iff \dim T_x V = n - 1 \iff df_x$ is a nonzero map.
2. x is singular $\iff \dim T_x V = n \iff df_x$ is the zero map.

Each irreducible component of a hypersurface is of dimension $n - 1$, hence for any $x \in V$, we have $\dim_x V = n - 1$. Hence

$$\text{Sing } V = \{x \in V : df_x = 0\}.$$

Going back to the definition of df_x , we can write this as follows:

$$\text{Sing } V = \left\{ x \in V : \frac{\partial f}{\partial X_i} \Big|_x = 0 \text{ for } i = 1, \dots, n \right\}$$

Note that this may look like the definition of singular points which you have seen before for curves in $\mathbb{A}^2$. For each i , $\frac{\partial f}{\partial X_i}$ is a polynomial. Therefore, for any hypersurface $V \subseteq \mathbb{A}^n$, $\text{Sing } V$ is a closed subset of V .

In the projective case, the analogous calculation holds. Let $V = \tilde{V}(f) \subseteq \mathbb{P}^n$ is a hypersurface cutout by a degree d homogeneous polynomial, then

$$\text{Sing } V = \{x \in V : \frac{\partial f}{\partial X_i} \Big|_x = 0, \text{ for } i = 0, \dots, n\}.$$

To see this, use homogeneous coordinates $[x_0 : \dots : x_n]$ for points of $\mathbb{P}^n$. The sets $\{x_i = 1\} \cong \mathbb{A}^n$, for $i = 0, \dots, n$ form an open affine cover of $\mathbb{P}^n$. For a fixed i , let

$$g(x_0, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

be the polynomial $g = f|_{x_i=1}$. One can easily check that for $j \neq i$ we have $\frac{\partial g}{\partial x_j} = \left(\frac{\partial f}{\partial x_j}\right)\Big|_{x_i=1}$. If we let S denote the set

$$S = \{x \in V : \frac{\partial f}{\partial x_i} = 0 \text{ for } i = 0, \dots, n\},$$

the above calculation shows that

$$S \cap \{x_i = 1\} = \text{Sing } V \cap \{x_i = 1\}.$$

Hence we have $S = \text{Sing } V$.

Going back in the affine case, we want to show that for a hypersurface V , $\text{Sing } V \subsetneq V$. This is a little harder in positive characteristic than in characteristic zero, so first we prove a lemma on derivatives in positive characteristic. Over a field of characteristic p , X^{ip} has derivative zero for any positive integer i . We prove that these span all the polynomials with zero derivative. $\square$

Remark 4.3.14 — A takeaway from this proof is that $\text{Sing } V$ can be found by searching for the points where all partials vanish.

Lemma 4.3.15

Let k be an algebraically closed field of characteristic $p > 0$. Let $f \in k[X_1, \dots, X_n]$ be a polynomial. Then, the following are equivalent.

- (i) $\frac{\partial f}{\partial X_i} = 0$, for all i .
- (ii) $f = g(X_1^p, X_2^p, \dots, X_n^p)$
- (iii) f is a p -th power itself.

Proof. (ii) $\implies$ (i) is clear, and (i) $\implies$ (ii) is just a simple calculation. For (ii) $\iff$ (iii), we first see that (iii) $\implies$ (ii) is clear. Thus, (ii) $\implies$ (iii) needs work, and this is where we use that k is algebraically closed. Every coefficient of f admits a p -th root; thus, f is a sum of p -th powers, so f is a p -th power itself, by Freshman's dream. $\square$

Proposition 4.3.16

If V is a non-empty hypersurface, then $\text{Sing } V$ is strictly contained in V .

Proof. Assume for contradiction that $\text{Sing } V = V$. Then

$$\frac{\partial f}{\partial X_1}, \dots, \frac{\partial f}{\partial X_n}$$

are all zero on V . Since f generates $I(V)$, this implies that f divides $\frac{\partial f}{\partial X_i}$ for each i . But $\frac{\partial f}{\partial X_i}$ has strictly smaller X_i -degree than f . This forces $\frac{\partial f}{\partial X_i} = 0$ for each i (as a polynomial in $k[X_1, \dots, X_n]$).

Over a field of characteristic zero, this implies that f is constant. But then V would be empty, contradicting the hypothesis.

In the case of a field of characteristic $p > 0$, Lemma 4.3.15 tells us that $f = h^p$, for some polynomial h . Therefore, f is reducible, which is a contradiction to f being irreducible. $\square$

The Singular Locus of an Irreducible Variety

We prove here Theorem 4.3.13. Consider V an irreducible quasi-projective variety.

Lemma 4.3.17

Let V be a quasi-projective variety. For any integer d , the set

$$\Sigma_d(V) = \{x \in V : \dim T_x V > d\}$$

is closed.

Proof. $\Sigma_d(V)$ is closed if and only if its intersection with an affine open cover is closed: hence we can consider that $V \subseteq \mathbb{A}^n$ is an affine algebraic subvariety. Choose polynomials $f_1, \dots, f_m$ which generate $I(V)$. Recall that

$$T_x V = \bigcap_{i=1}^m \ker(df_i)_x.$$

In other words, $T_x V$ is the kernel of the matrix

$$M_x = \left(\frac{\partial f_i}{\partial X_j} \Big|_x \right)_{ij}$$

which represents a linear map $k^n \rightarrow k^m$. By the rank-nullity theorem, $\dim T_x V$ is equal to $n - \text{rank} M_x$. Hence

$$\Sigma_d(V) = \{x \in V : \text{rank} M_x < n - d\}$$

By linear algebra, $\text{rank} M_x < n - d$ is equivalent to every $(n - d) \times (n - d)$ submatrix of M_x having determinant zero. The determinant of a submatrix of M_x is a polynomial, hence this gives us polynomial equations defining $\Sigma_d(V)$. $\square$

Lemma 4.3.18

Let V be an irreducible quasi-projective variety. Then the non-singular points of V are dense in V .

Proof. We consider a nonempty affine open subset U of V . By Proposition 1.7.34, U (and then V) is birational to a hypersurface $H \subseteq \mathbb{A}^{d+1}$; in particular, there exists an open subset U of V and an open subset U' of H that are isomorphic. By the previous subsection, the non-singular points of H form a nonempty open subset $H_{ns} \subseteq H$. Note that H is irreducible from the birational equivalence. Since H is irreducible, H_{ns} is dense in H and must intersect U' . Thus $H_{ns} \cap U'$ is open and dense in U' ; hence also in $V_{ns} \cap U$

is also dense in U . However, U is dense on V , so $V_{ns} \cap U$ is dense in V . Then, V_{ns} is a superset of this intersection, so V_{ns} is dense in V . Here, we pass through the following chain of isomorphism:

$$V_{ns} \cap U \cong U_{ns} \cong U'_{ns} \cong H_{ns} \cap U'.$$

□

Remark 4.3.19 — This tells us that the singular points of V are not all of V , since then the set of nonsingular points would be empty, which is certainly not dense. Thus, $\text{Sing } V \subsetneq V$ is attained, but the fact that it's closed is not yet given; this is what is show below.

Remark 4.3.20 — Note that the proof showed that $\text{Sing } V \cap U$ is closed in U .

Lemma 4.3.21

Let V be an irreducible quasi-projective variety. For every $x \in V$,

$$\dim T_x V \geq \dim_x V.$$

Proof. Since V is irreducible, $\dim_x V = \dim V$ for all $x \in V$, so we can work with $\dim V$ instead of $\dim_x V$. Note that this lemma is true for reducible V as well, but we do not have the tools to prove it when $\dim_x V$ is not constant.

Let $d = \dim V$ and consider the set $\Sigma_{d-1}(V)$ as in Lemma 4.3.17. By Lemma 4.3.17, $\Sigma_{d-1}(V)$ is closed. Every non-singular point of V is in $\Sigma_{d-1}(V)$, so Lemma 4.3.18 implies that $\Sigma_{d-1}(V)$ is dense in V . Since $\Sigma_{d-1}(V)$ is closed and dense in V , we conclude that $\Sigma_{d-1}(V) = V$. Thus, we're done by the definition of $\Sigma_{d-1}(V)$. □

Remark 4.3.22 — This is also true for non-irreducible V if we replace $\dim V$ with $\dim_x V$; this part requires Nakayama's lemma.

Theorem 4.3.23

Let V be an irreducible quasi-projective variety. Then $\text{Sing } V$ is a proper closed subset of V .

Proof. By Lemma 4.3.21, $x \in V$ is a singular point if and only if $\dim T_x V > \dim_x V$. Again since V is irreducible, we can replace $\dim_x V$ by $d = \dim V$. Thus:

$$\text{Sing } V = \Sigma_d(V).$$

Thus, Lemma 4.3.17 tells us that $\text{Sing } V$ is closed in V . Lemma 4.3.18 implies that $\text{Sing } V$ is properly contained in V . □

Remark 4.3.24 — The theorem above tells us then that $\dim_x \text{Sing } V < \dim V$, for all $x \in \text{Sing } V$.

Exercise 4.3.25. Let f be a non-zero polynomial in $k[X_1, \dots, X_n]$ with no repeated factors. Let $V \subseteq \mathbb{A}^n$ be the hypersurface defined by f .

1. Let x be a point in V and let u be a vector in k^n . Prove that the polynomial:

$$f(x + Tu) \in k[T]$$

has repeated root at $T = 0$ if and only if $u \in T_x V$.

2. Suppose that $\deg(f) = 2$. Prove that if x is a singular point of V and $L \subseteq \mathbb{A}^n$ is a line through x , then either $L \subset V$ or $L \cap V = \{x\}$.
3. Now let $\deg(f) = 3$. Prove that if V contains two singular points, then it contains the line joining them.

Solution: 1. The Taylor formula gives:

$$\begin{aligned} f(x + Tu) &= f(x) + T \sum_{i=1}^n u_i \frac{\partial f}{\partial X_i} \Big|_x + T^2 g(T) \\ &= T df_x(u) + T^2 g(T) \end{aligned}$$

where we have used the fact that $x \in V$, hence $f(x) = 0$, and the definition of df_x . Then $f(x + Tu)$ has repeated root at $T = 0$ if and only if $u \in \ker df_x$. But f has no repeated factor, hence f generates $I(V)$, and then $\ker df_x = T_x V$.

2. If V is singular at x , then $T_x V = k^n$ since V is a hypersurface. Consider the tangent vector u of the line L . The points of $L \cap V$ are the points $x + tu$ such that $f(x + tu) = 0$. Because $u \in k^n = T_x V$, then $T^2 \mid f(x + Tu)$; but f is of degree 2, hence $f(x + Tu)$ is of degree 2 as a polynomial in T , hence $f(x + Tu) = aT^2$ with $a \in k$. Then:

- Either $a = 0$, hence $f(x + tu) = 0$ for all $t \in k$, hence $L \subset V$.
- Or $a \neq 0$, hence $f(x + tu) = 0$ if and only if $t = 0$, hence $L \cap V = \{x\}$.

3. Suppose that V contains two singular points x, y and let L be the line joining x and y . Then $f|_L$ is a polynomial in one variable of degree at most 3. Since x is singular and V is a hypersurface, it follows that $T_x V = k^n$, because every irreducible component of a hypersurface has dimension $n - 1$. Therefore a vector u parallel to the line L is contained in $T_x V$. Therefore, by part 1, $f|_L$ has a repeated root at x . Similarly, $f|_L$ has a repeated root at y .

Thus the multiplicities of the roots of $f|_L$ add up to at least $2 + 2 = 4$. However, $f|_L$ has degree at most 3, this is only possible if $f|_L = 0$, i.e. $L \subseteq V$. $\square$

Exercise 4.3.26. Let $V \subseteq \mathbb{P}^4$ be a hypersurface. Suppose that V contains the 2-dimensional linear subspace L defined by the equations

$$X_3 = X_4 = 0.$$

Find two homogeneous polynomials g, h such that

$$\text{Sing } V \cap L = \{x \in L : g(x) = h(x) = 0\}.$$

Deduce that L contains a singular point of V .

Solution: Write $V = V(f)$, for some homogeneous polynomial f . Then, we note that since $L \subseteq V$, $f = X_3g + X_4h$. We note that $\frac{\partial f}{\partial X_i} = X_3\frac{\partial g}{\partial X_i} + X_4\frac{\partial h}{\partial X_i}$, for $i = 0, 1, 2$, which vanishes on L . Therefore, $\text{Sing } V \cap L$ is determined by $\frac{\partial f}{\partial X_3}$ and $\frac{\partial f}{\partial X_4}$. These are the polynomials desired in the problem. Now, we use Theorem 3.2.14 to deduce that the intersection is nonempty. We have that this intersection is a closed subset of $\mathbb{P}^2$. Thus, we can let $C = V(g)$ and intersect it with the hypersurface defined by $V(h)$, which is nonempty. We note that $V(h)$ certainly doesn't contain $V(g)$ because their defining polynomials are different. $\square$

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